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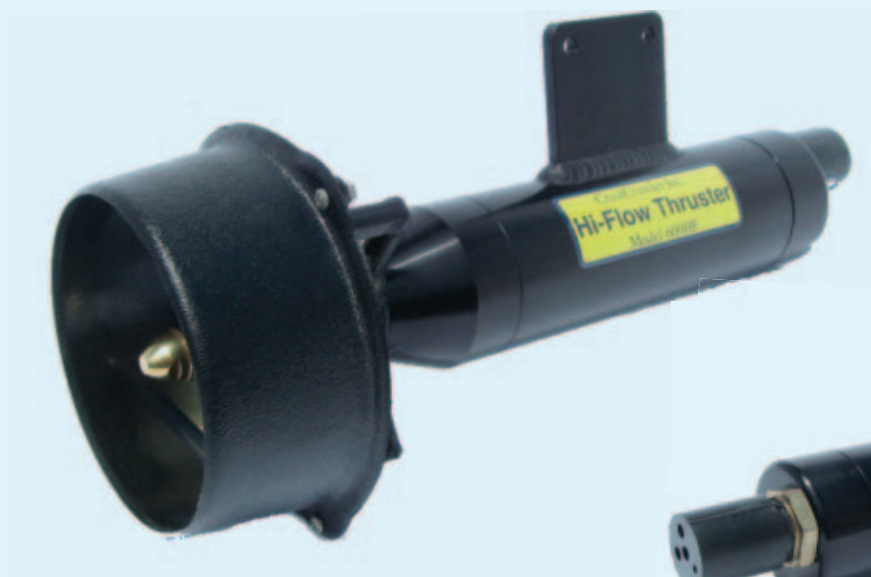
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The Combat Zone...



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SERVO Magazine (ISSN 1546-0592/CDN Pub Agree#40702530) is published monthly for \$24.95 per year by T & L Publications, Inc., 430 Princeland Court, Corona, CA 92879. PERIODICALS POSTAGE PAID AT CORONA, CA AND AT ADDITIONAL ENTRY MAILING OFFICES. POSTMASTER: Send address changes to **SERVO Magazine, P.O. Box 15277, North Hollywood, CA 91615** or Station A, P.O. Box 54, Windsor ON N9A 6J5; cpcreturns@servomagazine.com

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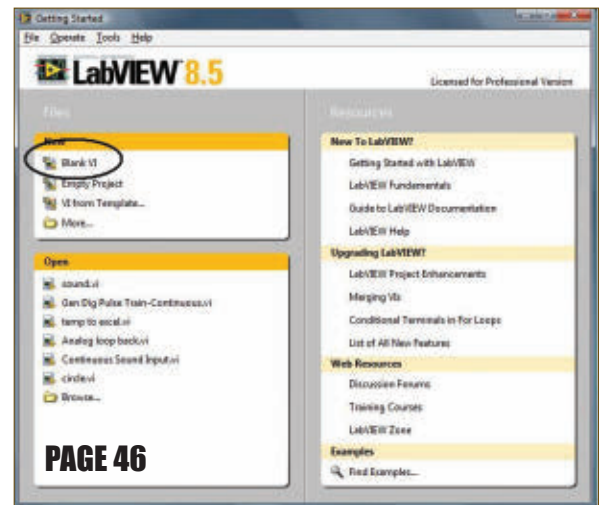
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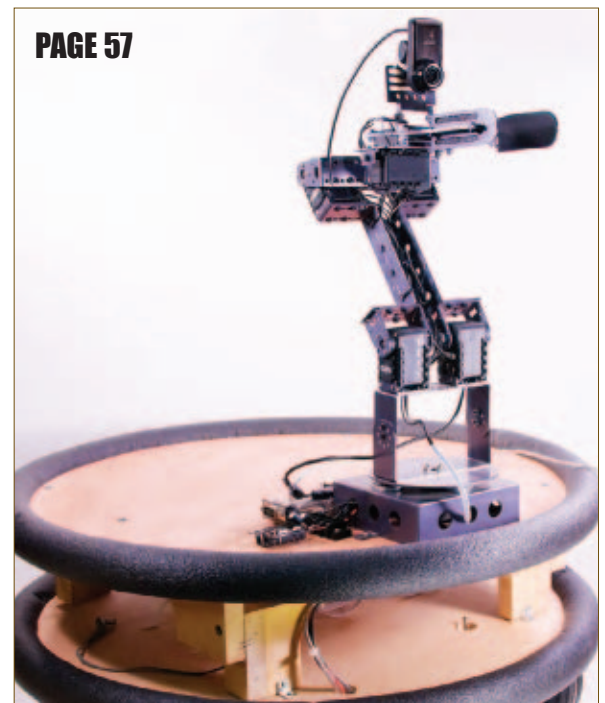
Give your TankBot the freedom to explore the environment all on its own.



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Mind / Iron



by Bryan Bergeron, Editor

Unintended Consequences

In the world of robotics, advances in one application area often have unintentional, beneficial consequences in other areas, regardless of whether the initial application is a commercial success. Case in point — the Japanese robotic strawberry picker, touted as one of the prominent engineering failures of 2008 (www.spectrum.ieee.org/jan09/7130). The autonomous robot is considered an economic failure by some because it's too expensive, too slow, and is restricted to a specially configured hothouse environment.

Despite this criticism, consider the engineering challenges that the developers had to address, and how the solutions might be applied to other fields. First, there's the image recognition challenge of identifying the strawberry that's red enough — but not overly so — for picking. Then, there's the 3D manipulation challenge of plucking a berry from the vine and placing it in a shipping container — all without crushing or even bruising the flesh. Consider that a target berry could be located in 3D space anywhere from the top to the bottom of a vine, on the far or near side of the main vine.

To appreciate the technical hurdles the developers had to successfully overcome, consider what you would need to replicate a robotic strawberry picker using off-the-shelf components. First, there's a moveable platform with control circuitry, battery charger, and sensors to detect (for example) when people or

animals are in harm's way. This could be a modest self-contained robot using the new heavy duty 12V motors from Parallax (www.parallax.com).

Then, there's the vision recognition software and hardware — a couple cameras and a laptop running the appropriate software would do. The multi-axis picking arm is a major decision. At the low end, I'd consider an aluminum CrustCrawler arm (www.crustcrawler.com) with pressure sensors. At the high end, I'd shop for a used commercial arm in the \$15K-\$20K range. For pressure sensors, my first choice is the new line of highly linear, 0 - 1,500 g piezoresistive force sensors from Honeywell (sensing.honeywell.com).

Given the relatively high cost of transporting and maintaining human



The gripper of my CrustCrawler AX-12 Smart Arm, posing as a berry picker.

Published Monthly By
T & L Publications, Inc.
430 Princeland Ct., Corona, CA 92879-1300
(951) 371-8497
FAX **(951) 371-3052**
Webstore Only **1-800-783-4624**
www.servomagazine.com

Subscriptions
Toll Free **1-877-525-2539**
Outside US **1-818-487-4545**
P.O. Box 15277, N. Hollywood, CA 91615

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laborers, robotic fruit and vegetable harvesters may be limited to future space craft and lunar or Martian bases. But before then, the advances in robotic technology made possible by funding from Japan's Institute of Agricultural Machinery's Bio-oriented Technology Research Advancement Institution (IAM-BRAIN) will certainly help propel the field forward to applications that the original developers likely never envisioned.

For example, could the image recognition and 3D arm movements be adapted to identifying and extracting shrapnel from a wounded soldier? Splinters from a youngster's hand? Cancerous cells in a patient's liver? What applications in your domain could benefit from the lessons learned and disseminated by the developers of strawberry pickers? **SV**



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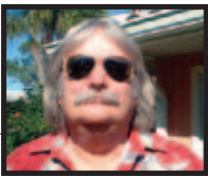


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Automated Blade Inspection



*A robot inspects a wind energy converter's rotor blades.
©Fraunhofer IFF.*

Back when oil was selling for \$4 per gallon and US banks still appeared to have money, T. Boone Pickens' announced a \$10 billion wind farm project that would have added 2,700 wind turbines to the grid and 4,000 MW of generation capacity. The old skinflint has since changed his mind, but as of the end of 2007, the USA already had nearly 17,000 MW of installed wind capacity, ranking it no. 2 in the world. Nr. eine was Germany, with better than 22,000 MW, so it is perhaps not surprising that the first robot designed specifically to inspect wind generation equipment came from engineers at Deutschland's Fraunhofer Institute for Factory Operation and Automation.

The new RIWEA autonomously pulls itself up a rope and then inspects rotor blades centimeter by centimeter, detecting cracks and delaminations caused by inertial forces, erosion, collisions with condors and ultralight aircraft, etc. In basic operation, the bot radiates heat into the blade surface and uses a high-res thermal camera to record temperature patterns and detect flaws. It also carries an ultrasonic system to spot things missed by the thermal equipment, making it much more accurate than a human eye.

According to Fraunhofer, RIWEA can perform its job on any wind energy converter, regardless of whether it

is large or small, on land or offshore. This is no small feat, given that rotors can span up to 200 ft (60 m). For details, visit www.iff.fraunhofer.de.

Pole Dancers to Replace Construction Workers



VaTech's pole-climbing serpentine robot. Courtesy of Virginia Tech.

Winning both the grand prize at the 2008 International Capstone Design Fair and this issue's Bloated, Labyrinthine Acronyms for Hype (BLAH) award are three pole-climbing inspection bots from Virginia Tech's Robotics and Mechanisms Laboratory (www.me.vt.edu/romela). The HyDRAS (Hyper-redundant Discrete Robotic Articulated Serpentine) Ascent I and II, along with the CIRCA (Climbing Inspection Robot with Compressed Air) raked in a cool 1,000,000 South Korean won (\$737.21 as of this writing) for the former honor and the equivalent of nearly 1,000,000,000 Zimbabwean

dollars for the latter.

The autonomous slitherers are "designed to climb scaffolding and buildings by wrapping around a pole or beam and then rolling upward via an oscillating joint motion. Using built-in sensors and cameras, the robots would then inspect the structures or handle other dangerous tasks now done by humans."

The overall aim is to save construction workers from dangerous tasks that can lead to precipitous dives from scaffolding and other raised structures. Such falls led to deaths in 2006, according to the US Bureau of Labor Statistics.

Robotic Deer Bags Hunters

Robotic decoys have been around for a few years, but if you're a gun-toting game hunter, be warned that they are becoming widely used for attracting more than just deer. Back in January, for example, Officer Jeff Babauta of the Florida Fish and Wildlife Conservation Commission (myfwc.com) set up a robotic deer somewhere in Leon County just before dawn. The decoy soon was spotted by a poacher who made two basic mistakes (in addition to being unable to tell the fake deer from the real thing): firing two shots at the



Reclining robotic deer moves head and tail. Courtesy of Custom Robotic Wildlife.

deerbot and commanding his two Labradors to attack Babauta.

The story ends happily, though, with the dogs dropped off at home, the hunter given temporary residence in the county jail, and a .223 Remington rifle seized as evidence. If you want your own deer or other species, one source is Custom Robotic Wildlife.

The basic standing deer with moving head and tail will run you \$1,200, or \$1,700 if you want one that moves a leg, as well. My favorite is the snarly-looking boar that slides through the brush on an eight foot track. It would be great for greeting salesmen at your front door. But the \$4,000 moose is nice, too. Download the complete color catalog at www.wildlifedecoys.com.

Jumbo 'MO



ASIMO replica in 2009 Rose Parade. Courtesy of Honda.

Things were tough for Honda last year, with a reported pretax

income drop of 78.8% from 2007. The company has even killed plans for the 500 hp, V-10 powered NSX supercar. Things are still looking up — literally — for ASIMO, the world-famous humanoid. A 49 ft (15 m) replica was on hand for the 2009 Rose Parade in California in celebration of Honda's 50th year of operation in America.

This version pays homage to California's ecologically conscious folks, being built entirely from all-natural, earth-friendly materials. These include lettuce seed, rice carnations, and strawflowers, so maybe he became a nice salad after the parade was over. Also on display was a Honda Super Cub motorcycle and an FCX Clarity hydrogen car.

Hug Your Killer Robot

Moving from the nonfunctional to the truly pointless, we arrive at the Huggable Killer Plush Robots, very



Stepper, the huggable killer robot.

loosely based on the "Terminator" movies. Like Big ASIMO, they don't actually do anything, but they are very soft and "perfect for kids of all ages to squeeze and throw and drag around and talk to and all sorts of other fun stuff." You have a choice between Stepper (shown) and Ninja Servo (darker, with one red eye and an "angry" mouth). You can get one for \$8.99 from www.thinkgeek.com, so at least it's a better deal than a Vermont Teddy Bear. **SV**

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GEER HEAD

by David Geer

Contact the author at geercom@alltel.net

What is a DAGSI Whæg? An adaptive wheel-leg robot!

Doctors Roger Quinn (engineering), Roy Ritzmann (biology), and colleagues at the Case Western Reserve University (Case) collaborate in the neuro-mechanical research of cockroaches. In 2001, their studies lead to the birth of the Whægs (wheeled legs) robots, a product of the Case Center for Biologically Inspired Robotics. Research (or, the Biorobotics Lab).

Alexander Boxerbaum developed the DAGSI (Dayton Area Graduate Studies Institute) Whægs robot — named for the Institute that funded the work — in Dr. Quinn's Biorobotics Lab. It is the largest and most recent robot in the Whægs series. Its creators are in the process of patenting its body flexion joint, detailed below. One of the more promising potential applications is otherworld surface exploration.

The design of the Whægs robot — which combines the utility of legged transport to the wheel — has some precedent in other robots, such as the

RHex and the PROLERO. While those both each have a single leg per wheel, Whægs have multiple legs per wheel — typically three — as with the DAGSI Whægs.

The PROLERO rotated its legged wheels at an even, unchanging speed. The body of the robot connected with the terrain when the legs did not. This is partially a function of having only single legs. But, the RHex was different. Because its wheels rotate faster when the legs are not touching the ground and the legs use an alternating tripod gait (see sidebar), the body remains in a reasonably stable, level position compared with the vertical motion of the PROLERO.

Both the PROLERO and RHex use independent motors; one per each of the six wheel-attached, leg-like appendages. Each robot requires an onboard control system for leg and gait coordination. They both use skid steering just like a bulldozer.

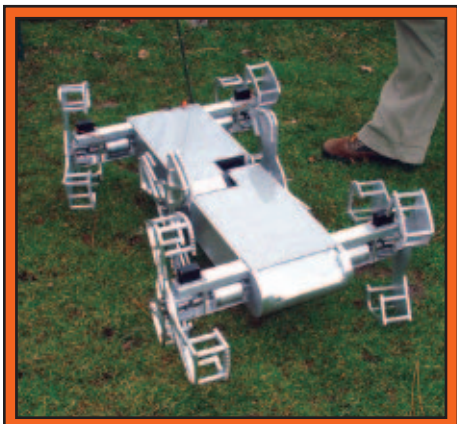
efficient than cyclically accelerating them as done in RHex," says Dr. Roger D. Quinn, director of the CWRU Biorobotics Lab. The single motor design improves the power-to-weight ratio over multi-motor designs because the single motor platform scales well.

A mechanism of chains and sprockets connect and drive them all in a tripod gait. This is a crucial imitation of the capabilities of the cockroach. Whægs have the advantage that when other legs lack traction, the system is able to transfer power to the leg(s) with traction to move the robot forward.

"Cockroaches change their gait when they run on rugged terrain and when they climb obstacles. Whægs changes its gate passively without the need for an active control system. It has torsional (clock) springs in each of the axles that permit its legs to passively comply with the terrain," explains Quinn.

When a leg meets a larger obstacle, the motor attempts to rotate the wheel and leg. But, because the load on the leg is bigger, the motor instead winds the torsional spring in the axle. The leg does not move at this point. However, the other legs — which have not met a large barrier — continue to move. In this way, the

The DAGSI Whægs (wheeled-legs) robot sporting six wheel-legs.



What is Different About Whægs?

Whægs use a single motor which drives all six wheels and all 18 legs at a constant rate. "Running motors at constant speed is more energy

robot's gait responds to the terrain passively. "There is a mechanical stop so that the spring only winds 70 degrees. At that point, the leg will begin to rotate again," he adds.

For steering, the Whegs use asymmetrical motor activity. Whegs turn left or right via the following mechanical cooperation of its wheeled legs. When turning to the right, for example, it turns its front wheel-legs to the right and its rear ones to the left.

This imitates how the cockroach (or how a car, for example) turns.

The robot uses Ackerman steering to ensure that the wheel-legs cannot work against each other during a turn. They work in agreement as to which way to go as they steer the robot together. "This keeps the robot from being extremely inefficient, or even potentially ripping itself apart," says Alexander Boxerbaum.

The Middle Joint is a Defining Point

Whegs employ one of its most unique features as it prepares to climb. It has a front and a rear body segment that is connected at the middle wheel-leg axle; the robot uses this to its advantage, rotating its front segment up or down relative to the rear segment using a motor that drives the middle joint, according to Dr. Quinn. The robot uses this central, motor-driven, body flexion joint to flex its body to avoid high centering/ scraping the middle of its body on the edge of the obstacle during a climb, Dr. Quinn explains. This enables the robot to climb up and down tall objects and varied size sets of stairs.

"At the beginning of a climb, it rotates the front of its body up so that it reaches its front feet on top of a tall obstacle. During the climb, it rotates the front of its body down, to avoid high centering.

The body joint mechanism itself

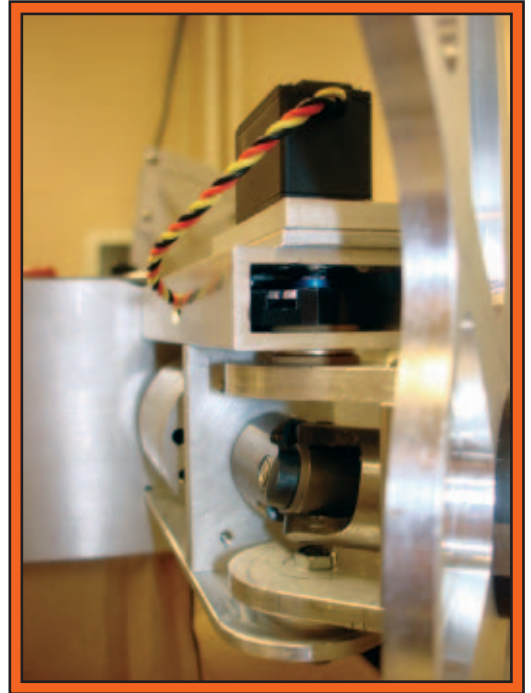


Robot walking in the snow.

consists of a modified worm gear. Worm gears have various applications, including elevator lift motors. "In a worm drive system, the motor is attached to a shaft that is shaped like a screw. The screw presses against a gear, so that when the screw rotates, it causes the gear to rotate very slowly. There is a large reduction in speed and a large increase in torque. The device is non-backdrivable, which means the motor can move the joint, but the joint can't move the motor," says Boxerbaum.

This setup keeps the motor from working all the time to maintain joint position. But, this worm gear is different from other such gears. "It is springy (instead of rigid). When something tries to drive it backwards, it passively complies and absorbs impact loads instead of possibly breaking," continues Quinn.

During a dynamic simulation of the robot, Quinn and Boxerbaum discovered that the weight distribution on the robot affects its climbing behavior, as they had suspected. "We found that for climbing up obstacles, it helps to have the center of gravity near the front. We verified this with the actual robot. It could



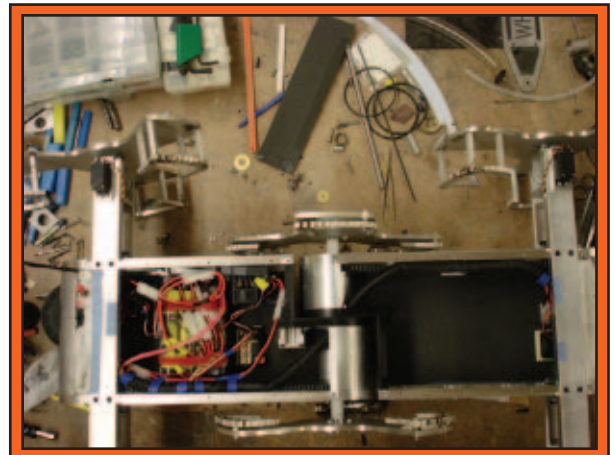
The steering axle for WHEGS robots.

climb taller obstacles when more of its weight (the batteries) was moved forward," says Quinn.

Part Selection

The robot's parts and materials included aluminum and carbon fiber, which the researchers selected for their light weight and strength. They used steel for smaller (yet higher) load components in the drive train. Because the robot must be water

A top view of the WHEGS robot without its coverings.



resistant, it employs tight-fitting matting surfaces with gaskets in its internal components.

The drive motor is a Maxon 150 watt, 12V model. The gear transmission is also Maxon, with a 48:1 ratio and a 2:1 spur gear, and 3/8th inch drive shafts. The wheel spokes are 7.5 inches in length, enabling the robot to climb full standard height stairs despite its total body weight of 40 lbs.

The speed controllers are Victor 885s with fans; one for the joint motor and one for the drive motor. In R/C control mode, a Futaba eight channel receiver and 9CAP transmitter are used. The drive batteries are 7.4V 4,200 mAh Nickel Metal Hydride batteries. There are two of them, wired together in series.

Conclusion

The DAGSI Whegs are the culmination of two decades of investigation

into the neuro-mechanics of insects, and eight years of progress in the Whegs robot series. The bug-inspired bot is exceptionally autonomous for its kind. It adapts to and scales broader extremes in terrain than its forebearers with no requirement for control software for leg coordination. Moreover, its creators derived it all from the critical dissection of the despised, yet unstoppable cockroach. **SV**

The Tripod Gait

In a tripod gait (or walk), the first and last leg (wheel-leg, in this instance) on one side of a hexapod robot move (rotate) in unison with the middle leg on the other side, and vice versa. When the Whegs are on the ground, they establish a tripod beneath the robot's body, supporting it and all its weight. This tripod alternates with the other tripod formed by the other three wheel-legs.

Resources

Case Western Reserve University's
Center for Biologically Inspired
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<http://biorobots.cwru.edu>

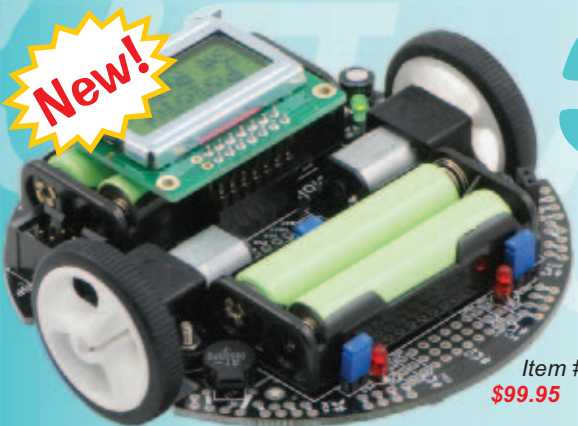
Various iterations of the WHEGS
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Other important links for
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The PROLERO
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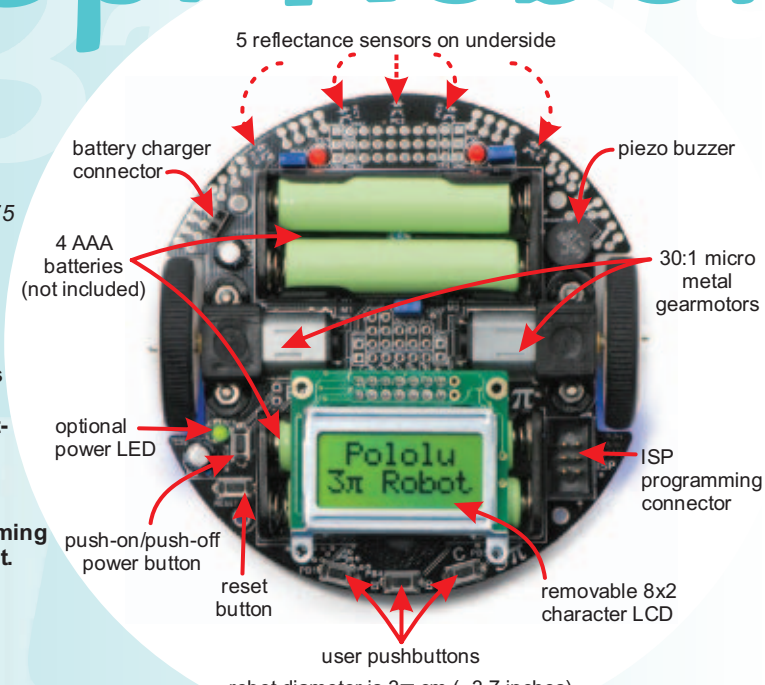
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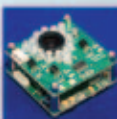
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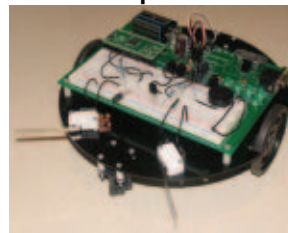
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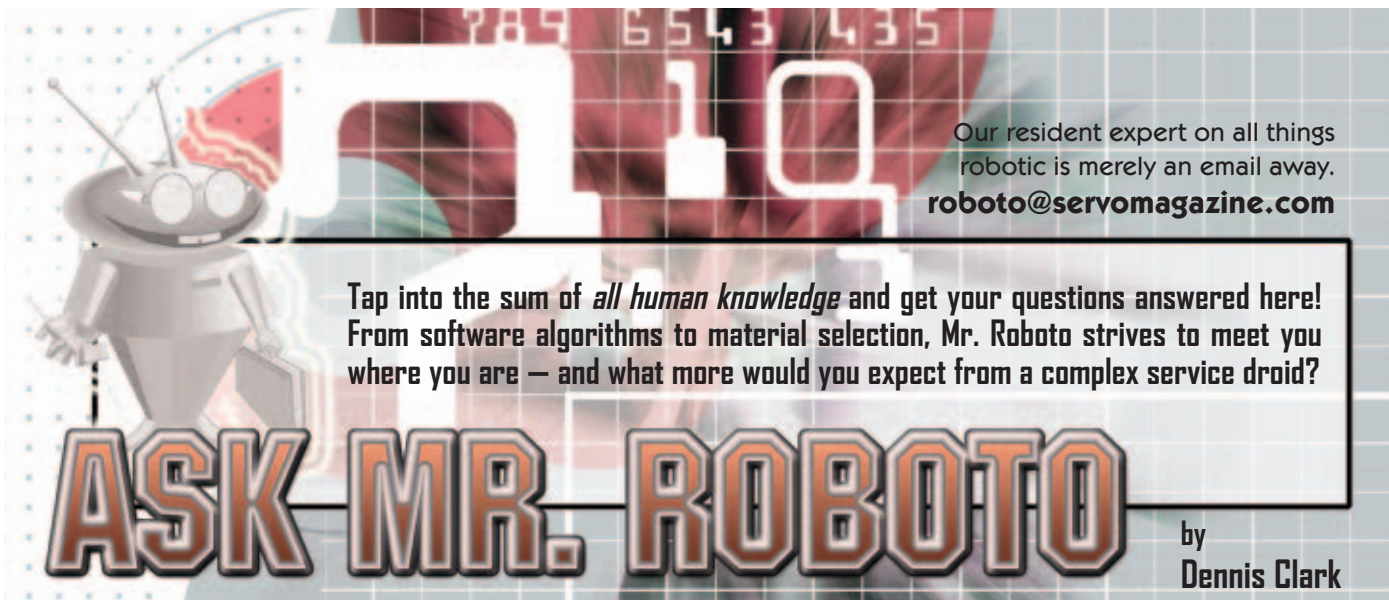
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Q. I'm interested in the Dinsmore electronic compass circuit and I want to build one for myself. Can you provide more information about how I can get the parts for it?

— Goran
via email

A. This question is in reference to a post that I made several years ago detailing how to use a Dinsmore 1490 compass module. Between then and now, the manufacturing for that module has moved from the Dinsmore company to Robson Company, Inc. (www.robsonco.com/Dinsmore/Untitled_6.html). It is still the same sensor and unbelievably, it is still about the same price (\$15). The Dinsmore 1490 compass has four digital outputs; each corresponding to one of the cardinal points on the compass. This means that by looking at all of the outputs you can differentiate eight compass points: N, NE, E, SE, S, SW, W, and NW. The device is basically a heavily

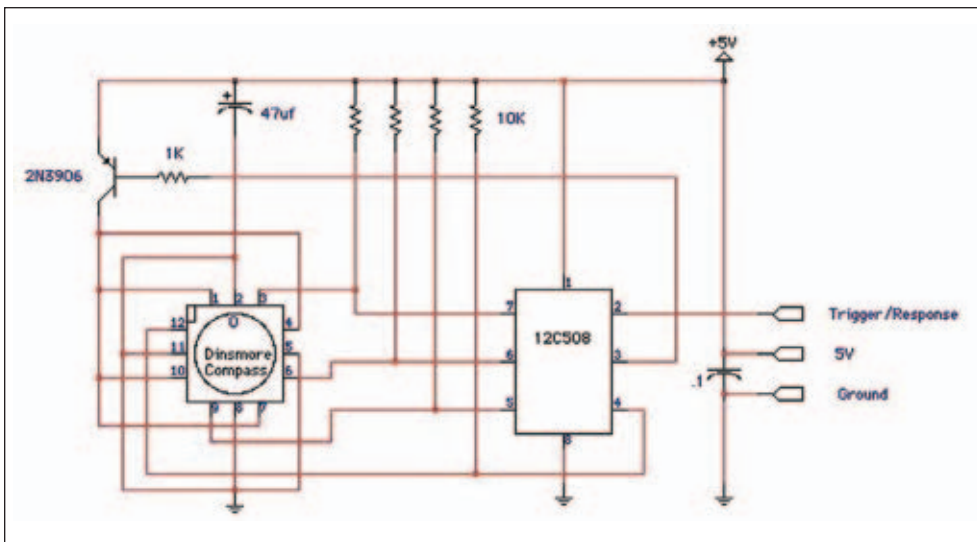
damped magnetic compass with Hall Effect sensors that detect where the needle is pointing. The 1490 is a bit of a power hog at 30 mA of current draw when it is turned on so I created a circuit that only powered it up when I wanted a reading. Rather than have my main controller deal with the fiddly details, I wrote some assembly for a PIC12C508 processor to handle turning the compass module on, taking readings, turning it off, and reporting back what the heading was. To trigger a reading, you toggled an I/O line and then waited about two-thirds of a second for the data to be sent back at 2400 baud 8N1 (8 bits, no parity, 1 stop bit.) So, a \$15 compass sensor module and another \$2 or so will get you a basic compass for your robot. That was a good idea several years ago and it's still a good deal today if those eight compass positions are good enough for your navigation and you don't mind that it isn't a very fast sensor. See the schematic in Figure 1 for how to use this sensor.

You can download this program at www.servomagazine.com under Mr. Roboto as Dinsmore.zip; the file is compass.asm. It is PIC assembly for the PIC12F508, but will work with the 12C508, as well. Read the comments carefully and be sure to understand the OSCAL value issue with this PIC (it's in the comments).

Parts needed include:

- 1 - Dinsmore 1490
- 1 - PIC12F508 microcontroller
- 1 - 2N3904 PNP transistor
- 1 - 47 μ F 16V electrolytic capacitor
- 1 - 0.1 μ F bypass capacitor
- 4 - 10K 1/4 watt resistors
- 1 - 1K 1/4 watt resistor

Figure 1. Dinsmore 1490 interface.



Of course, you'll need to have the PIC programmed. You can look at what I did in the source code and write your own for your favorite microcontroller, or simply integrate the process into your main brain on your robot. It is an easy part to use. Figure 2 shows what my board looked like. I used www.expresspcb.com to make a quick and dirty prototype board for it.

Table 1. LEGO NXT motor pin-out.

Wire	Function
White	Motor 1
Black	Motor 2
Red	GND
Green	4.3 Volts
Yellow	Encoder 1
Blue	Encoder 2

Q . We were discussing closed-loop motor controllers at our club meeting last night. It occurred to me that the LEGO NXT motors are good quality motors with built-in encoders, and a good deal at about \$18. They're easy enough to use with LEGO's NXT controller, but how would I use them (including the encoder feedback) from my own microcontroller, such as the ATmega168?

— Joe

A . LEGO's newest motors for their robot system are much cooler than the ones they used in the "old" RCX Mindstorms days! I went and looked up how these motors are wired and will impart said wisdom to you. Originally, I thought that the motor drivers would be in the motor and the NXT just sent control signals — but no! It is even easier than that. Table 1 shows how you wire this motor up. Figure 3 shows a LEGO shot of the NXT motor (which LEGO calls a servo, for good reason).

You will PWM the white and black wires. The red wire is actually ground, and apparently the green wire is the power to the wheel encoders. I suspect that 5V will work fine here. The Encoder1 and Encoder2 outputs are the wheel encoder quadrature encoder outputs. With these, you can not only tell how fast the motor is going but which direction it is going. When the Encoder1 square wave leads Encoder2, then the motor is spinning one way. When Encoder2 is leading Encoder1, then the motor is spinning the other way (see Figure 4). This is just about the perfect motor and encoder system, all in one box.

One full transition of an encoder cycle is two degrees of rotation, so from that you can get speed. I don't know how good the motor is in the NXT motor, so I don't know the PWM frequency that is best for it. Knowing LEGO, it is a good one, though.

Q . I would like to configure a robot so that it can run in a straight path for some distance without following a line or guidance beam. I'm sure the concept is fairly simple, but being an amateur hobbyist I'm at a loss on the technique involved. I was thinking of

Figure 2. Dinsmore compass sensor board.

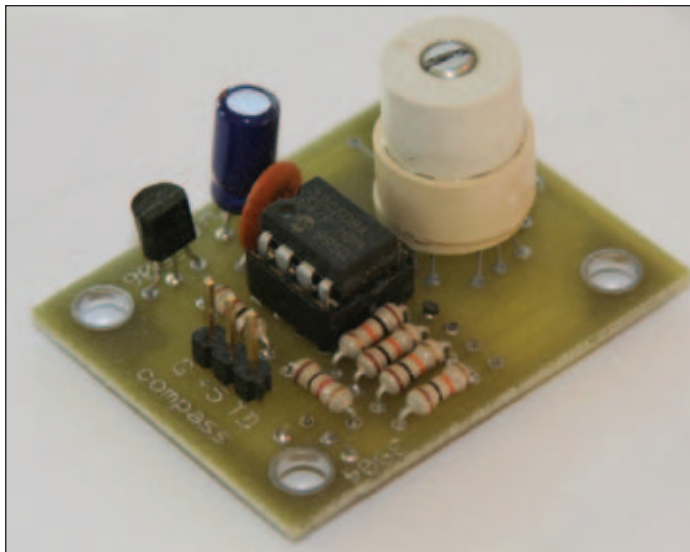
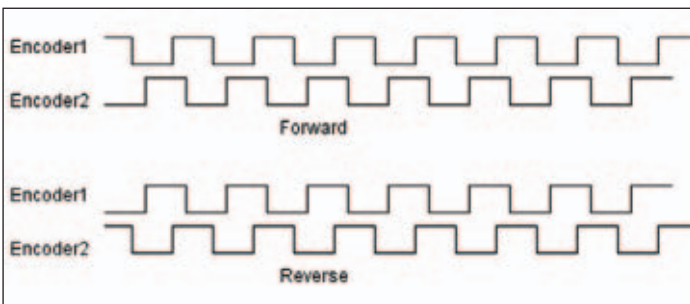


Figure 3. LEGO NXT servo.



Figure 4. Quadrature motion encoding.



generating a pulse train from each of the two drive wheels by means of simple wheel encoders. I figured that if I could compare the two pulse trains and use the frequency difference between them as a correction factor to synchronize one motor with the other so that they both turn at the same rate, I could therefore avoid a long, curving path. I hope to use this as a key aspect in navigation for indoor or outdoor Robo-Magellan type activities. Any guidance you can offer in this will be greatly appreciated.

— Tom W.

A. You are indeed on the right track. I recently worked over my Trinity Fire Fighter and I wanted it to be able to accurately map out the house on the first pass so that I didn't have to sort out everything every time through. To do this, I really needed to accurately assess my direction, as well as distances. I used wheel encoders mounted to my hacked hobby servos (one of my favorite small robot gear boxes). The encoders that I used came from Nubotics — the Wheel Watchers; you can find them at www.nubotics.com/products/ww01/index.html.

These give full quadrature encoder outputs at sufficient resolution to get my bot to return to within two inches of its starting point after traversing an entire Trinity Fire Fighting house. My installation is shown in Figure 5 on my old Budget Robotics Scooter bot. My robot brain is an Atmel ATMEGA8535. What I did was set a 10 μ s interrupt to go out and poll for one channel of the quadrature encoding. I used the output that delivered a pulse at any transition of either quadrature phase. This effectively doubled my rotational resolution, but I lost the ability to check for direction. See Listing 1 for my ISR.

The ISR simply got the encoder pulses; it didn't fix any

Listing 1. Encoder ISR polling wheel encoder pulses.

```
//*****
// ISR routines
//*****
ISR(TIMER0_COMP_vect)
/*
 * 10 microsecond ISR
 */
{
    static uint8_t    t_10us=0;
    //Wheel encoders
    static uint8_t    leftOne=1;
    static uint8_t    rightOne=1;

    t_10us++;
    if (t_10us > 100)
    {
        t_1ms++;
        t_10us = 0;
    }

    if (LEFT_ENCODER != leftOne)
    {
        leftOne = LEFT_ENCODER;
        leftTics++;
    }
    if (RIGHT_ENCODER != rightOne)
    {
        rightOne = RIGHT_ENCODER;
        rightTics++;
    }
}
```

wheel speeds or directions. For that, I had a function that ran in my normal activity loop. This function — called *go()* — simply tracked the encoders of either wheel when going forward and stopped the fast wheel until the slow one caught up. Yeah, it's pretty crude, but very effective since it was looking every few milliseconds and you never notice the motor cycling. See Listing 2 for how I implemented this.

You still have to turn corners, but you can do the same thing by knowing how many tics it takes to make various turn angles. I'll leave the turn algorithm as an exercise for the student. <grin> It worked for me because my robot has its motors and wheels in the very center, and uses differential or "skid" steering. In case you were wondering, this program is written to be compiled with gcc-avr — the open source AVR C compiler.

I hope that you've learned something that you wanted

Listing 2. The "go straight" checking code.

```
uint8_t Go(int16_t tics)
/*
 * Go the number of cms specified, measured in
 * 3.4cm per tic. Will return 1 when done, and
 * 0 otherwise. Continually call this; it won't
 * block other routines.
 *
 * USES: rightTics, leftTics
 */
{
    static uint16_t    doneTicR;
    static uint16_t    doneTicL;
    static uint8_t     goState = GSTART;
    uint8_t            res = 0;

    //Set motor speed
    MotorsGo(leftMotor,rightMotor);

    switch (goState)
    {
        case GSTART:
            rightTics = 0;
            leftTics = 0;
            doneTicR = rightTics + (tics*3);
            doneTicL = leftTics + (tics*3);
            goState = GMOVE;
            break;

        case GMOVE:
            //Try to keep the thing going straight!
            if (rightTics > leftTics)
                UpdateMotors(leftMotor,0);
            else if (leftTics > rightTics)
                UpdateMotors(0,rightMotor);
            else
                UpdateMotors(leftMotor,
                               rightMotor);
    }
```

Listing 2. continued.

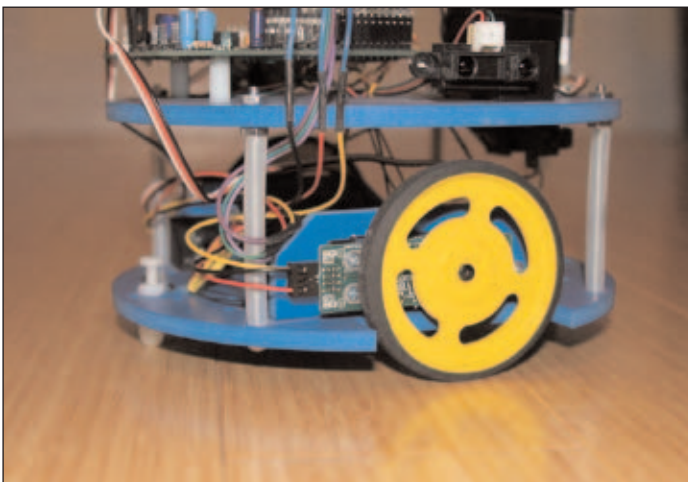
```

if (doneTicR <= rightTics)
// We are done
{
    UpdateMotors(leftMotor,0);
}
if (doneTicL <= leftTics)
{
    UpdateMotors(0,rightMotor);
}
if ((doneTicR <= rightTics) &&
    (doneTicL <= leftTics))
{
    goState = GSTART;
    res = 1;
    MotorsStop();
}
break;
}
return (res);
}

```

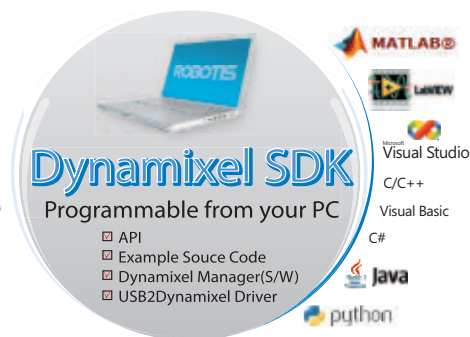
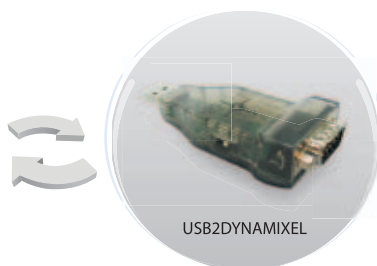
to with this column. As usual, I can be reached for questions, comments, and criticisms at roboto@servo-magazine.com and I'll be happy to work on it! Until next time, keep on building those robots! **SV**

Figure 5. Robot with Nubotics wheel encoder.



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Weight(g)	53.5	

RX-28 / DX-117



- Full metallic gear
- Control network: RS-485
- Gear Ratio : 193
- Position Sensor : POT

	RX-28		DX-117	
	at 12V	at 16V	at 12V	at 16V
Holding torque(Kgfc.m)	28.3	37.7	28.3	37.7
Speed(sec/60deg)	0.167	0.126	0.167	0.126
Weight(g)	72		67	

RX- 64



- Full metallic gear
- Control network: RS-485
- Gear Ratio : 200
- Position Sensor : POT

	at 15V	at 18V
Holding torque(Kgfc.m)	64.4	77.2
Speed(sec/60deg)	0.188	0.157
Weight(g)	125	

EX-106



NEW

- Full metallic gear
- Control network: RS-485
- Gear Ratio : 164
- Position Sensor : Encoder

	at 14.8V	at 18.5V
Holding torque(Kgfc.m)	84	106
Speed(sec/60deg)	0.182	0.143
Weight(g)	155	

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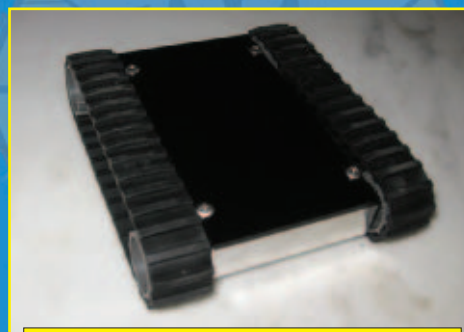


FIGURE 1. THE SURVEYOR DRIVE BASE KIT.

This month, we have the pleasure of presenting the Surveyor Drive Base kit from Inertia Labs. Readers may recall our adventure with an already assembled Surveyor robot from a few months ago, which sported a unique method of teleoperation over a wireless network. The drive base, on the other hand, is a *tabula rasa* as clean and pristine as Descartes could have imagined. A kit with such potential is both exciting and intimidating, with the only limit to the possibilities being your imagination.

As an extra challenge and as a dictum of circumstance, we were also

determined to make something out of the drive base using only parts that we had available — we would be living off the land. Our garage is pretty fertile land, but it would be an exciting challenge nonetheless.

Robinsonade With Robots

To kick-start our own imaginations, we took a look at the drive base to see what we did indeed have to start with. The plastic top is fastened to the aluminum base with four screws, and when we popped open the bot we were met with a desolate

and silvery landscape filled with nothing but possibility. The only inhabitants of the aluminum wasteland were four small motors and their associated gearboxes. Each motor/gearbox assembly is fastened into the drive base with a solitary screw, and even though this doesn't sound very secure it was a bit of a challenge to remove them. We eventually discovered that the motors were actually very easy to remove after the treads were taken off. The treads themselves are quite secure, and have to be finessed by pulling them off while spinning the wheels. After the drive base is relieved of its treads and the

motor unburdened of its fastening screw, the motor is much more cooperative when it comes to removal.

The small and compact nature of the drive base offers both opportunity and challenge. Most serious tinkerers would likely be inclined to use ant bot parts to bring their drive base to life; many of which could be acquired from Inertia Labs. This might be a very viable

FIGURE 2. TABULA RASA.



FIGURE 3. THE SRV-IQ AND THE DRIVE BASE.



option for some tinkerers, because the Surveyor drive base comes with a price tag of \$120; much less than the asking price of \$475 for the complete Surveyor SRV 1Q. The necessary ant parts to bring the drive base to life might run you in the range of \$16 for a rechargeable battery pack to an affordable \$64 for a radio controller and receiver, so it is very likely that a custom bot could come in a much lower price tag than the premade one. We were roughing it, however, so to keep our price tag at zero dollars we decided to live off of the land in our garage (known as Robot Central in tales of yore) and the fruits of the soil being the shells of former FIRST robots and the relics of our stint in the combat robotics circuit.

We elected to tackle the task of making the drive base into a radio-controlled driving platform, and to make this entrée palatable we had to find some essential ingredients.

The Robot Family Surveyor

Our first order of business was to find a brain for our robot. Because of the small scale of the Drive Base, a microcontroller from an ant robot or perhaps an OOPIC from a Mark III Sumo robot would be a good fit, but we were determined to use what was available. We were a bit more careful in selecting our raw materials than young Frankenstein's assistant, and instead of Abbey Normal we ended up using EDU Robot. The EDU Robot was the preparatory project from FIRST competitions past, and even though it was not nearly as compact as an OOPIC controller, the EDU Robot brain was not unreasonably large. It also sported plenty of PWM inputs, digital in/outs, and the advantage of being something that we were familiar with. The sleek black and white color scheme also matched rather nicely with the sleek black and silver color scheme of the drive base. Beggars can't be choosers, but that doesn't mean we don't appreciate a well-coordinated robot.

The brain may have been the

most essential of the components for our drive base, but a microcontroller without a receiver is like a brain in a jar. We needed to find a radio receiver and the accompanying radio if we hoped to have the drive base trundling around and doing our bidding.

Our combat robots Troublemaker and Twibill Trouble have been some of our most agile radio-controlled robots, and their maneuverability certainly achieves its full potential thanks to our Futaba radios. We happened to have an extra PCM (Pulse-Code Modulation) receiver that accompanied one of our combat robots, and the relatively unobtrusive size made it an attractive option for the drive base. Once again, true ant parts make our chosen receiver look like a Gulliver in the land of Lilliputians, but we were doing our best to pull a Robinson Crusoe.

Our final concern was finding a power source, but our choice of the EDU Robot brain had fortuitously already seen to that demand. The EDU Robot brain came with its own power source: a rechargeable AA battery pack. Of course, the ant battery packs make our battery pack look like a Cyclops to Odysseus, but the price was right and it was still emblazoned with the emblem of our FIRST team, number 1079 (see the August, October, November, and December 2004 issues of *SERVO* for more on those adventures and the reasons why reusing the battery pack might conjure up sentimental memories of exercise balls and the music of the band Boston).

Outwit. Outplay. Cannibalize.

With the EDU Robot microcontroller, battery pack, and Futaba radio and receiver we had all of the raw materials for our radio-controlled drive base with one notable exception. Gulliver might have disdained cables, but without them our drive base would remain permanently dismembered. The inputs to the EDU Robot brain demanded some PWM cables, and even though

FIGURE 4. A MARK III SPORTING AN OOPIC.



FIGURE 5. THE EDU ROBOT BRAIN.



FIGURE 6. A SENTIMENTAL BATTERY PACK.



FIGURE 7. A FUTABA PCM RECEIVER.



we were convinced that we had some readily at our disposal, the fickle whims of nature saw it fit to deny us such a convenience. With no cables readily available, we found that we

had no other option but to rely on the generosity of robotics projects past. One of our former projects that was actually quite a glutton for PWM cables was Dreimo, Team 1079's labor of love for the 2006 FIRST season and game Aim High. Dreimo hoarded so many PWM cables because of its trusty CMU camera, and as bad as we felt about blinding our old friend, sacrifices need to be made so that new projects can be given life.

FIGURE 8. OUR T6XA FUTABA RADIO.



FIGURE 9. ITSY BITSY MOTORS AND GEARBOXES.

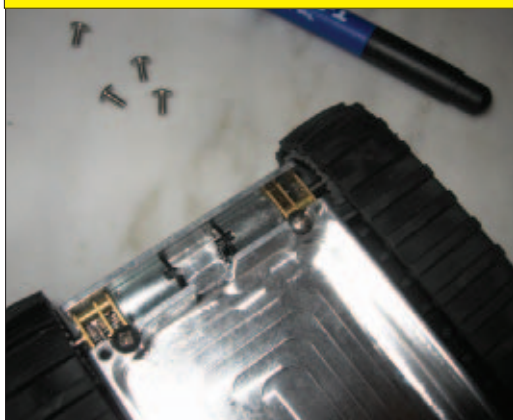
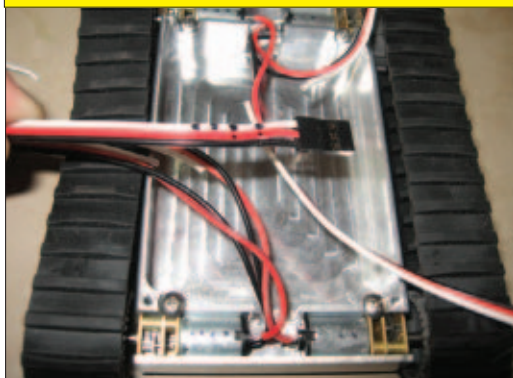


FIGURE 9A. WIRING UP THE MOTORS.



The PWM cables that we cut from our friend Dreimo — albeit a bit on the long side — were exactly what we needed to connect to the naked motors. The process of connecting the cables required some delicate surgery, however. The extreme tininess of the motors meant that they also had miniscule leads, and soldering them was a task that demanded delicate dexterity. At first, we thought that we would be forced to solder the motors in place, but then we discovered the trick of removing the treads before messing with them.

Once removed, the motor/gearbox assembly was easily clamped into a small vise for easy soldering. The small motor leads still demanded extraordinarily steady hands, and we would strongly recommend generously tinning both the wires and the motor leads before attempting to solder them. The motors certainly seem very robust, but it is never a bad thing to minimize the amount of extreme heat that you expose them to.

PWM cables, of course, have three leads, and our miniscule motors had only two. We decided to solder the red and black wires to the appropriate motor leads (the positive was marked with a clear plus sign), and simply cut the white signal wire short and tuck it out of the way where it wouldn't cause any trouble. During our wiring process, we made sure to include a valuable trick that is useful for any project involving a lot of similar looking wires. We numbered each of the drive motors 1 through 4 by dotting them with a permanent marker, and we made corresponding lines at the ends of the PWM cables.

Even though we couldn't see our motors once we reattached the acrylic top plate, our simple numbering system would allow us to remember which motor was wired where so during testing we could easily identify

any troublemakers without being forced to pop the whole robot open.

After soldering the borrowed PWM cables to the motors of the drive base, we needed to give them a chance to escape from the belly of the beast. The black acrylic top plate is so pristine that we were a bit reluctant to cut it up, but after cannibalizing so many other robots it seemed only fitting that the drive base would need to shed some shavings. The good news is that the black acrylic top cover is easily machined.

To allow room for the PWM cables to escape, we used a wheel grinder to put two notches in the side of the plate and sandpaper to deburr the edges. We bundled up the PWM cables and reattached the top plate as we prepared to wire up the bot.

The extra length of the PWM cables made laying out the circuit easy, and with careful reference to documentation we connected the PWM cables from the drive base motors to the EDU Robot brain and another cable from the brain to our PCM receiver. The battery easily connected to the EDU Robot brain, and we had a functional circuit on our hands.

Desert Island Robot

Or, at least we hoped that we did. To ensure that our little creation didn't try anything funny, we placed it on top of a small box so that the treads could spin freely without resulting in an unexpected trip off the table. We turned on the EDU Robot brain and then the receiver, and we were simultaneously pleased and perplexed to see the motors rev to life. We were pleased because we had given the robot and its motors life when it had none before. We were perplexed because the motors seemed to act of their own accord; we had given no commands to the bot through our Futaba radio, and yet they spun with alacrity.

At first we thought that it could be an issue with the trim on the radio channels — something that had caused similar mysterious movements

with our other Futaba controlled robots. We adjusted the trim, however, and were met with no change in the odd behavior of the bot. Specifically, that odd behavior was a vigorous spinning of the two treads in opposite directions. Had we left the robot on the table with plenty of traction it would not have taken a dive off of the edge, but it certainly would have caused some dizziness (most likely on behalf of whoever would have been watching).

Our first troubleshooting instinct was to check if we had made a mistake in connecting the power and ground wires of the PWM cables to the motors instead of using the ground and signal wires. Instead of giving in and resoldering everything, we took another one of the PWM cables generously donated by DreiMO, attached the connector to a cable emanating from one of the drive base motors, and carefully touched the stripped ends of the ground and signal wires to the corresponding pins on a PWM input on the EDU Robot brain.

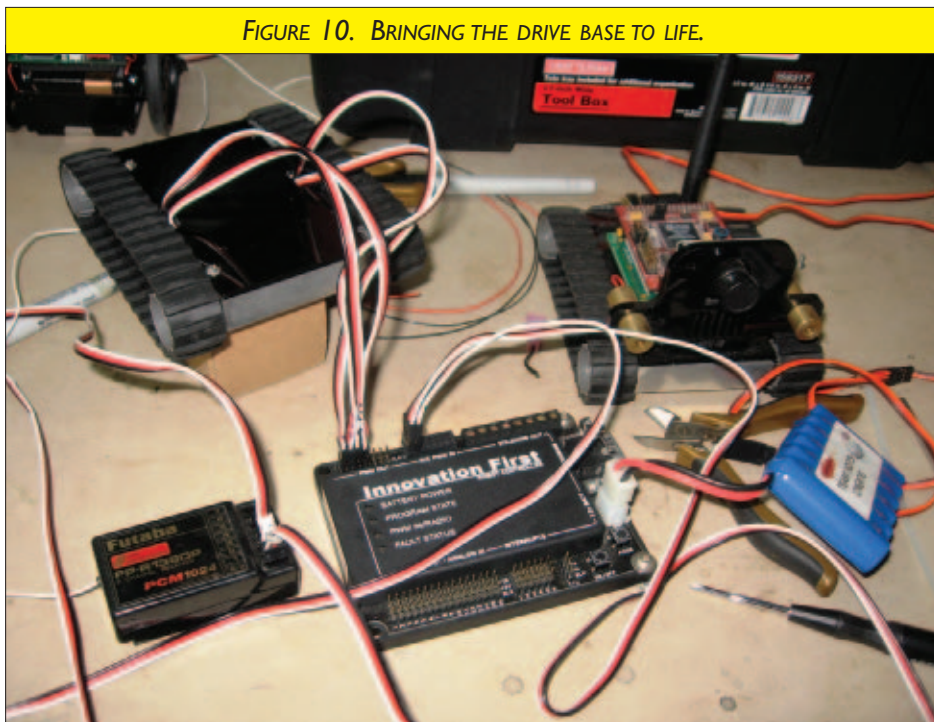
Normally, we would recommend using a lot more caution when testing connections to the microcontroller, but we were as eager to get our robot working as Tom Hanks was to escape the tropical island with his friend Wilson. Our investigation led us to conclude that the signal wire did not hold the key to the mystery, and we were sent back to the documentation (which is readily available online for anyone else that happens to have an EDU Robot brain) to find the answer.

After some reading, we were ready to go back to our mess of wires armed with some new strategies that involved the digital in/outs on the EDU Robot brain and the actual programming of the Futaba radio.

Lord of the Zip Ties

After making some progress with the circuit of the Frankensteined drive base, we wanted to clean it up a bit. The primary way in which we wanted to spiff the bot up was to make a shelf to mount the components onto. We once again found ourselves

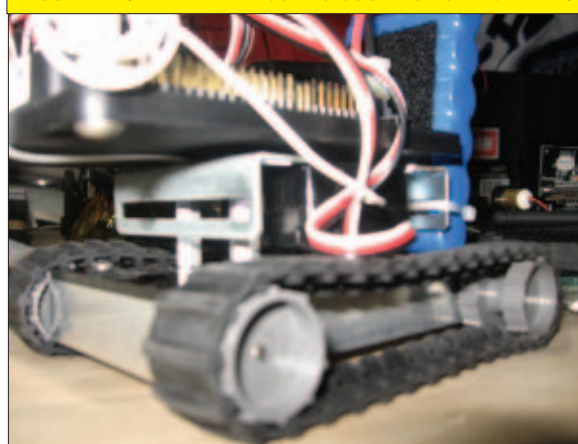
FIGURE 10. BRINGING THE DRIVE BASE TO LIFE.



borrowing parts from another robot; this time we had our eyes on some bits from a Vex robot. The threaded posts from the Vex kit came in a variety of sizes, and one midsize post seemed to be the perfect scale for our application. A couple of angled frame members would give the shelf some form, and we used the notches in the Vex frame to eyeball the location of the holes to fasten the posts.

Drilling holes in the acrylic top of the drive base didn't seem so bad now that we had gotten over our initial trepidation, and a drill press made quick work of the holes. We had to place the postholes near the middle of the plate to avoid the motors and gearboxes which was a little less than ideal, but the small size of the motors, gearboxes, and their associated mounts still gives you plenty of real estate to work with. The relative thinness of the plate also means that we didn't need terribly long mounting screws, which was perfectly compatible with the screws that we had stolen from the Vex kit in the

FIGURE 11. THE FRAME COMES COURTESY OF A VEX KIT.



process of removing the posts.

The EDU Robot brain did have holes for mounting in the corners, but these holes would have positioned our posts right above the motors and gearboxes. In the same way that fate did not smile kindly upon Crusoe, we were forced to make do with the situation and eventually turned to the indefatigable ally of the zip tie.

With the brain secured to the robot using zip ties, we now only had to attach the receiver and battery pack. Unfortunately, the geometry of our shelf did not accommodate the battery pack on the bottom level, but the receiver fit in quite securely. We

FIGURE 12. THE DRIVE BASE IS RADIO CONTROLLED AND READY TO GO.

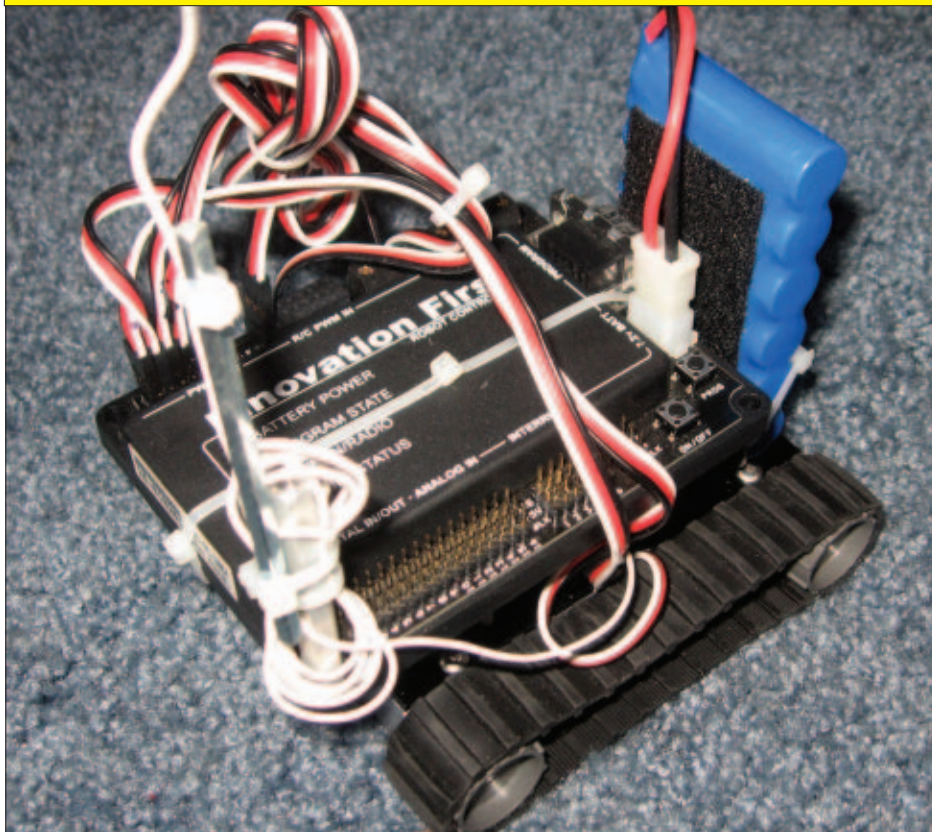
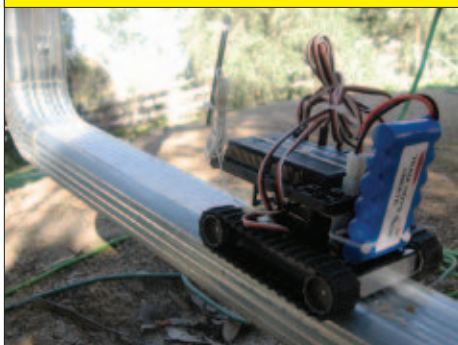


FIGURE 13. TACKLING THE OUTSIDE WORLD.



eventually zip tied the battery pack to the back of the robot in a regrettably unfashionable way, but at least we did make use of one of the EDU Robot brain's mounting holes by fastening a mast to it for the receiver antenna. And with that, we had our radio-controlled garage survivor.

Back to Civilization

For our project, we treated the wealth of commercially available ant parts as a luxury, but we're sure that there are plenty of intrepid tinkerers

out there that might even think of the drive base as unnecessary. A basic chassis for a small scale can certainly be improvised economically, but the excellent caliber of the drive base makes it an attractive option for prospective bot builders that want an astoundingly solid foundation for their robot without spending money like the federal government. And while a small chassis may be easy to improvise, the miniature 1:100 gearbox that comes on the stout motors would be excruciatingly difficult to replicate without some nice machines to do the precision work for you. At \$120, the drive base is certainly reasonable, and there are plenty of affordable parts available from folks like Inertia Labs. The Inertia Labs website even advertises how the drive base can be used to make a rather nasty looking combat robot. The drive base would certainly be a robust and stylish start to any project, even if it does leave the details completely to your imagination.

Our experience of living off the

land in the garage points to what we think is something that all roboticists can relate to. We often make do with what we have, whether it's improvising a fix during a competition or using something less than ideal while trying to stay within budget for a project. Sometimes that might involve cannibalizing past projects, as we did here.

Having experience with other robot projects gives us much more than a potential (albeit sentimental) last resort for robot parts. The experiences, tricks, and skills that we scavenge from each project are something we take into every new endeavor. Everything from labeling wires to debugging with a multimeter to using shop tools like drill presses, lathes, and mills were all skills that we had to learn somewhere along the way. Even if those projects were recent or a long time past, by plundering the experience of previous projects we are not actually diminishing their standing but increasing it.

Every time Evan solders up a PCB, he is reminded of working on leg modules at PARC, and each hole made with a drill press brings back fond memories of long nights in the garage with the FIRST team. Every time we reach back to an experience while using the lessons learned there on a new project, our appreciation for our former ventures grows.

For veteran roboticists, the drive base is a great kit to use a wealth of experience to bring to life. The wide open possibility of a chassis empty save for motors and gearboxes is a great opportunity to use established strategies to defray the intimidation of tackling something where you are so much on your own.

For novice tinkerers, the drive base might be a bit of an intimidating project, but the blank slate is a great field for experimentation so that you can learn the sort of tricks that will inevitably help you in later projects. **SV**

RECOMMENDED WEBSITES

For more information, go to:
www.inertialabs.com

NEW PRODUCTS

ADHESIVES

Thermally Conductive Adhesive

Duralco 132 thermally conductive adhesives combine Cotronics' unique, high temperature resins with highly conductive fillers to form thermally conductive, adhesive bonds with continuous service up to 500°F. It's ideal for any industrial, electrical, or electronic high power application including: bonding and assembling heating coils, cooling coils, heating elements, heatsinks, reaction vessels, semi-conductors, rectifiers, power supplies, replacement for soldering and welding, etc.

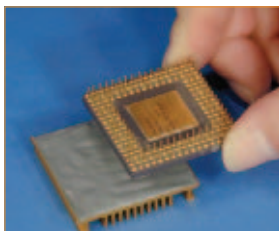
The adhesive is easy to use; simply mix and apply Duralco 132 for adhesion to metals, glass, ceramics, and plastics. It offers excellent resistance to chemical solvents and moisture and provides the heat dissipation required for many high temperature electronic and industrial applications; 100% solids formulations; no volatiles; and no VOCs.

Duralco 132P is commonly used as a heat tracing adhesive which is used to bond heating or cooling tubing to equipment. Its non-sag formulation and room temperature curing provides for efficient placement of heating or cooling coils.

For further information, please contact:

Cotronics Corporation

131 47th Street
Brooklyn, NY 11232
718 • 788 • 5533 Fax: 718 • 788 • 5538
Email: sales@cotronics.com
Website: www.cotronics.com



CABLES

Flexible Servo Cables

The new Series SF flexible servo cables from Alpha Wire offer increased flexibility for connecting power

between supply and servo motors, drives, and controllers. A flexible thermoplastic elastomer (TPE) jacket and PVC/nylon insulation make the cable suitable for flexing applications that require continuous movement. Superior flexibility and a tight bend radius of ten times



the cable's diameter also make installation easier.

The UL listed cables meet the requirements of NFPA 79, allowing them to be used in industrial machinery applications such as machine tools, pick and place equipment, Cartesian and articulated motion control, and automated material handling systems.

To meet a range of power requirements, Series SF cables are available over an 18 to 8 AWG (0.96 to 8.5 mm²) conductor size range. The cables are available in both power and composite cable configurations. The Series SF power cable has four conductors for power and ground. The Series SF composite power cable adds two twisted pairs for use in braking control, temperature monitoring, or other control/monitoring uses. To provide excellent EMI protection in noisy electrical environments, the cables are shielded with a combination foil and tinned copper braid.

"The new Series SF gives users a more reliable and easy-to-use way to connect servo power," says Mike Dugar, Senior Product Manager at Alpha Wire. "Users want more flexibility to make installation and startup faster, as well as increase long-term reliability. Series SF does all of that, while maintaining the rugged performance needed for industrial machinery."

Series SF flexible servo cables are rated for 600 volts and meet the UL requirements for TC-ER and WTTC applications and CSA for AWM I/II A/B. They are oil and sunlight resistant. The new Series FL flexible servo cables are members of the Alpha Wire Communication, Control, and Industrial product family.

For further information, please contact:

Alpha Wire

Tel: 800 • 52 • ALPHA
Website: www.alphawire.com

MOTOR CONTROLLERS

14 kiloWatt, Dual Channel DC Motor Controller

Roboteq, Inc., now offers an intelligent dual channel DC motor controller capable of directly driving up to 120 amps on each channel at up to 60V. The AX2860 is targeted at designers of mobile robotic vehicles including Automatic Guided Vehicles (AGV), Underwater Remote Operated Vehicles (ROVs), and mobile



continued on page 72

EVENTS CALENDAR



Send updates, new listings, corrections, complaints, and suggestions to: steve@ncc.com or FAX 972-404-0269

If you happen to be in Dallas, TX, be sure to catch the Dallas Personal Robotics Group at the annual All-Con science fiction convention on March 13, 14, and 15. I'll be there, too, so stop by and say hello.

Know of any robot competitions I've missed? Is your local school or robot group planning a contest? Send an email to steve@ncc.com and tell me about it. Be sure to include the date and location of your contest. If you have a website with contest info, send along the URL as well, so we can tell everyone else about it.

For last-minute updates and changes, you can always find the most recent version of the Robot Competition FAQ at Robots.net: <http://robots.net/rcfaq.html>

— R. Steven Rainwater

March

- 7 CIRC Central Illinois Bot Brawl**
Lakeview Museum, Peoria, IL
Events this year include line following, line maze, 500g mini Sumo, 3 Kg Sumo, LEGO Sumo, and Ant-class RC combat.
<http://circ.mtco.com/>
- 7-8 METU Robotics Days**
METU Culture & Convention Center, Turkey
Events include line following, mini Sumo, standard Sumo, trash hunting robots, stair climbing, slalom, and freestyle.
www.roboticsdays.org/
- 13-14 AMD Jerry Sanders Creative Design Contest**
University of Illinois at Urbana-Champaign, IL
Robots must navigate a 3D course, retrieve balloons, and use them to play a game of tic-tac-toe against and oppose a team's robot.
<http://dc.cen.uiuc.edu/>
- 13-15 DPRG at All-Con**
Crowne Plaza Hotel, Addison, TX
DPRG hosted events include traditional RoboRama events such as quick trip, line following, T-time, and possibly a demo of a new event called square dance that's based on the Borenstein Square UMBmark metric. There will also be other demos and panel discussions.

www.all-con.org/

- 21 RobotChallenge**
Vienna, Austria
Standard Sumo, mini Sumo, micro Sumo, Slalom, and parallel Slalom.
www.robotchallenge.at/
- 21-22 Manitoba Robot Games**
TecVoc High School, Winnipeg, Manitoba, Canada
Mini Sumo, prairie Sumo, Tractor Pull, Super Scramble, line following, and Robo-Critters for the kids. Sumo events are held for both autonomous and tethered robots.
www.scmb.mb.ca/
- 21-22 Roboticon**
University of Guelph, Ontario, Canada
A robot soccer event for LEGO robots.
www.collegeroyal.uoguelph.ca/
- ## April
- 4-5 Trinity College Fire-Fighting Home Robot Contest**
Trinity College, Hartford, CT
The well-known championship event for fire-fighting robots.
www.trincoll.edu/events/robot/
- 14-16 DTU RoboCup**
Technical University of Denmark, Copenhagen, DENMARK
Line following and wall following events for autonomous robots.
www.robocup.dtu.dk/
- 16 Austrian Hexapod Championship**
FH Hagenberg, Austria
Events include Hexapod dancing and a Hexapod race.
www.fh-ooe.at/campus-hagenberg/aktuelles/events.html
- 16-18 FIRST Robotics Competition**
Georgia Dome, Atlanta, GA
National Championship for the regional

EVENTS CALENDAR



FIRST winners.
www.usfirst.org/

- 17 Carnegie Mellon Mobot Races**
CMU, Pittsburgh, PA
The traditional Mobot slalom and MoboJoust events.
www.cs.cmu.edu/~mobot/

- 17-18 BlimpDuino Aerial Robotics Competition**
Georgia Dome, Atlanta, GA
Autonomous blimps must complete five tasks.
<http://robots.net/article/2739.html>

- 17-18 National Robotics Challenge**
Marion, OH
Student competition designed to complement classroom instruction.
www.nationalroboticschallenge.org/

- 18 Penn State Abington Fire-Fighting Robot Contest**
Penn State Abington, Abington, PA
Based on the Trinity College Fire-Fighting contest. Autonomous robots must locate and extinguish a flame in a scale model of a home.
www.ecsel.psu.edu/~avanzato/robots/contests/

- 18 RoboRodentia**
Mott Gymnasium, California Polytechnic, San Luis Obispo, CA
Autonomous micromouse-like robots must navigate a maze while picking up and moving small balls.
<http://tiedye-srv.csc.calpoly.edu/~jseng/robotics.html>

- 18 UC Davis Picnic Day MicroMouse Contest**
University of California, Davis campus, CA
Standard micromouse contest.
www.ece.ucdavis.edu/umouse/

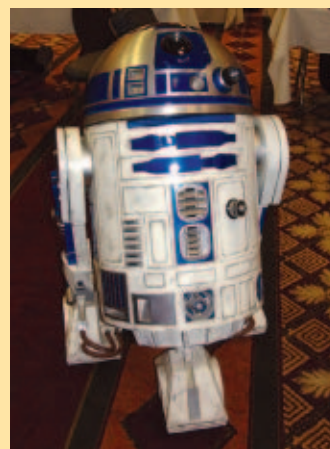
- 25 Historical Electronics Museum Robot Festival**
Linthicum, MD
Events include Fire-fighting, FIRST, robot Sumo.
www.robotfest.com/

- 25 Penn State Abington Mini Grand Challenge**
Penn State Abington, Abington, PA
Autonomous outdoor ground robots must navigate around the campus — both on and off-road — avoiding obstacles.
www.ecsel.psu.edu/~avanzato/robots/contests/outdoor/

For Your Information ...

The Dallas Personal Robotics Group will have a large presence at the 2009 All-Con science fiction convention in Dallas, TX, March 13-15. Fans of fictional robots like R2D2 will have the opportunity to see and learn about real robots.

The DPRG will host a one hour panel discussion on robotics at noon on Saturday. At 8 pm on Saturday evening, the



DPRG will host a two hour robot petting zoo where visitors can get hands-on time with real robots built by DPRG members. On Sunday, the DPRG will put on a full scale demonstration of their twice annual robot competition known as the RoboRama. The group will also be showing off robot technology throughout the convention on the demo stage in the vendor room. And, finally, the group will have two tables full of robots in the display area throughout the convention. If you get tired of looking at robots, All-Con boasts a wide range of other unusual activities and sights ranging from local burlesque group demos to workshops on Do-It-Yourself vacuforming.

Dallas Personal Robotics Group (DPRG)
www.dprg.org/

All-Con 2009
www.all-con.org/

Full All-Con 2009 Schedule
www.all-con.org/2009-Crowne/forms/Schedule.pdf

COMBAT ZONE

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BUILD REPORT

An Introduction to Wedges

● by Thomas Kenney

Since the inception of robotic combat, the wedge has been viewed as one of the most successful and easy to build designs, though it has also become among the most hated. The combination of reliability and simplicity has yet to be bested by any other combat robot design. Since the original wedge was introduced by US Robot Wars champion "La Machine" almost 14 years ago, many different designs of wedges have developed, all of which have their own unique quirks and advantages. In addition to their common use as a robot's main weapon, they are also often used as a backup weapon in case the primary weapon fails because of how easy it is to implement a simple wedge into a robot's design.

Overall, the wedge can be divided into two basic categories: wedges that drag along the arena floor and those that don't. The most obvious advantage the dragging wedge brings to the

table is that if the arena floor is smooth enough, it should have no problem getting under other robots and non-dragging wedges. Non-dragging wedges — though obviously not as low as dragging ones — are usually more durable and with adequate ground clearance, should never become caught up on the floor like dragging wedges sometimes do. Besides these two categories, one other important design variable is the number of wheels that the robot uses to drive. Two-wheeled wedges, for instance, are usually designed completely different than robots with four or more



FIGURE 1. Antweight "Inkspot", an example of a two-wheeled dragging wedge. Photo courtesy of Steve Judd.

FIGURE 2. A top view of beetleweight “Cloud of Suspicion” showing its hinged wedge assembly.

wheels, which provide superior traction and maneuverability.

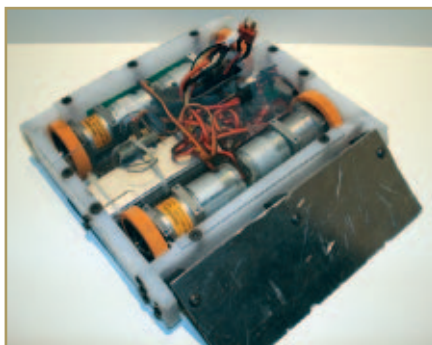
The most commonly seen types of dragging wedges are the two-wheeled type. Typically, the two drive wheels will be mounted in a position so that the center of gravity is placed in front of them. The wedge is normally on the front of the robot, and because of this wheel placement, a portion of the robot’s weight is used to hold the wedge to the ground. One variation of this type of wedge robot is what I like to call a “cheese block wedge.”

This robot is normally two wheeled and is shaped like a cheese block lying on its side, so that the entire body of the robot acts as the wedge. In most cases, the entire top armor acts as the wedge surface. Usually in the lower weight classes, usually the robot’s side is cut out into an acute triangular shape, and the top, bottom, and back armor are end tapped into this. In large bots, the same design technique can be used, though most builders simply weld frames together and screw on the armor. Most of these bots have large exposed wheels and are fully invertible, being able to operate equally well upside down — an advantage that two-wheeled wedges without this chassis shape do not have.

The other commonly used type of dragging wedge robot uses hinges and normally has four or more wheels; though some two-wheeled robots with hinged wedges exist. These robots hold the advantage of still having a wedge dragging along the ground while holding the maneuverability advantage four-wheel drives brings.

When the wedges are used as primary weapons — unlike two-

FIGURE 4. Lightweight “The Brown Note’s” wedge proved so tough and reliable at RoboGames 2008 that it was able to keep its wooden chassis from being shredded by the numerous spinners it fought, until it made it to the finals against “K2”, where it was promptly destroyed. The main reason K2 was able to defeat The Brown Note so handily in the end was that its sharpened titanium wedge easily slid under the dull steel wedge of The Brown Note.



wheeled dragging wedges — these robots do not normally have a wedge shape built into their frame. Instead, the wedges are simply screwed or welded to the robot. Although many builders use easy to implement, commercially available hinges that can be bought from vendors such as McMaster-Carr and other hardware and home improvement stores, I (among many other builders) prefer to make my own hinge assemblies. When I tried some commercially manufactured hinges on my 30 pound featherweight, I found that the weak steel used in their construction could easily be torn in half when hit by another robot’s weapon.

Similarly, I could see that the smaller aluminum hinges sometimes used on insect class robots were made of a soft alloy and bent easily. In the end though, nearly all hinged wedge assemblies are less durable because they normally aren’t as solidly attached to the bot.

When building all of my various robot’s hinged wedge assemblies, I would simply run a shaft from an extension of the sidewall of each

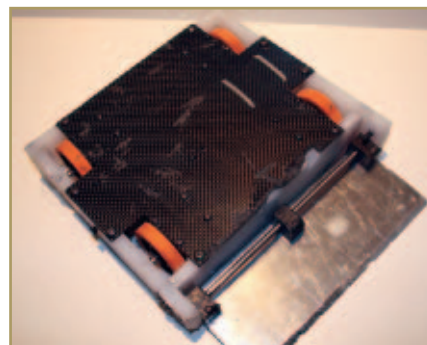


FIGURE 3. An underside view of beetleweight “Cloud of Suspicion” showing the bottom of its hinged wedge assembly, including its UHMW blocks and the 3/16” titanium shaft that the wedge rotates on.

side of the robot. I would then screw the metal wedge piece itself to two or three pieces of UHMW (a material with a very low coefficient of friction, making it very slippery), which have holes drilled for the size of the shaft. I would then slide the wedge with the UHMW blocks onto the shaft. This created a durable and simple hinged wedge assembly which can be seen in Figures 2 and 3. (For more details on the robot shown here and its construction, please see the December ‘08 issue of *SERVO*.)

Finally, the last common type of wedge is the fixed, non-dragging wedge. Overall, whether the robot has two wheels or more does not matter with this design (even if the robot is two wheeled) because it will normally have a caster or something else to keep the wedge off the ground. Some two-wheeled, non-dragging wedges may look the same as a two-wheeled dragging wedge, other than the previously mentioned caster.

Since they are solidly connected



FIGURE 5. Middleweight “Professor Chaos”, a vertical spinner, uses a hinged rear wedge as a reliable secondary weapon.

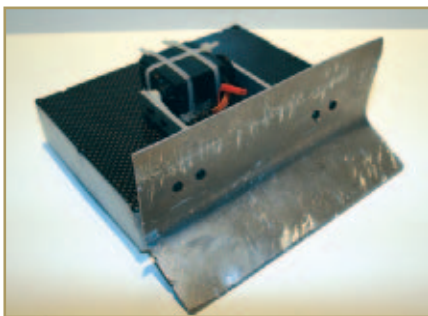


FIGURE 6. Antweight "Misdirected Pedestrian's" steep anti-spinner lifting wedge plow shows how wedges can be used effectively in other designs. This robot's wedge plow takes about 95% of spinner hits directed at the robot, and only a few small scratches can be seen on the rest of the robot's body.

to the robot's chassis through screws or welded joints, these types of wedges are generally more durable and less likely to be torn off of the robot. Shock mounting fixed wedges has also become popular in the 12 pound and up weight classes. This process involves mounting the wedge to the robot's chassis through rubber sandwich mounts. This isolates the robot's chassis from shock and allows the wedge to take more of the impact from the blows.

Beyond the way the wedge is mounted, there are two final variables I'm going to mention: angle and material. Both of these aspects go hand in hand. A wedge can serve different purposes, depending on the angle. A shallower wedge can slide under spinners and other robots with ease, and hits from the

aforementioned spinner will simply bounce off of the wedge because of the shallow angle. Wedges with steeper angles are meant to be much more of an attack method against spinners. Instead of sliding under the weapons, the objective is normally to simply slam into the weapons and have them bounce off in a dramatic manner — possibly causing damage to the spinner instead of its intended target.

As stated previously, the thickness needed for the wedge's material is directly proportional to the angle of the wedge. A shallow wedge that is designed to simply slide under an opponent will take many less solid hits than one intended for use as a much more offensive wedge with a steep angle. Thus, the steeper wedge would require much less thickness from the same material.

In conclusion, I'd like to say this — I hate slow, boring wedges — even the brightly colored ones — just as much as everyone else, but

contrary to what it may seem, there are a good amount of fast wedges that can be just as exciting as any good spinner. I know this for a fact, as I personally own one that has actually damaged the arena and disabled spinner's weapons on occasion.

If you still think that wedges are bad for the sport, why not add something else to the table? Wedges can be effective with clamping arms, lifters, and spinning weapons mounted to the robot, usually in which case the wedge will help lead the opponent up to the weapons. Also (as stated earlier in this writing), wedges always make an effective backup weapon when constructed reliably. When your robot's main weapon fails during a fight, it's much more effective to be able to get under the opponent and bully them around as opposed to simple pushing.

Finally, I'd like to note one tip that's constantly overlooked by many builders of dragging wedges — Sharpen your wedge and do your best to get it as close to the ground as possible! The robot with the lowest wedge always wins ... at least in wedge fighting. **SV**

MANUFACTURING: Attaching Wheels to Your Robot's Drill Motors

● by Ken Brandon

Drill motors are often used in 12 and 30 lb robots. One of the hard parts about using them is finding a way to attach a wheel, since the shaft is threaded and uses a lefthanded screw to keep the chuck from coming off. Here, I will show you a simple way to make press fit hubs to mount Colson wheels to your drills.

You will need a 0.75" diameter aluminum stock, as well as a lathe, 3/8"-24 tap, Q size drill bit (0.332"), and a 0.25" drill bit.

• **STEP 1:** Chuck the stock in the lathe chuck,



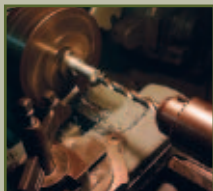
leaving 1.25" or so exposed. Then, mount a left-handed cutting tool into the tool holder.

• **STEP 2:** Turn on the lathe and begin removing metal to 1" from the end



of the stock, checking your progress until the stock is turned to a diameter of 0.61".

- **STEP 3:**
Chuck that size Q drill bit into the tailstock then drill a hole 0.75" deep from the end of the stock.



- **STEP 4:**
Unchuck the drill bit from the tailstock and chuck a 3/8"-24 tap into the tailstock, leaving the chuck loose in the tailstock to allow the chuck to be turned by hand.



- **STEP 5:** With the lathe off, begin

tapping the hole, two turns clockwise, one turn counter-clockwise, etc., to allow the chips to be broken up. Remember to use plenty of lubrication.



- **STEP 6:**
Change the cutting tool out for a cutoff tool. Turn the lathe on and cut off the hub 1" from the tapped end.



- **STEP 7:** Now chuck the hub into the lathe chuck with the threaded end facing the



chuck, then chuck the Q drill bit into the tailstock. Drill a hole 0.25" deep from the end of the stock.

- **STEP 8:**
Next, chuck a .25" drill bit into the tailstock, then drill all the way through the piece.



- **STEP 9:** Press the hub into the wheel, twist onto the drill shaft, and tighten up the left-handed screw. Now you have a simple method to attach wheels to your robot's drill motors. **SV**



EVENTS

Event Results and Upcoming Events

Event Results for December 8, 2008 to January 9, 2009

Rumble at the Rock was presented by BotsIQ and Boston Tooling on December 13th at Plymouth North High School.

RUMBLE @ THE ROCK!

Upcoming Events for March-April 2009

MegaCon RobotBattles will be held in Orlando, FL on March 1st. For more information, go to



www.robotbattles.com.

Robots Live will hold an event in Burgess Hill, England on March 28th and 29th. For more information, go to www.robotslive.co.uk.

Mad Metal Machines 7 will be held in Bochum, Germany on March 6th through 8th. FCC Featherweight 2009 will be held in the Netherlands on March 22nd, and the Tijdens Antweight World Series will also be held in the Netherlands on April 10th. For more information, go to www.dutchrobotgames.nl.



Central Illinois Bot Brawl 2009 will be held in Peoria, IL on March 7th. For more information, go to

<http://circ.mtco.com>.



Ideas Festival 2009 will be held in Brisbane, Australia on March 27th through 30th. For more information, go to www.ideasfestival.com/au.

Seattle Bot Battle 7 will be held in Seattle, WA on April 12th. For more information, go to www.westernalliedrobotics.com. **SV**



HISTORY REPORT: Rob tic Competition, Southern Style

● by Kelly Lockhart

There have been many articles and books written over the years trying to untangle the murky pre-history of robotic combat in the United States. Many people trace the roots back to engineering competitions at MIT, CalTech, and other universities in the 1970s, while others look to the San Francisco, CA, performance artists of the 1980s and their monstrous mechanical creations. For most enthusiasts, though, their first exposure to the sport was in 1994 when Marc Thorpe organized the first Robot Wars competitions in San Francisco.

But Thorpe was far from being the first. In fact, it was eight years earlier in Denver that the seeds of robotic combat were first sown. An engineer by the name of Bill Llewellyn got together with a group of like-minded friends and formed the "Denver Mad Scientists Society" and organized a competition where people constructed autonomous robots to complete a simple race.

He called it the "Critter Crawl" and it was met with great interest. As Bill recalls, "Everyone came up to us afterwards and said that as cool as the event was, it would be so much cooler if the robots actually fought each other."

So, in 1987, Bill and his mad scientist cohorts developed a simple set of rules for cable and R/C-controlled robots and launched the very first "Critter Crunch" at the MileHiCon science fiction convention. The event has been held at the convention every year since, making it easily the oldest and longest running robotic combat event in the world. But the story doesn't end there. In fact, it takes a strange — and decidedly southern — twist when in 1991, a disc jockey by the name of Kelly Lockhart got a copy of the rules and talked it over with the then chairman of the Dragon*Con science fiction convention to see if they thought it would be possible to stage a similar event in Atlanta, GA.

They decided to go for it, had the rules published in a convention flyer, and then set aside a clear space near the loading docks of the Atlanta Hilton & Towers hotel for the competitors that showed up to fight. Two showed, and they fought their robots against each other until one could no longer function. Even with just two robots, a crowd of nearly 200 watched the action and wanted to know how they could get involved the next year. And thus, "Robot Battles" was born.

The following year, the event was moved to a ballroom with a Sumo-style stage set-up (which is used to this day) and an even-dozen competitors showed up to compete in front of an audience over twice as large as the year before. Intrigued by the response of both Robot Battles and what Marc Thorpe was doing out west in San Francisco, in 1994 Thorpe was invited to Dragon*Con to co-host the event with Lockhart. While the Robot Wars events were on a much larger scale, Thorpe appreciated the enthusiasm that the Robot Battles competitors brought to the event. So much so, that he returned to Atlanta in 1995 to co-host the competition for a second year.

In 1997, Robot Battles moved from the Atlanta Hilton into a full theater complex in the Atlanta Apparel Mart, which coincided with a dramatic increase in attendance and participation. This can be directly linked to the spread of the Robot Wars publicity and several television shows that took a direct look at what Marc was doing in San



Francisco. The following year, Robot Battles not only moved into a larger hall — the 800 person capacity Regency Ballroom of the Hyatt Regency Atlanta — but was also staged on a smaller scale at the January Chattacon convention in Chattanooga, TN, marking the first time that the event had been held separate from Dragon*Con. The move into a larger facility and the continued spread of awareness about robotics sports made for a record event, with more robots and a larger crowd than ever before.

1998 was also the first year that Lockhart started his tradition of wearing his signature bizarre costumes on stage as he hosted the event. Over the years, audiences have been subjected to everything from full Roman Centurion dress armor to the skinniest Elvis impersonator they've ever seen, to Monty Python's "Spamalot" to this past year's bizarre melding of Willy Wonka and The Joker.

One of the more unusual aspects of Robot Battles that makes it stand out amongst the other competitions (aside from its longevity and Sumo-style rule set) is the nature of how it is presented.

Many longtime attendees refer to the event as a combination robotic competition and nightclub comedy act. There are many people who come to the events as much for the on-stage antics as for the competition itself.

But what really kicked off the growth and popularity of Robot Battles was when BattleBots debuted on Comedy Central in 2000, and the heretofore underground sport became a national obsession. Robot Battles responded by adding weight categories and reaching out to schools and universities to bring in new contestants — especially younger ones. The result was incredibly positive, and continues to this day. There are more teenage — and younger — contestants participating in Robot Battles than any other robotic combat event (not counting the FIRST events, obviously, since they are not considered robotic combat). And they are competing quite well. In fact, the current record for youngest tournament winner is nine years old, winning the 12 pound weight class in 2007 at North Carolina's StellarCon competition.

In addition to expanding weight categories, Robot Battles also began expanding from its Atlanta base in the early part of the decade. Coming full-circle to their roots in Denver, in 2003, Robot Battles came to the Denver Opus Fantasy Arts convention with assistance from Llewellyn and several other of the mad scientists. It was that year that Lockhart was granted full membership in the Denver Mad Scientists Society, something of which he is quite proud of and finds very amusing to add to his resume.

Today, 19 years and 35 events after its inception, Robot Battles is larger and more popular than ever before. In addition to dealing with standing-room-only crowds at nearly every event, Robot Battles is very pleased to continue its expansion this year by hosting an event at the Orange County Convention Center in Orlando, FL in February as part of the MegaCon convention.

If you'd like to learn more about Robot Battles, where the next events will be, and read one of the most concise (and amusing) rule sets in the entire sport, head on over to their website at www.robotbattles.com. **SV**

PARTS IS PARTS: Flipper Calculator Turn "Cut and Try" (nearly) into Science

● by Kevin Berry

I recently ran across a couple of very well written websites that show how to design pneumatic flippers. Most of us use very empirical, "try it and see" techniques, but these teams from the UK have turned flipper design into

science. The "Hassocks Hog" team — veterans of Robot Wars — has set up a step-by-step guide to design your flipbot (www.hassockshog.co.uk/flipper_calculator.htm).

First, you download a spreadsheet. After picking the

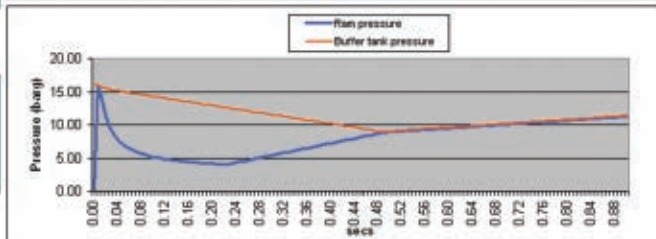
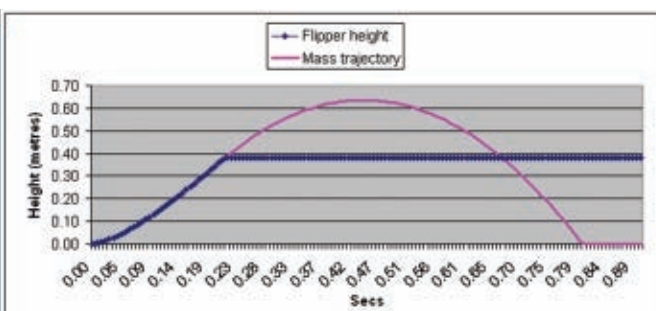
geometry of the mechanism, proposed dimensions are entered. Next, the specifics of your pneumatic setup are entered. The outputs are graphs showing height and trajectory of your victim, and the pressure profile of your system.

System variables	
Gas reservoir size	1.1 kg
Regulated pressure (Pb)	10 bars
Regulator flow (Fr)	900 litres/min
Buffer tank volume (Vt)	1 litres
Combined valve flow (Fv)	1722 litres/min
Pipe outside diameter	12 mm
Pipe inside diameter	mm
No of pipes feeding ram	1
Pipe length	0.1 m
Ram bore	100 mm
Ram stroke	200 mm
Number of rams	3
Ram operation	single sided
Back pressure (PQ)	0.5 bars
Flipper geometry type	Type B
Mass of flipper (Mf)	10 kg
Mass to throw (M)	100 kg
Regulator variation	5 %

Narrower graph
Wider graph

Max velocity	2.33 m/sec
Max height above flipper	0.28 m
Max height above ground	0.66 m
Max weight lifted	636.06 kg
Time to lift off	0.22 sec
Lift off to max height time	0.24 sec
Max flips per bottle	22

Maximum actuator flow rate is
1722 litres/min per feed pipe.



Hassocks Hog's calculator outputs both graphical and tabular information.

to document his own research, then polished it up for others to use, exemplifying the spirit of cooperation in the sport.

Another British team, "Onslaught", has a very understandable tutorial on pneumatic flipper system design (www.teamonslaught.fsnet.co.uk/pneumatics.htm).

The writer, Alan Wood, leads you through the purpose of each component. Then, there is an embedded calculator

where you can enter various options for your design, and the model calculates throw height, acceleration, gas flow parameters, and the forces involved. A second page has lots of interesting info on valves and max flow rates.

In a recent forum exchange, Woody extensively refined his models and tools to help another builder with his design; again, a fine example of the professional attitude of combat bot builders. Alan says "I enjoy trying to help out others by passing on what I know ... or think I know!" He is in the process of developing a more complex set of models and tools that will be available soon.

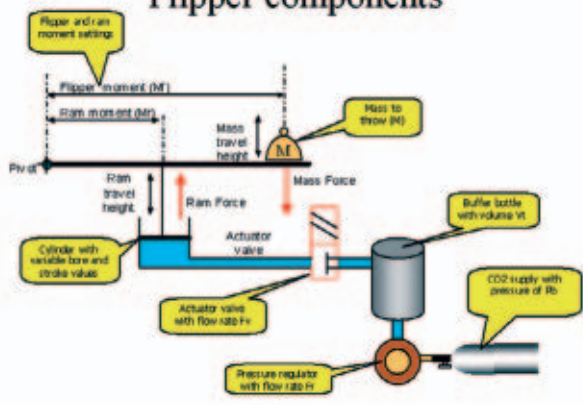
Both authors are careful to point out that their models are idealized and incomplete, and your mileage *WILL* vary! **SV**

Team Onslaught exposes the soft, creamy filling of one of their flip-bots.



Team Onslaught's model calculates forces, distance, and gas flows.

Flipper components



The Hassocks Hog spreadsheet includes a tab with this easy to understand graphic of a typical flipper design.

Another cool feature is an animation of the flipper's action, graphing the tip position at each point in the throw. Noel Poncelet, creator, generated this tool in an effort

Maximum Required SYSTEM FLOW for Pneumatic Flippers

Ram bore (mm)	Ram stroke (mm)	Pressure (bar)	Mass to move (Kg)	Leverage (ratio x 1)
100	300	50.00	105.00	4.0
Force (Kgf)	Force/leverage Newtons	Acceleration (metres/sec/sec)	Velocity (metres/second)	Throw (metres)
4000	9800.0	83.53	14.16	10.23
Ram extension time (seconds)	Ram Volume litres	Flow SCFM	Flow Litres/min	
0.17	2.36	1497.13	42394.08	
Maximum required CV for co2	Maximum required Kv (M3/hr) for co2	Qn Air (L/Min)	Qn CO2 (L/Min)	Maximum required CV for air
21.79	18.8	18157.5	21789.0	18.16

All the above assumes a constant gas supply with a 10 psi pressure drop -- Delta P.
Any increase in flow beyond the calculated Max gives no extra return in terms of throw
...it only increases the unloaded ram extension speed.

Suggested pressure drop in psi -->>	109
Giving a recommended Cv for co2 of --	7.1

CHEAP SPEED

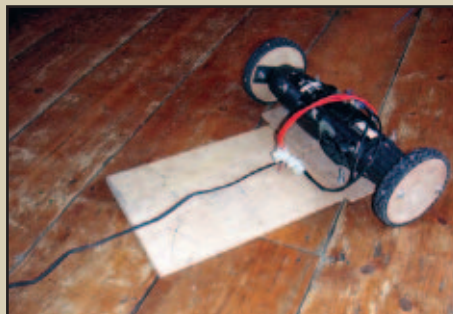
● by John Frizell

You can control your robot with a simple relay H-bridge that gives you just three settings: off, full forward, and full reverse. These controls — sometimes called “bang bang” for the way they drive your machine — are cheap and easy to make. Although they are unlikely to get you into the top leagues, they provide surprisingly good control for a fighting machine since you are likely to be running it at full speed most of the time.

An H-bridge is based on the single pole double throw (SPDT) switch or relay. These have a common terminal and two contacts: normally open (NO) and normally closed (NC). The NO contact acts like an ordinary switch — off until the switch is pressed or the relay coil energized. The NC is the reverse — on until the switch is pressed. You need two of these to make an H-bridge.

Figure 1 shows an H-bridge. This is the easiest circuit you will ever wire. Connect the two NO terminals to positive, the NC terminals to negative, and connect the motor to the two common terminals. No current flows until one side or the other is triggered by pressing a microswitch or energizing a relay coil. Triggering one side runs current through the motor in one direction. Triggering the other side runs current in the other direction, reversing the motor. If you have built your H-bridge properly, it will not be possible to trigger both sides at once. However, if you somehow did, it would simply turn the motor off.

You can build an H-bridge from a servo and two microswitches. Photos 1 and 2 show the beginning and end of a build. You will need a drill and a flat file. Get the servo arm as close to the middle as it



will go and use the transmitter's trim control to center it. In this example, the finished product weighs about 25 grams and can handle three amps. With bigger microswitches, it could handle 10 amps. Photo 3 shows various microswitches.

If you want more current than you can get from microswitches, use the NO contacts of the microswitches to trigger a relay H-bridge. Cheap 12V automotive relays will handle 30–40 amps. Figure 2 shows the schematic for this type of bridge.

Photo 4 shows a pair of small microswitches operating a relay H-bridge to provide a 30 amp controller. The base is made of wood because it is light, cheap, and easy to work with; 5 mm machine screws serve as terminal posts. This

example has the relay coils at 12V and the motor at 24V (or whatever you want to put through the relay contacts), but you could operate it all at 12V or 24V with suitable relays; you would only need two terminal posts.

Servo-based controls have one disadvantage: they don't have a failsafe. If the transmitter signal or power is lost (say because your aerial has been ripped off or battery disconnected), the servo stays at its last position — which may well be full on. No venue will accept this for a big fighting robot, but for smaller robots or non-combat ones, this may be acceptable. Check with the venue where you plan to run the robot. You can buy failsafes (Photo 5), intended for model planes that

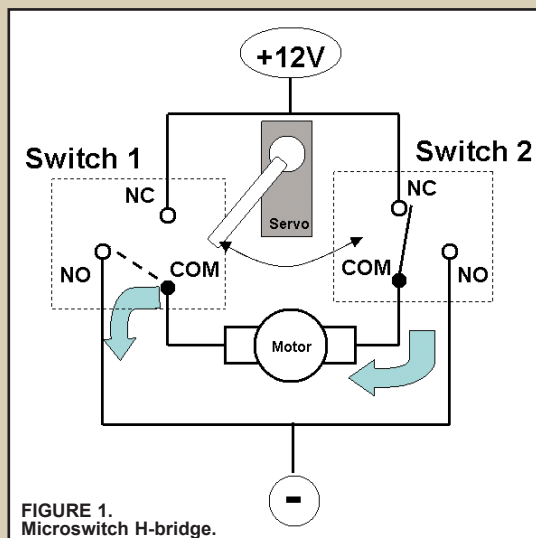
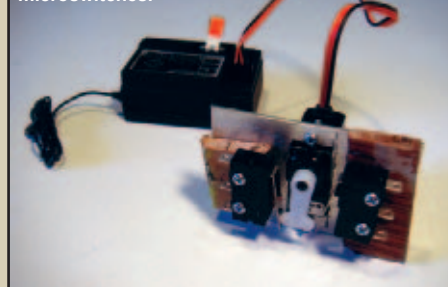


FIGURE 1. Microswitch H-bridge.

PHOTO 1. Piece of aluminium angle drilled to form mounting hole for servo.



PHOTO 2. Finished servo H-bridge using microswitches.



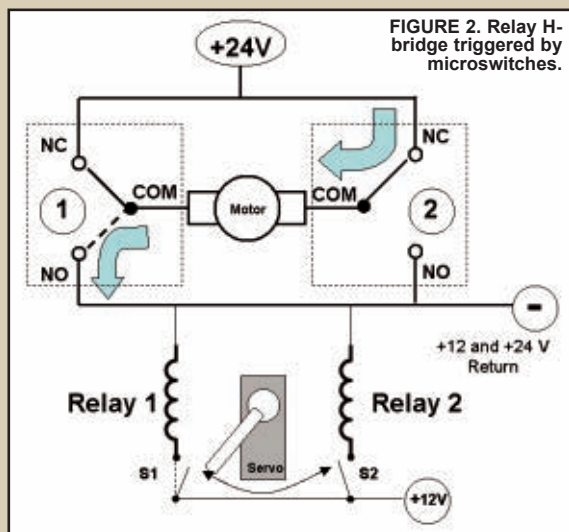


FIGURE 2. Relay H-bridge triggered by microswitches.

PHOTO 3. Various microswitches.



solve this problem by returning the servos to the off position if the signal is lost. Or you could buy a Team Delta (www.teamdelta.com) dual ended switch which has built-in failsafes, and wire up its relays as an H-bridge.

When you drive a robot with H-bridge controllers, you use the transmitter sticks like switches. You push one forward a bit, nothing happens until a bit further, then suddenly the motor comes on full. Moving the stick further does nothing more. To get proportional control — where a bit of stick gives you a bit of speed with the speed increasing as the stick is moved up — you need speed controllers.

If you use drill motors, you need speed controllers designed for brushed motors. There are two ways to go with speed controllers: you can buy a two channel one intended for robots; or you can use two single channel ones intended for

model cars or boats. Model plane controllers — which have forward only — aren't much use. Most model car or boat controllers will only handle up to 12 volts; specialized robot controllers like the Scorpion or Sabertooth will handle up to 24 volts or beyond.

A model car controller will have seven wires coming out of it: two thick wires that go to the battery, two more thick wires that go to the motor, and three thin wires that end in a plug that fits into your receiver.

Photo 6 shows some typical single channel controllers. The power cables are almost always red and black, and keeping the polarity correct is absolutely critical. If you connect the speed controller to the battery the wrong way around, it will blow instantly. Most speed controllers have a battery eliminator circuit (BEC) which supplies 5V to your receiver, eliminating the need for a receiver battery. But you only need one BEC. Running two will

cause problems. Disconnect the second one by cutting or unplugging the middle wire — it is usually red. You can see in Photo 6 that one of the speed controllers — an Msonic 25 amp model, made for model boats — is provided with an extra pair of wires and a switch for disabling the BEC. On the second controller — a general-purpose 100 amp model from China — the BEC has been disabled by taking the middle wire out of the plug.

DC motors draw their maximum current at stall so be sure to get controllers that have overload protection or thermal cutouts to protect them from excessive current draw. That way, if you stall your motors in a fight (say by getting into a pushing match), you won't burn out your controllers. Buy the best controllers you can afford — they tend to follow you from one robot to the next. **SV**

All photos are courtesy of Michi Mathias. All figures are courtesy of Kevin Berry.

PHOTO 4. Relay H-bridge operated by microswitches.

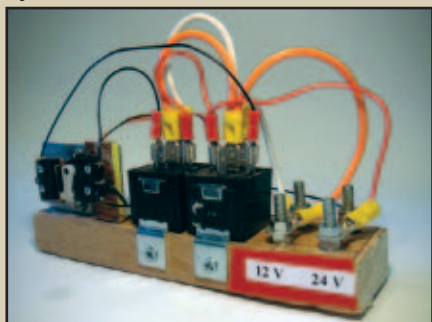
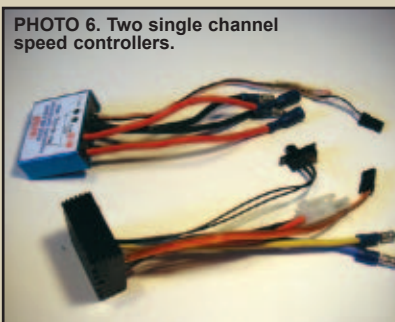


PHOTO 5. A failsafe.



PHOTO 6. Two single channel speed controllers.



ROBOT PROFILE

TOP RANKED ROBOT THIS MONTH

● by Kevin Berry

Top Ranked Combat Bots

History Score			Ranking		
Weight Class	Bot	Win/Loss	Weight Class	Bot	Win/Loss
150 grams	VD	26/7	150 grams	Micro Drive	7/2
1 pound	Dark Pounder	44/5	1 pound	Dark Pounder	13/1
1 kg	Roadbug	27/10	1 kg	Roadbug	7/4
3 pounds	3pd	48/21	3 pounds	Yeti	8/0
6 pounds	G.I.R.	17/2	6 pounds	G.I.R.	8/0
12 pounds	Solaris	42/12	12 pounds	Tourinho	13/1
15 pounds	Humdinger 2	29/2	15 pounds	Humdinger 2	29/2
30 pounds	Helios	31/6	30 pounds	Touro Feather	8/2
30 (sport)	Bounty Hunter	9/1	30 (sport)	Upheaval	8/4
60 pounds	Wedge of Doom	43/5	60 pounds	K2	14/2
120 pounds	Devil's Plunger	53/15	120 pounds	Touro	23/3
220 pounds	BioHazard	35/5	220 pounds	Original Sin	13/4
340 pounds	SHOVELHEAD	39/15	340 pounds	Ziggy	8/2
390 pounds	MidEvil	28/9			
History Score is calculated by performance at all events known to BotRank			Current Ranking is calculated by performance at all known events, using data from the last 18 months		
Data as of January 9, 2009					

Ziggy – Currently Ranked #1



Class: Super Heavyweight

Team: CM Robotics

Builder(s): Mark Demers, Tim Bayne, Vincent Chiu, Jerome Johnson, Michael Worry, Robert Frankovich, and Murat Ozkan

Location: San Jose, CA and Ottawa, Canada

BotRank Data	Total Fights	Wins	Losses
Lifetime History	23	15	8
Current Record	10	8	2
Events	9		

Ziggy has competed at RoboGames 2008, RoboGames 2007, RoboGames 2006, Battle Beach 4, Game Developers Conference 2006, February Fun Fest 2006, ComBots Cup I, RoboGames 2005, and MMR. Details are:

- **Overall configuration:** Lifter.
- **Frame:** Aircraft aluminum (including 2024, 7075, 6061).
- **Drive:** Electric drive MagMotor, 4.5 hp per side; 9 hp total.
- **Wheels:** 6" Colsons.

- **Drive ESC:** MC1.
- **Drive batteries:** NiMh, with a new LiFe option.
- **Weapon type:** Pneumatic flipper, four bar mechanism.
- **Weapon power:** Compressed gas, up to 14,000 lbs push force.
- **Weapon controller:** Freescale 9RS08 microprocessor.
- **Armor:** Titanium, 6Al-4V ... 120 lbs in total.
- **Radio system:** Spektrum DX6.

- **Design philosophy:** Make it entertaining for the audience.
- **Builder's bragging opportunity:** Most proud of the wins at RG07 where Ziggy was able to compete effectively against Vladiator and The Judge, a couple of machines from BattleBots which served as inspiration for our team to build a super heavyweight. We also love the positive response Ziggy gets from the fans. **SV**

Photos are courtesy of Annie Blumenberg. Information is courtesy of Mark Demers.

ROBOT PROJECTS

BY FRED EADY

Robotics and electronics go hand in hand. If you're also a *Nuts & Volts* reader, you know that we assembled a working Silicon Laboratories-based USB microcontroller interface in the *Nuts & Volts* Design Cycle column. This month, we're going to pull that *Nuts & Volts* USB project into RoboLand and put its resources to work in a mechatronic kind of way.

The Vehicle

The transport mechanism in our case is a printed circuit board (PCB). The fuel for our robotronic vehicle is composed of silicon and floobydust. The silicon is supplied by Silicon Laboratories and Microchip. The floobydust is a

product of our minds and the binary output we coax from a PIC C compiler. The metallic, silicon, and fiberglass components of our logical conveyance are shown in Photo 1.

The circuitry you see in Photo 1 was originally intended to introduce the feature set of the Silicon Laboratories CP2102 USB-to-UART bridge IC. The CP2102 is a drop-in replacement for legacy Maxim and Sipex RS-232 interfaces. The major selling point of the chip is that the programmer is not required to have indepth knowledge of the inner workings of the USB protocol to use it. The CP2102 takes care of the embedded USB intricacies with the help of a factory-supplied VCP (Virtual Comm Port) driver at the personal computer end. Thus, the USB programmer can assemble a complete embedded USB-based application without having to code any USB firmware. We went into the CP2102's details in the *Nuts & Volts* Design Cycle column. So, if you want to learn more about the CP2102 and what it takes to assemble your own embedded USB interface project board, take some time to look at the Design Cycle piece. After all, a combination of nuts, volts, and servos can be found in many a robotic device.

In this discussion, we'll treat the CP2102/PIC18LF4620 circuit you see in Photo 1 as a black box programmable module. The circuit layout for the PIC18LF4620-based black box is displayed graphically in Schematic 1. All of the programming will be aimed at the PIC. I'll compile our applications using the HI-TECH C PRO Compiler for the PIC18 MCU Family.

Think Big ... Start Small

Let's code up a simple USB communications link, send a message across it, and use MPLAB and the new Microchip ICD 3 debugger/programmer to view the results. To make things a bit easier, I've already coded up a PIC18LF4620 EUSART driver. You can see the details of this by examining the code in the download package available on the *SERVO* website (www.servomagazine.com). The driver is

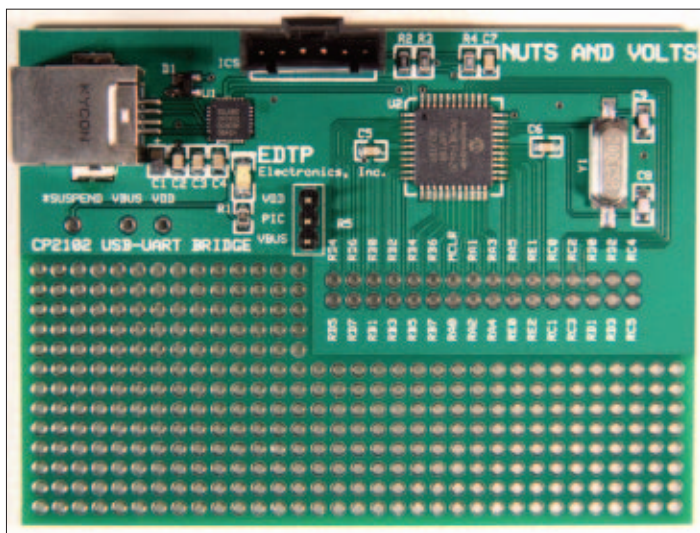
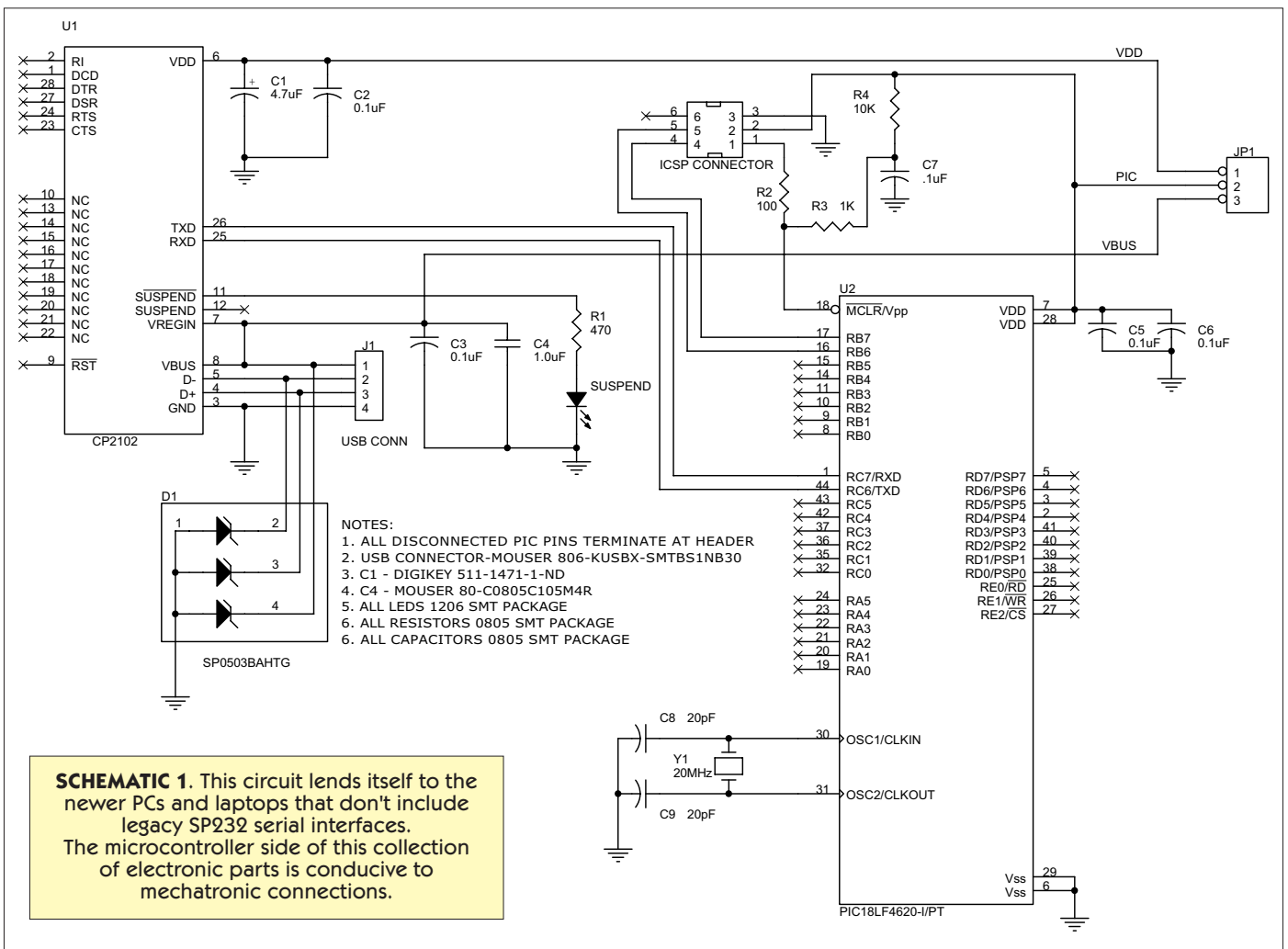


PHOTO 1. As a robotician and *SERVO* reader, you are used to supplying the NUTS of a project. This month, I'm supplying the VOLTS with this USB-enabled project board. A Silicon Laboratories CP2102 enables USB connectivity to the on-board PIC18LF4620 microcontroller. Many a mechatronic dream can be realized as 99% of the PIC18LF4620's I/O is available to the robotic user/programmer.

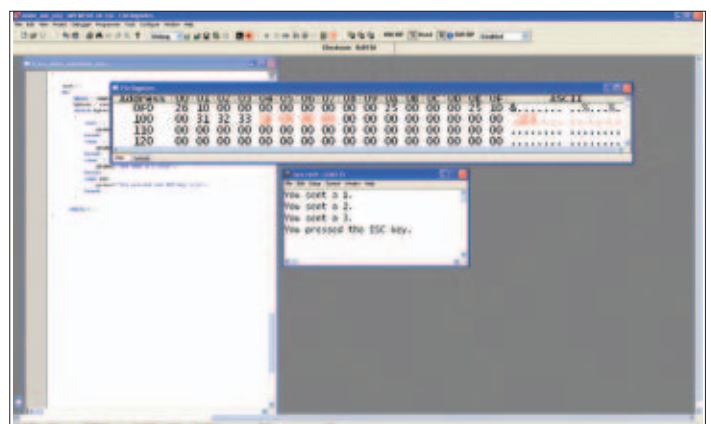


composed of a set of functions that enable transmit and receive operations. For now, just consider the EUSART driver as part of the programmable module.

A standard USB cable attached to my Lenovo S10 NetBook is the sole carrier of signals and power to the PIC-based programmable module. As long as we don't exceed a current draw of 100 mA, we can power the programmable module's electronics directly from the CP2102's built-in 3.3 volt regulator. The USB-fronted programmable module can also be configured to power its PIC and electronic payload directly from the USB VBUS (+5 volt) supply. In the case of the link test, we're not using the USB power to drive any external circuitry other than the PIC18LF4620 itself. So, now that we have the electrical portion of the link test application under control, here's the link test code snippet:

```
void main(void) {
    init();                // init the EUSART
    do{
        //wait for a character
        while(!(CharInQueue()));
        bytein = recvchar();    // get it
        switch(bytein)          // process
```

```
{
    case '1':
        printf("You sent a 1.\r\n");
        break;
    case '2':
        printf("You sent a 2.\r\n");
```



SCREENSHOT 1. This screen capture was taken after I had entered 1, 2, 3, ESC in the Tera Term Pro emulation window. The entries and responses match up with the code snippet we devised.

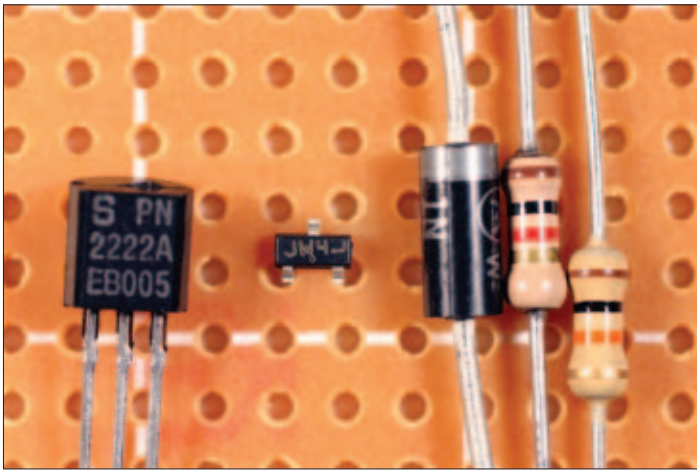


PHOTO 2. The NUD3105 is dwarfed by the PN2222A transistor alone. The rest of the components consist of the biasing/safety resistors and the free-wheeling diode.

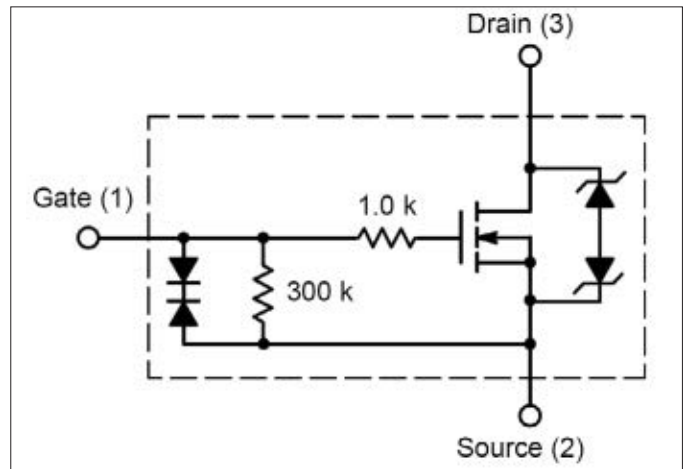


FIGURE 1. The NUD3105 replaces the resistors and diodes under the lens in Photo 2 and brings ESD protection and a zener clamp to the relay driver party in a tiny SOT-23 package.

```
break;
case '3':
    printf("You sent a 3.\r\n");
break;
case esc:
    printf("You pressed ESC.\r\n");
break;
}
```

```
}while(1); // do this forever
}
```

Once the EUSART is kicked into gear, we spin waiting for a character to be processed by the EUSART and placed into the receive buffer, which is a slice of the PIC18LF4620's internal SRAM. We use the EUSART driver's `recvchar()` function to extract the incoming byte from the receive

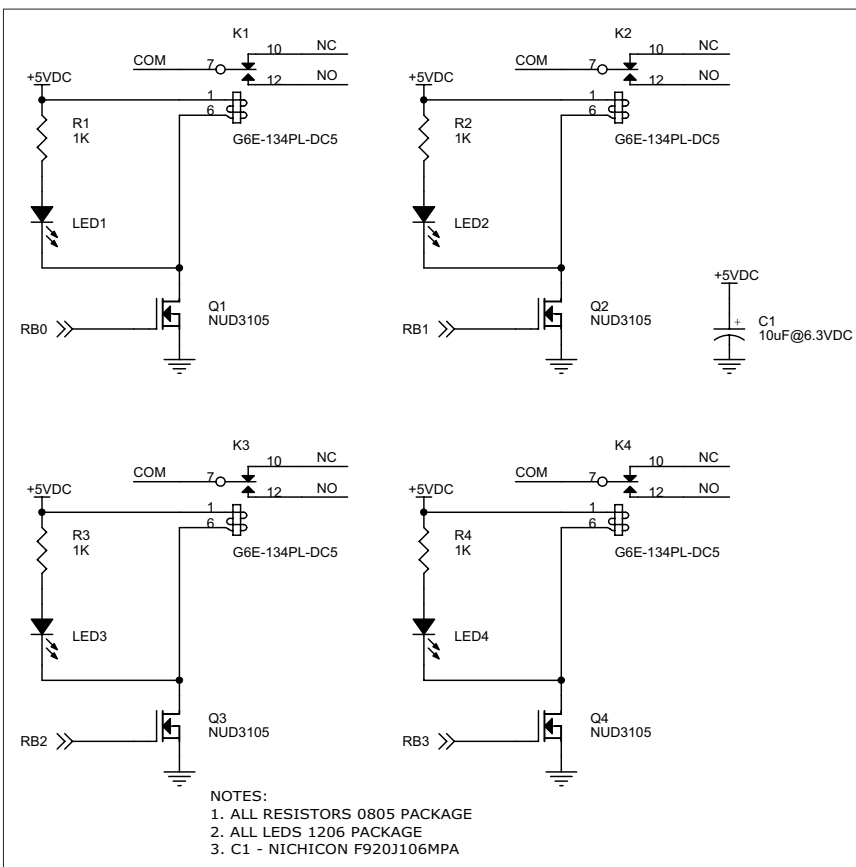
buffer. If the incoming byte matches any of the `switch(bytein)` statement cases, a unique message is returned to the Lenovo NetBook, representing the receipt of the byte.

The File Registers window in Screenshot 1 shows that the PIC18LF4620's EUSART received a sequence of four bytes (0x31, 0x32, 0x33, 0x1B) from the Tera Term Pro terminal application running on the Lenovo NetBook. Translating the hex bytes into ASCII characters, I sent 1, 2, 3, ESC to the PIC18LF4620's EUSART. The PIC responded by returning the messages you see in the screenshot of the Tera Term Pro window.

The results of the link test drive home some important points:

- The USB interface has the ability to provide power, as well as a data path to the PIC18LF4620-based programmable module.
- We didn't have to write any specific USB code for the PIC18LF4620 or the Lenovo NetBook.
- We were able to use a legacy RS-232 PC application to interface with the legacy EUSART application via a USB connection.

As Karen Carpenter once said, "We've only just begun .."



SCHEMATIC 2. The NUD3105 is very versatile. Small signal relays are just one type of device it can control.

Driving Mechanical Relays

Photo 2 consists of a collection of the electronic components that can be used to drive small signal relays. Take a look at the relay driver PCB layout in Screenshot 2. Note that the only resistors (R1-R4) in the relay driver design are servicing the quad of LED indicators. The free-wheeling diodes that normally straddle the relay coils are also absent. In this relay driver design, the four parts that are normally needed to drive a small signal relay coil are all contained within a tiny three-pin part: the NUD3105.

Figure 1 is a schematic representation of the NUD3105. Relay coils that operate between 3.0 volts and 5.0 volts DC and consume less than 400 mA can be driven with the NUD3105. A logical high level applied to pin 1 turns it ON. Our design places a NUD3105 (Q1-Q4) between each relay coil and common ground. Thus, when a logical high is applied to the Gate (pin 1), the relay coil's path to ground is established.

The ExpressPCB relay driver PCB you see in Screenshot 2 is designed to plug into the I/O and power connectors of the PIC18LF4620-based programmable module. Energizing all of the Omron G6E five volt relays simultaneously will exceed the 100 mA limit of the CP2102 voltage regulator as each relay coil draws 40 mA. To provide enough power to drive the relay coils and the PIC18LF4620, an external five volt supply connector has been included in the relay driver design. The relay board's basic electrical layout is shown in Schematic 2.

I put together a small application that allows the user to control the relays from a Tera Term Pro window over the USB interface. As you can see in the code snippet that follows, pressing the ESC key before entering the relay number turns the selected relay OFF:

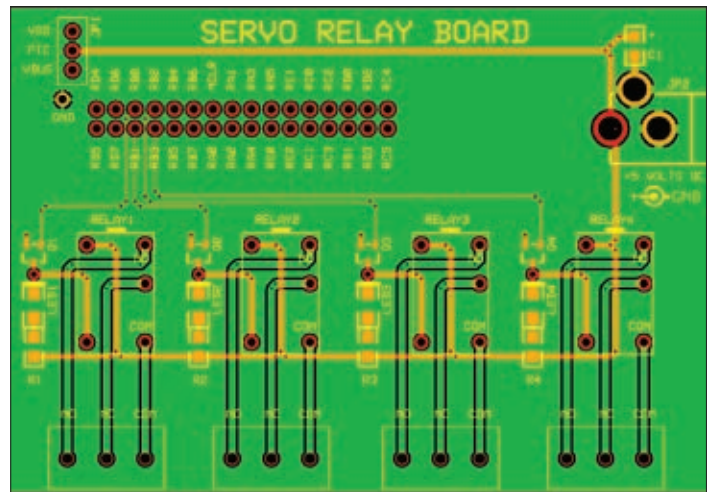
```
void main(void) {

    init();                //init the EUSART
    RELAY1 = OFF;
    RELAY2 = OFF;
    RELAY3 = OFF;
    RELAY4 = OFF;
    do{

        //wait for incoming character
        while(!(CharInQueue()));

        //extract first incoming ON byte
        bytein = recvchar();

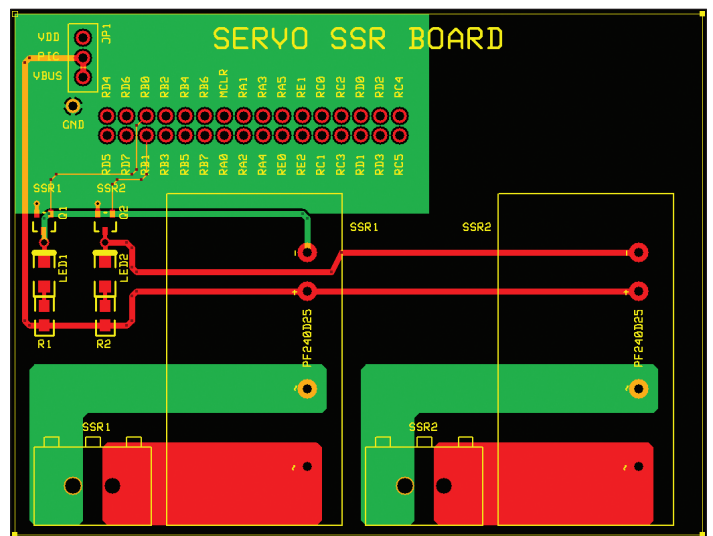
        switch(bytein) {
            case '1':
                RELAY1 = ON;
                printf("RELAY 1 IS ON.\r\n");
                break;
            case '2':
                RELAY2 = ON;
                printf("RELAY 2 IS ON.\r\n");
```



SCREENSHOT 2. The GND pin matches up with the breadboard pin that is nearest to the ground plane in that area. I used OMRON G6E 5.0 volt relays in my design and any 1206 LED will work here. The relay termination points are laid out to take a three position Phoenix contact screw terminal (Mouser 651-1729131).

```
break;
case '3':
    RELAY3 = ON;
    printf("RELAY 3 IS ON.\r\n");
    break;
case '4':
    RELAY4 = ON;
    printf("RELAY 4 IS ON.\r\n");
    break;
case esc:
    //wait for incoming OFF character
    while(!(CharInQueue()));

    //extract OFF character
```



SCREENSHOT 3. This AC SSR design switches a pair of AC loads. If you need to add more SSRs, be sure to take the SSR's control current consumption into account. Each SSR control interface needs 15 mA to operate.

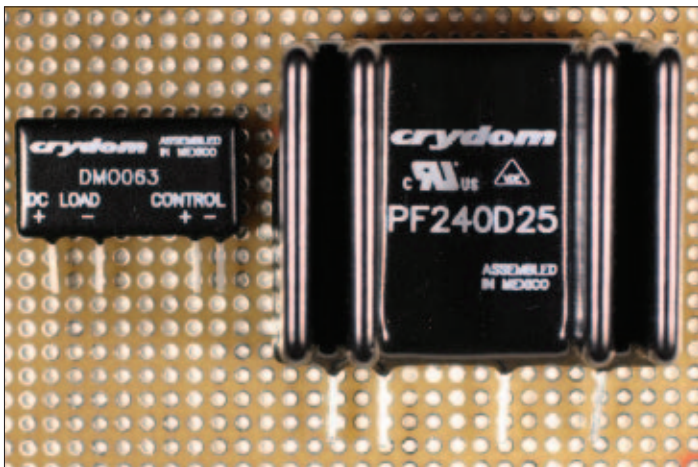


PHOTO 3. The large finned SSR is designed to switch AC loads. Its sidekick is a lower amperage variant that switches DC loads.

```
OFF.\r\n");
    break;
}
break;
}
}while(1);
}
```

Driving SSRs

There are times when you'll need to switch an AC load. If you choose to use SSRs (solid-state relays), you can switch most any AC load using the basics of the mechanical relay code and the PCB design you see in Screenshot 3.

The finned Crydom PF240D25 SSR shown in Photo 3 can switch AC loads up to 10 amperes at voltages up to 240 VAC. If you force enough air across its integral heatsink fins, the PF240D25 can switch a 25 amp AC load. The PF240D25 control inputs require 3.0 to 5.0 volts DC. The nominal current consumption of the PF240D25 control inputs is 15 mA. The low drive level requirement allows the PF240D25 to be driven directly from a PIC I/O pin. However, I've chosen to buffer the PIC-to-SSR I/O connection with a NUD3105. Using the NUD3105 multiplies the current capacity of the PIC's I/O pin allowing me to safely add an indicator LED for each SSR. I can also drive multiple SSRs from a single PIC I/O pin by incorporating a NUD3105 buffer. As you can see in Photo 3, Crydom also offers smaller lower amperage AC SSRs and DC SSRs. I've included ExpressPCB footprints for the Crydom SSRs in the download package.

Note that I'm powering the PF240D25 SSRs directly from the USB VBUS (+5 VDC) line. The trace connecting the PIC and VBUS power inputs also sends +5 VDC to the PIC18LF4620 power rail. In this situation, we could power the SSRs and the PIC from the VDD (+3.3 VDC) source, as well.

Spinning Motors

When it comes to robotics, two motors are most always better than one. With that sage advice in mind, I drew up the dual-motor PCB design you see in Screenshot 4. The board has a ground plane on both sides. The ground planes double as a massive noise quench and heatsink for the pair of Allegro A3953 full-bridge PWM motor drivers (U1-U2).

The power of the A3953 can be easily accessed by folding the following A3953 definitions and macros into your application:

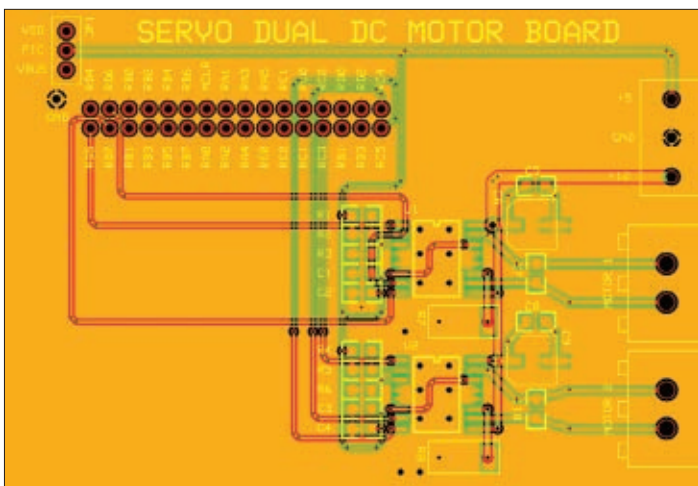
```
/**
 * *****
 * // DUAL MOTOR DRIVER DEFINITIONS
 * *****
 */
//brake1      LATD5
//phasel      LATD6
//enable1     LATD7
//brake2      LATC3
```

```
bytein = recvchar();
switch(bytein) {
    case '1':
        RELAY1 = OFF;
        printf("RELAY 1 IS
OFF.\r\n");

        break;
    case '2':
        RELAY2 = OFF;
        printf("RELAY 2 IS
OFF.\r\n");

        break;
    case '3':
        RELAY3 = OFF;
        printf("RELAY 3 IS
OFF.\r\n");

        break;
    case '4':
        RELAY4 = OFF;
        printf("RELAY 4 IS
```



SCREENSHOT 4. Don't let the +12 VDC motor input voltage stop you from using this circuit design with hungrier motors. The A3953 can handle a continuous load of 1.3 amperes at 50 volts.

```
//phase2      LATC4
//enable2     LATC5
#define fwd_1      0b01100000
#define rev_1      0b00100000
#define fwd_2      0b00011000
#define rev_2      0b00001000
#define mtr_1      LATD
#define mtr_2      LATC
#define cntl_mask_mtr1 0b00011111
#define cntl_mask_mtr2 0b11001111

#define brk_mtr1  mtr_1 &= cntl_mask_mtr1;

#define fwd_mtr1  mtr_1 &= cntl_mask_mtr1; \
                 mtr_1 |= fwd_1;

#define rev_mtr1  mtr_1 &= cntl_mask_mtr1; \
                 mtr_1 |= rev_1;

#define brk_mtr2  mtr_2 &= cntl_mask_mtr2;

#define fwd_mtr2  mtr_2 &= cntl_mask_mtr2; \
                 mtr_2 |= fwd_2;
```

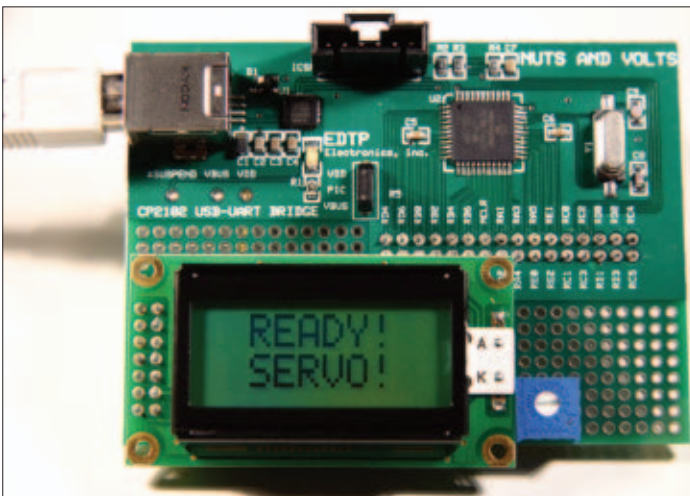
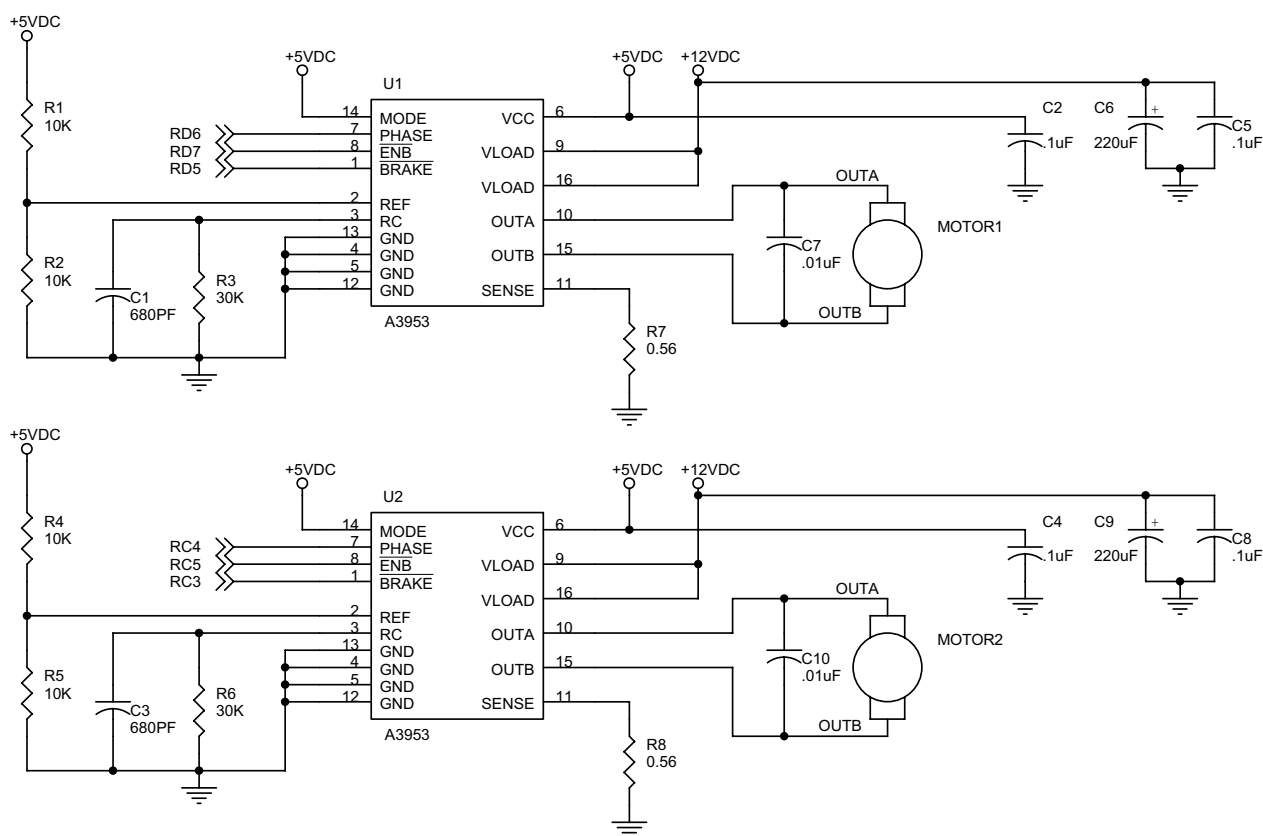


PHOTO 4. I purposely mounted the header on the solder side to facilitate assembly using wire wrap techniques. I entered "SERVO!" from the Tera Term Pro session running on the Lenovo NetBook.

```
#define rev_mtr2  mtr_2 &= cntl_mask_mtr2; \
                 mtr_2 |= rev_2;
```



NOTES:

1. C6 AND C9 NICHICON UWT1C221MCL1GS
2. ALL RESISTORS AND CAPACITORS 0805 UNLESS SPECIFIED OTHERWISE.
3. R7 AND R8 PANASONIC ERJ-1TRQFR56U
4. MOTOR CONNECTORS MOLEX 1586512-2
5. POWER CONNECTOR PHOENIX CONTACT MOUSER 651-1729131

SCHEMATIC 3. Need more than two motors? No sweat! Just expand upon the layout you see here. You only need three I/O lines per motor.

To demonstrate just how easy it is to use the A3953 hardware, here's all of the code it takes to spin MOTOR 1's shaft in a forward direction:

```
fwd_mtr1;
```

Want to quickly stop MOTOR 1? Code this:

```
brk_mtr1;
```

I can't show you spinning motor shafts in the confines of these pages. However, I can tell you that controlling small DC motors with the A3953 is a walk in the park. A breakdown of the componentry that supports the A3953s is shown in Schematic 3.

Driving a Tiny LCD

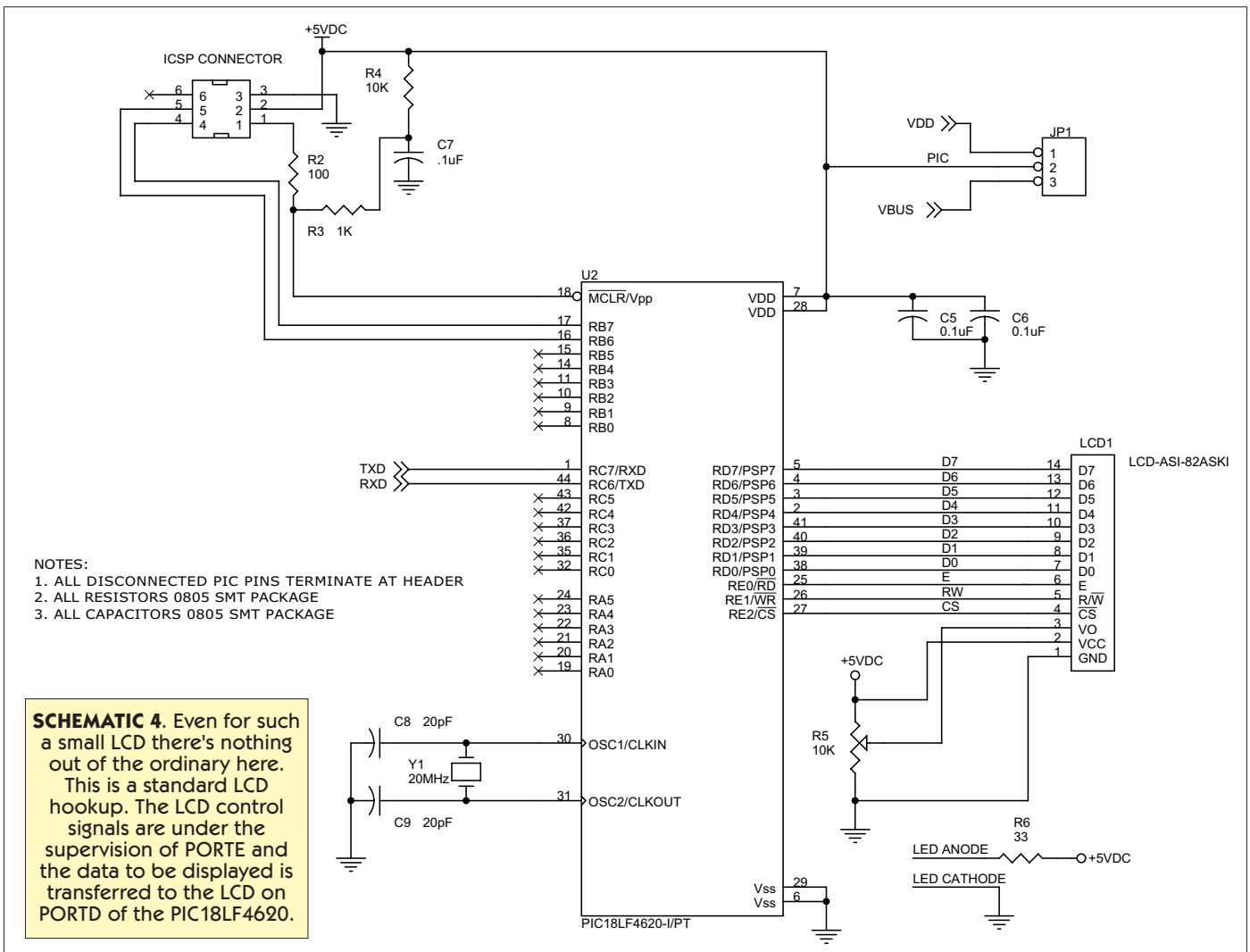
Believe it or not, you can convey a large amount of information with a 2 x 8 LCD. To prove my point, I've permanently attached such an LCD to the PIC18LF4620-based programmable module in Photo 4. I mounted the 0.1 inch pitch headers for the LCD and the PIC I/O on the

component side so that I could easily use a RadioShack wire wrap tool to wire the LCD into the PIC18LF4620's I/O subsystem. The connection details are shown in Schematic 4.

The LCD function library we're about to write can be applied to any LCD that uses the popular Hitachi HD44780 LCD controller. Our LCD driver library enables the programmer to initialize the LCD, send a byte to the LCD, and position the cursor of the LCD. There's also a function to scroll a message. Here are the prototypes for the LCD function calls:

```
//*****  
//*      FUNCTION PROTOTYPES  
//*****  
void lcd_init(void);  
void lcd_send_byte( char address, char n );  
void lcd_send_nibble( char n );  
void lcd_gotoxy( char x, char y);  
void scroll_SERVO(void);
```

To use the LCD driver library, we must configure the PIC's TIMER1 to tick at a 1 mS rate. That is easily accomplished by inserting the following code into the



interrupt handler routine:

```
if((TMR1IF && TMR1IE))
{
    TMR1IF = 0;
    TMR1H = 0xEC;
    TMR1L = 0x77;

    ++tmsecs1;
    if(++imsecs1 == 1000)
    {
        imsecs1 = 0;
    }
}
```

The value of 0xEC77 will cause TIMER1 to overflow every millisecond and trigger an interrupt. As far as the LCD driver is concerned, we won't code any delays of less than a millisecond. However, we will need delays that consist of multiple milliseconds. I've assembled a macro to accumulate milliseconds and produce the desired delay:

```
#define mdelay1(msecdelay) \
    TIMER1OFF; \
    TMR1IF = 0; \
    imsecs1 = 0; \
    tmsecs1 = 0; \
    TIMER1ON; \
    while(tmsecs1 < msecdelay);
```

The *mdelay1* macro simply turns off TIMER1 and resets the counter variables to zero before restarting TIMER1 and looping for the desired number of milliseconds. The PIC can easily outrun the LCD electronics. So, it's very important to be able to slow it down so that the LCD can keep up.

Now that we can throttle the LCD data and control signals issued from the PIC18LF4620, let's associate the LCD hardware with the LCD driver firmware:

```
/* *****
/* LCD DEFINITIONS
/* *****
#define databus LATD
#define lcdcntrl LATE
#define E 0x01
#define RW 0x02
#define RS 0x04
#define clrRW lcdcntrl &= ~RW
#define setRW lcdcntrl |= RW
#define clrRS lcdcntrl &= ~RS
#define setRS lcdcntrl |= RS
#define clrE lcdcntrl &= ~E
#define setE lcdcntrl |= E

#define lcdcls lcd_send_byte(0,0x01)
#define line1 lcd_gotoxy(1,1)
#define line2 lcd_gotoxy(2,1)
```

```
char LCD_INIT_STRING[5] =
{0x28,0x08,0x01,0x06,0x0E};
```

```
// LCD CHARACTER LOCATER "12345678";
char lcdmsg_ready[] = " READY! ";
char lcdmsg_clr[] = " ";
char lcdmsg_servo[] = " SERVO SERVO SERVO ";
```

The LCD data bus is driven by PORTD of the PIC18LF4620 while the LCD control signals are driven by PORTE. A look at Schematic 4 explains the LCD *E*, *RW*, and *RS* bit assignments you see in the LCD definitions.

An LCD operates according to the states of its three control signals. I've written a macro to set and clear each of these. For instance, to toggle the LCD *E* control signal we only need to code *clrE* to drive the *E* control signal logically low and *setE* to send the *E* signal to a logical high level.

As of yet, you have not seen the LCD functions we prototyped. So, let's work backwards to understand what's behind the *lcdcls*, *line1*, and *line2* macros. The *lcdcls* macro clears the LCD display and is built upon the *lcd_send_nibble* and *lcd_send_byte* functions:

```
void lcd_send_byte( char address, char n )
{
    clrE;

    switch (address)
    {
        case 0:
            clrRS;
            break;
        case 1:
            setRS;
            break;
        default:
            setRS;
            break;
    }

    mdelay1(1);

    lcd_send_nibble (n);
    lcd_send_nibble (n << 4);
    mdelay1(1);
}
```

The LCD *RS* line must be low with a 0x01 on the data bus to initiate the HD44780 internal instruction clear display. We're in no particular hurry here, so I have allotted an ample setup time for the data with a 1 mS delay (*mdelay1(1)*). The *lcd_send_nibble* function not only places the byte on the data bus, it toggles the *E* control line to clock the instruction into the LCD controller:

```
void lcd_send_nibble( char n )
{
    databus &= 0x0F;
```

```

        databus |= (n & 0xF0);
        mdelay1(1);
        setE;
        mdelay1(1);
        clrE;
    }

```

The *line1* and *line2* macros also depend on the *lcd_send_byte* function to issue the *Set DDRAM Address* instruction to the LCD controller. Each address variable within its respective case statement represents the beginning of an LCD row. For instance, row 2 begins at address 0x40. To position the cursor within a row, we simply add a column offset value. The *lcd_gotoxy* function can access individual character positions in LCDs that contain up to four rows with 20 characters per row:

```

void lcd_gotoxy( char x, char y)
{
    // where x = lcd row (1,2,3,4) and
    // y = column (1 thru 20)
    char address;
    switch (x)
    {
        case 1:
            address = 0;
            break;
        case 2:
            address = 0x40;
            break;
        case 3:
            address = 0x14;
            break;
        case 4:
            address = 0x54;
            break;
        default:
            address = 0;
    }
    address += (y-1);
    lcd_send_byte(0,0x80|address);
}

```

Now you understand how the *line1* and *line2* macros position the cursor at the beginning of rows 1 and 2, respectively. However, before we can do anything at all with the LCD, we must initialize it. This is the magical sequence that the HD44780 controller wants to see:

```

void lcd_init(void)
{
    char j8;
    clrRS;
    clrE;
    clrRW;
    mdelay1(16);
    lcd_send_nibble(0x30);
    mdelay1(6);

```

```

    lcd_send_nibble(0x30);
    mdelay1(1);
    lcd_send_nibble(0x30);
    mdelay1(2);
    lcd_send_nibble(0x20);
    for(j8=0;j8<5;++j8)
    {
        mdelay1(2);
        lcd_send_byte(0,LCD_INIT_STRING[j8]);
    }
}

```

Once we've issued the mumbo jumbo within the *lcd_init* function, we can use the remaining LCD driver functions to put a canned message on the LCD:

```

//*****
//*      INITIALIZE LCD
//*****
    lcd_init();
    lcdcls;
    line1;
    for(init8=0;init8<8;++init8)
    lcd_send_byte(1,lcdmsg_ready[init8]);

```

If everything goes as coded, the LCD is cleared, the cursor is positioned at row 1 column 1, and the canned message "READY!" is centered in row 1 of the 2 x 8 LCD. We have a visual indication that we can converse with the LCD. So, let's write some characters in row 2:

```

    line2;
    do{
        //wait for character from laptop
        while(!(CharInQueue()));

        //get character from receive buffer
        bytein = recvchar();

        //echo the character back to laptop
        sendchar(bytein);

        //look for carriage return (CR)
        // and print character to LCD
        // as long as character is not CR
        if(bytein != 0x0D)
            lcd_send_byte(1,bytein);
    }while(bytein != 0x0D);

```

When a carriage return character (0x0D) is received, let's scroll *SERVO* across row 1 of the LCD. Here's the scroll function:

```

//*****
//*      SCROLL SERVO MESSAGE
//*****
void scroll_SERVO(void)
{

```

```

char g8;
for(g8=window_bot;g8<window_bot + 8;++g8)
    lcd_send_byte(1,lcdmsg_servo[g8]);
mdelay1(150);

if(++window_bot == 23)
    window_bot = 0;
}

```

This is the mainline code that calls the scroll function:

```

window_bot = 0;
//start at beginning of canned message
do{
    line1;
    scroll_SERVO();
}

```

Here's what the complete *main* function looks like:

```

//*****
/** MAIN FUNCTION
//*****
void main(void)
{
    init();                //init the EUSART
    line2;
    do{
        while(!(CharInQueue()));
        bytein = recvchar();
        sendchar(bytein);
        if(bytein != 0x0D)
            lcd_send_byte(1,bytein);
    }while(bytein != 0x0D);

    window_bot = 0;
    do{
        line1;
        scroll_SERVO();
    }while(1);
}

```

Your Turn

You are now the potential proud owner of a USB-enabled microcontroller project board that is based on the Microchip PIC18LF4620 microcontroller. In addition to the PIC18LF4620 board, you have access to a trio of ExpressPCB layout files that will enable the PIC-based project board as a relay controller, an SSR controller, and/or a DC motor controller. Whether you build up the hardware or not, you are the proud owner of a set of firmware drivers for each of the three hardware modules, plus an LCD function library

SOURCES

Project Printed Circuit Boards
ExpressPCB
www.expresspcb.com

NUD3105
ON Semiconductor
www.onsemi.com

PF240D25
DM0063
Crydom
www.crydom.com

CP2102
Silicon Laboratories
www.silabs.com

A Microchip ICD3 was employed as the debugging device.

ICD3 Programmer/Debugger
PIC18LF4620
Microchip
www.microchip.com

The code for the PIC18LF4620 is compiled using HI-TECH PRO for the PIC18 Family.

H-TECH Software
www.htsoft.com

that can be adapted to any LCD that is based on the popular HD44780 chip set. The black box firmware also includes a full-blown interrupt-driven EUSART driver.

I plan on using the PIC18LF4620-based programmable module as a USB-enabled test stand for future projects. I'm sure that some of the ideas we've explored this month will end up in some of your projects, as well. **SV**

Fred Eady can be reached via email at fred@edtp.com.

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COMPUTER CONTROL and DATA ACQUISITION

by David A. Ward

Part 1: An Introduction to National Instruments LabVIEW Software

The ability to automate equipment and processes with a computer is what makes many of today's machines appear to have a degree of intelligence. One of the foremost companies in the world in computer control software and hardware is National Instruments. You can learn more about them at www.ni.com. NI has a wide range of computer interfacing software and hardware options. However, this series will only deal with the most affordable of these and is intended to get the user started in the right direction.

This month, we will get you started in computer interfacing using NI's LabVIEW software and hardware. The next article will introduce NI's USB-6008(9) DAQ (data acquisition) units, and the third will cover connecting the USB units up to real world circuits. The fourth will cover the analog-to-digital and digital-to-analog features of the USB units. The fifth and final article will tie everything together demonstrating how to build and program a thermal cycling system.

Intro to LabVIEW

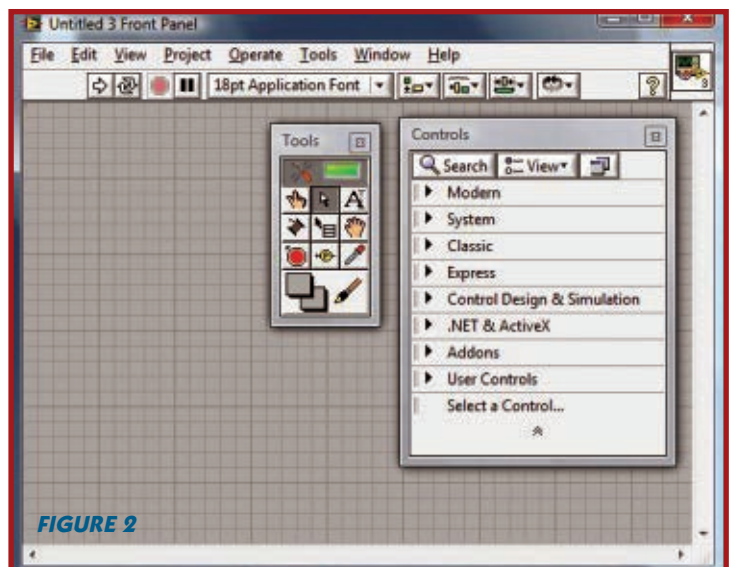
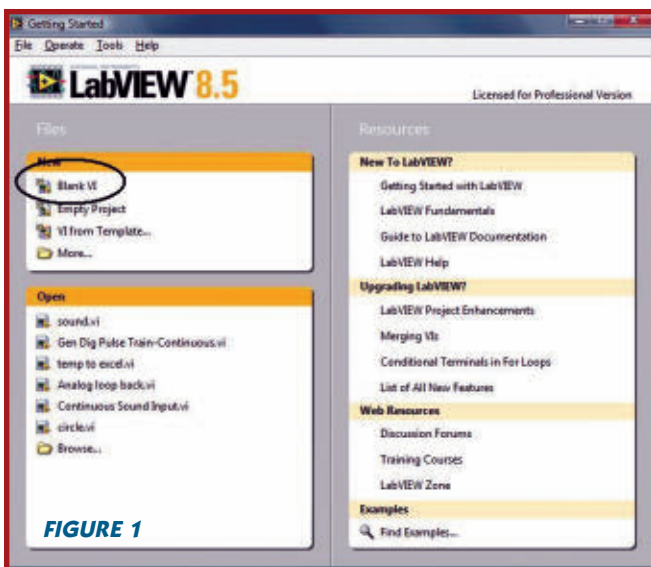
NI's LabVIEW is their main software product. The basic version retails for around \$1,250. However, they do allow a free test drive for 30 days. A less expensive way to get LabVIEW is to purchase the USB-6008(9) student kit for

around \$170 for the 6008 unit and \$280 for the 6009 unit which includes the USB hardware unit and a student copy of LabVIEW (part #779320-22 for the USB-6008 and part #779321-22 for the USB-6009). "LabVIEW 8 Student Edition" by Robert H. Bishop (which includes a student copy of LabVIEW; ISBN-10: 0131999184) can be purchased in the *Nuts & Volts* webstore. Discounts are offered for subscribers. Although you would still need to purchase the USB-6008(9) unit, you'd have an excellent text as well, to help you learn how to use LabVIEW. We'll compare the USB-6008 and USB-6009 units in the second article.

LabVIEW is a powerful graphical programming language that can be programmed by the user selecting and placing icons on the work areas and then "wiring" them together to get the results he or she desires from the software. NI refers to programs built in LabVIEW as "VIs" or virtual instruments. These VIs allow you to gather and store data and display information in graphs, charts, and through other methods. The VIs also allow the user to control external devices, turning them on and off as desired.

Hello World!

Let's go through the process of making a very simple VI at this time. You won't need any external hardware connected to your computer right now. The purpose of this



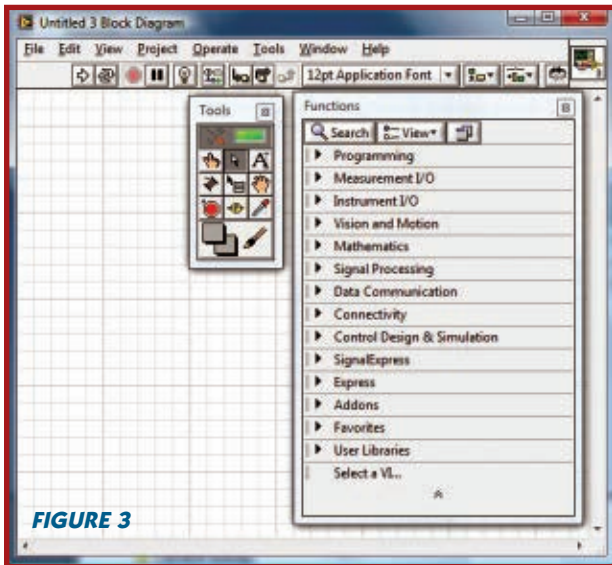


FIGURE 3

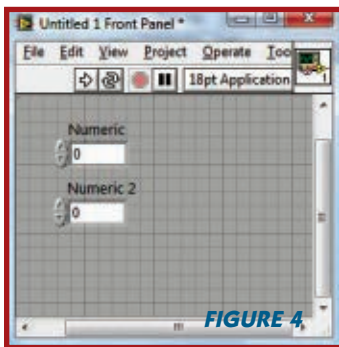


FIGURE 4

first VI is to simply add two numbers together and display their sum.

First, open LabVIEW and from the Getting Started screen, (see Figure 1) select: Blank VI. A VI consists of two screens or work areas: the front panel and the block diagram.

The front panel is what the user interacts

with when the VI is running providing inputs, such as pushing a virtual pushbutton to turn something on; it has a gray background (see Figure 2). The front panel is also where the user will see the outputs or results of the VI as graphs, virtual LEDs changing colors, and numerical data in text boxes, etc.

The block diagram screen is where the icons are wired together and most of the actual graphical programming occurs; it has a white background (see Figure 3). You can maximize either screen to fill the entire PC display or you can tile both displays. For all of the demonstrations in this article, we'll tile them horizontally by selecting from the top menu on either screen: Window -> Tile left and right.

In order to place icons on either the front panel or the block diagram, you select them from the Controls palette in the front panel window and from the Functions palette in the block diagram window. If these are not visible select View and either the controls palette or the functions palette, depending on which window is currently active. Also, if all of the menu choices do not appear on the controls or the functions palette, notice that there are arrows at the bottom of each palette to expand what's visible.

It will also be helpful for you to have the tools palette visible; select View and then the tools palette, as well as the context help window. Select Help and show context

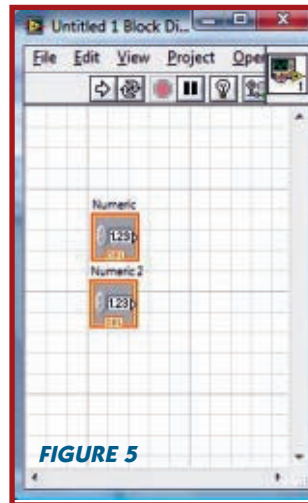


FIGURE 5

help. If all of these items are visible, we can begin building a simple VI to add two numbers together and display the results. First, place two numeric controls down on the front panel for the user to enter the two numbers to be added together. From the controls palette, select Express -> Num Ctrls -> Num Ctrl (see Figure 4). Notice as you place these two numerical controls down on the front panel that icons for them will also appear on the block diagram (see Figure 5). Now let's add an indicator that will display the results of the addition. From the controls palette on the front panel, select:

Express -> Numeric Indicator -> Num Ind. Again, notice that when placing the numeric indicator on the front panel an icon for it will show up on the block diagram.

We are done with the front panel for now, so go to the block diagram and place the addition icon. From the functions palette, select : Express -> Arith & Com... -> Numeric -> Add. After placing the addition icon on the block diagram, you are ready to "wire" them together. To wire icons



FIGURE 6

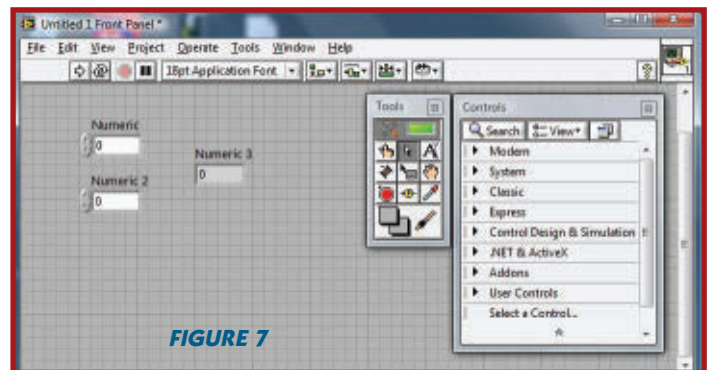


FIGURE 7

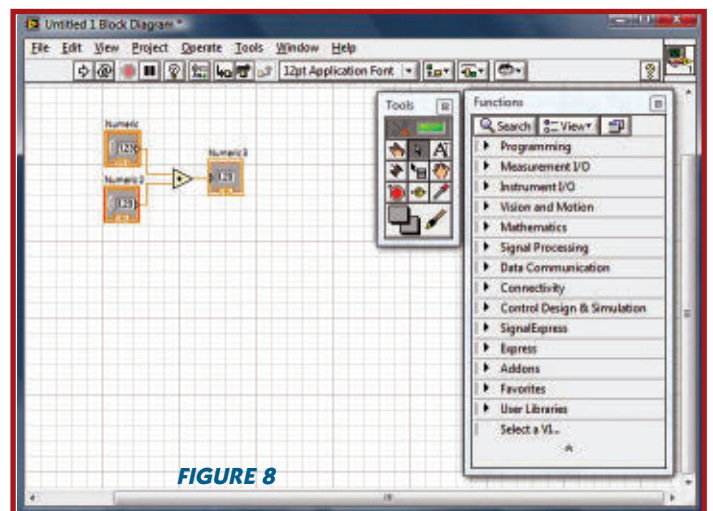


FIGURE 8

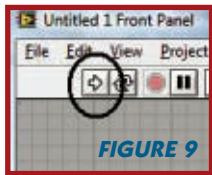


FIGURE 9

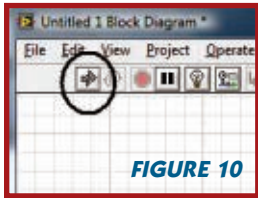


FIGURE 10

together, select the wiring tool icon on the tools palette; it looks like a tilted bobbin of wire with a piece of wire extending out from it (see Figure 6). As you bring the wiring tool next to each icon, you should see little wiring dots appear; these dots are where you connect the wires.

On some icons, there are several wiring dots. However, on the two numerical controls and the numerical indicator, there should only be one wiring dot per icon. The addition icon has three wiring dots: two for inputs on the left and one for an output on the right. Wire each numerical control to one of the inputs on the left-hand side of the addition icon and then connect the output of the addition icon to the numeric indicator as shown in Figures 7 and 8.

Time To Play

If there are no errors in your VI, the run or play button (which looks like a white arrow at the top of either screen) will be white (see Figure 9). If it is grayed and appears "broken," there is an error and pressing it will show you what the error is. Clicking on that error message will move your cursor over it (see Figure 10). If there are no errors, you can run your VI, but first go to the front panel, and with the pointer finger cursor (select this from the

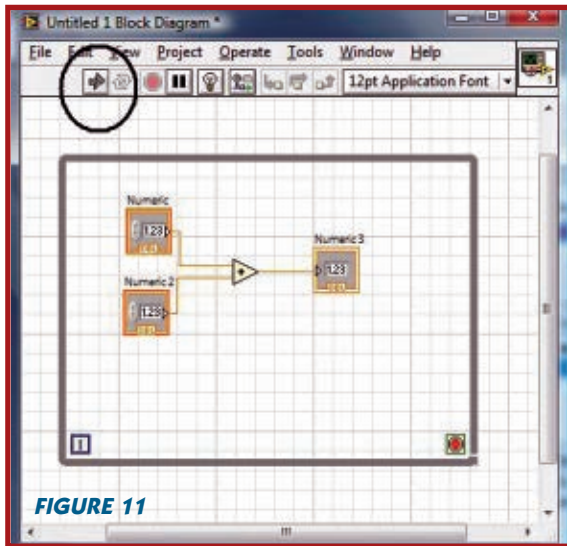


FIGURE 11

tools palette), click on both numeric control's up and down arrows to get the two numbers that you want added together. Or, you can highlight the text boxes and type in the numbers.

Now, press the run or play button and the VI should run one time and show the sum in the numeric indicator text box. It would be more convenient to see the sum of the two numbers without having to first enter the values to be added and then having to press the run button each time. To do this, we can add a looping structure of some type to the block diagram.

In the block diagram window, add a "while loop" by selecting: Programming -> Structures -> While Loop from the functions palette. Drag the while loop outline over the

three icons on the block diagram (see Figure 11). Notice that the run button is broken now. If you press it, the error that is displayed will be: *While loop: conditional terminal is not wired*. This refers to the "stop sign" icon on the lower right-hand corner of the while loop shown in Figure 12.

To correct this, add a stop button to the front panel so the user can terminate the VI when they are finished. From the front panel, place a stop button by selecting: Express -> Buttons -> Stop Button. Now, go back to the block diagram and wire the stop button icon to the stop sign icon in the while loop; this should fix the error (see Figure 13).

You'll notice a stop sign icon up to the right of the run button along the top menu of either screen (see Figure 14). It is recommended that this only be used to terminate or stop a VI that does not respond to anything else. This stop icon "slam dunks" your VI no matter what it is in the middle of. On the other hand, pressing the stop button that stops the while loop will stop the VI after the while loop has completed its operations. NI recommends that you use stop buttons wired to conditions in your block diagrams rather than using the stop sign (abort execution) on the top menu of each window.

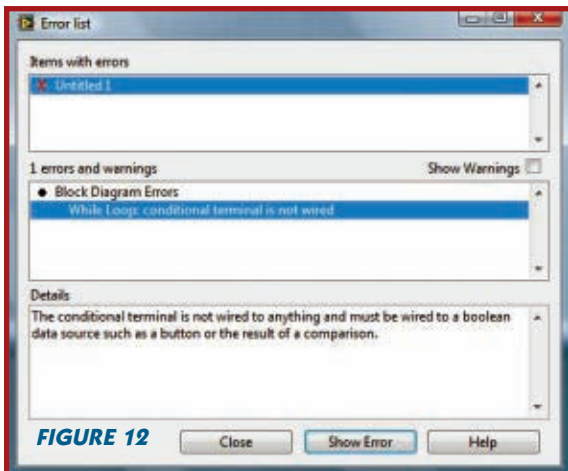


FIGURE 12

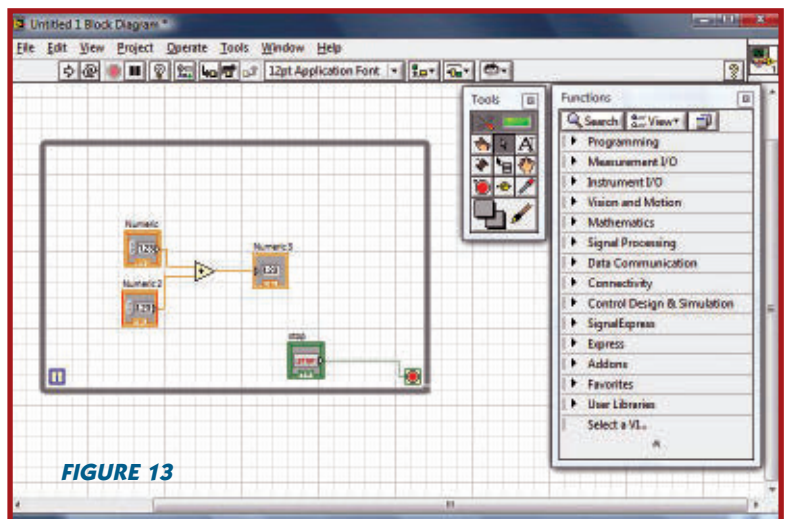


FIGURE 13

Next Steps

Now that you have a running VI you can experiment with, you might replace the addition icon with other arithmetic functions and see how they operate. You'll notice that the front panel window visibly changes from the editing mode to the run mode; grids are visible in the editing mode but not in the run mode. Also notice that some of the icons on the block diagram have different colors; these colors signify the types of data that the icons can interact with. The two numeric controls and the numeric indicators are orange which indicates they are floating point numbers. If you select one of these items, right click, and then from the pop-up menu select properties, under the data tab you will then see that it is a DBL or double number. You can also enter decimal values to be added together such as $25.33 + 44.56$.

Notice that the stop button you placed on the block diagram is green; this indicates a Boolean or true/false type of data. If you attempt to wire items together of different data types, you will probably get an error or an "X" in the middle of the wire. Placing your cursor over the X will cause a pop-up message to appear that will tell you what that error is. For example, try wiring a stop button — which is Boolean — to a numeric indicator — which is a double (see Figure 15). There are ways to convert data types from one type to another, but this is something that will be covered later on.

Wrap-Up

LabVIEW is a very powerful tool with many features. If you can't remember how to find a menu icon, you can run a search from either the controls or functions palette and

type in key words. Also, some of the examples given in this article can be accomplished through other menu choices.

All of the icons and menu choices may seem a bit overwhelming at first, but as you get more and more familiar with the program, you'll feel more comfortable and confident. You can find many examples at www.ni.com and from the Getting Started screen when LabVIEW is first opened. Robert H. Bishop's book I mentioned earlier has many excellent LabVIEW examples and tutorials that will cover many more aspects of LabVIEW than what can be covered here.

Next time, we will introduce the NI hardware that we'll be interfacing with. **SV**

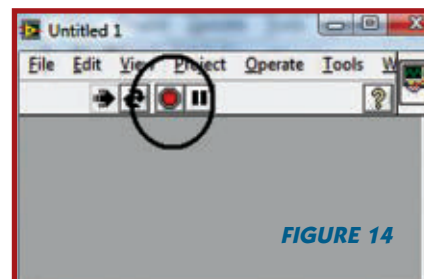


FIGURE 14

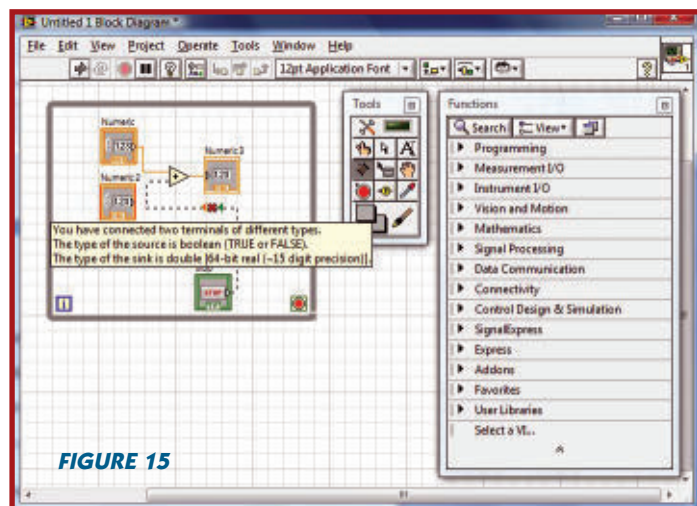
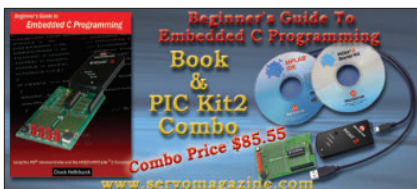


FIGURE 15

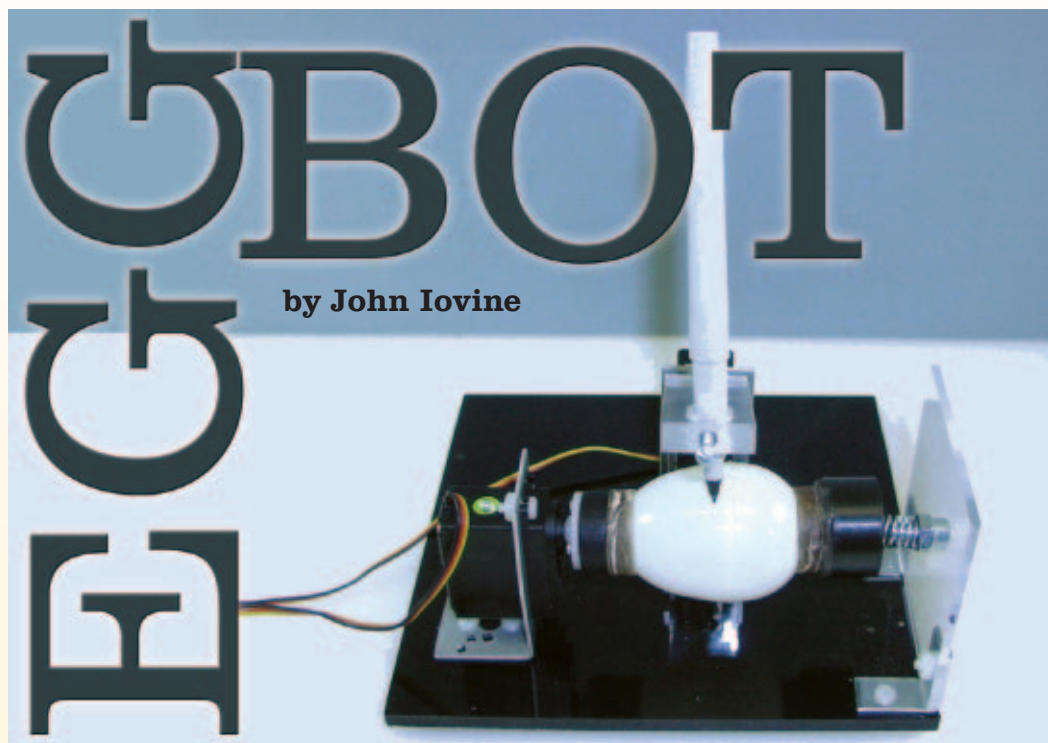


Egg-Bot is an egg marking and decorating robot, just in time for Easter.

It uses two standard Hitec HS-322 servos.

The brain for Egg-Bot is the SMC-04 USB servo controller. However, if you have a programmable servo controller

that can control two Hitec hobby servos simultaneously, you can use it instead. Graphic designs and/or text is programmed into Egg-Bot using the GUI interface of the SMC-04 USB controller.



by John Iovine

The inspiration for Egg-Bot began with a visit to Bruce Shapiro's motion control website at www.taomc.com/home.htm. If you have a chance, visit this site to see his excellent work.

Egg-Bot may be purchased as a kit or built from scratch, while the key components may be purchased separately. In this article, we will assume you are building it from scratch and the instructions are based on that premise. The kit version of Egg-Bot is shown in the Parts

List. Egg-Bot is built on a square 6.75" x 6.75" 3/4" thick wood platform. Component placement locations are shown in Figure 1. Brackets (see Figure 2) ease positioning and placing the servos on the platform. Location of the brackets are taken from one corner of the platform, so if you have a larger base you can still locate the positions accurately. Corner 0,0 is the lower right-hand corner as shown in Figure 1. Use that to transfer the measurements onto your wood platform.

The back plate is made from 3" x 3" x 1/8" thick acrylic plastic. It is held to the platform with two 1/2" brackets. Mount these brackets on the platform using two #4 x 3/4" length wood screws. Position the plastic back plate up against the brackets to mark the hole locations. Drill holes in the back plate and mount it to the brackets using two 6-32 machine screws and nuts (see Figure 3).

Next, mount a Hitec servos inside a bracket using 6-32 machine screws, nuts,

FIGURE 1. Platform layout drawing.

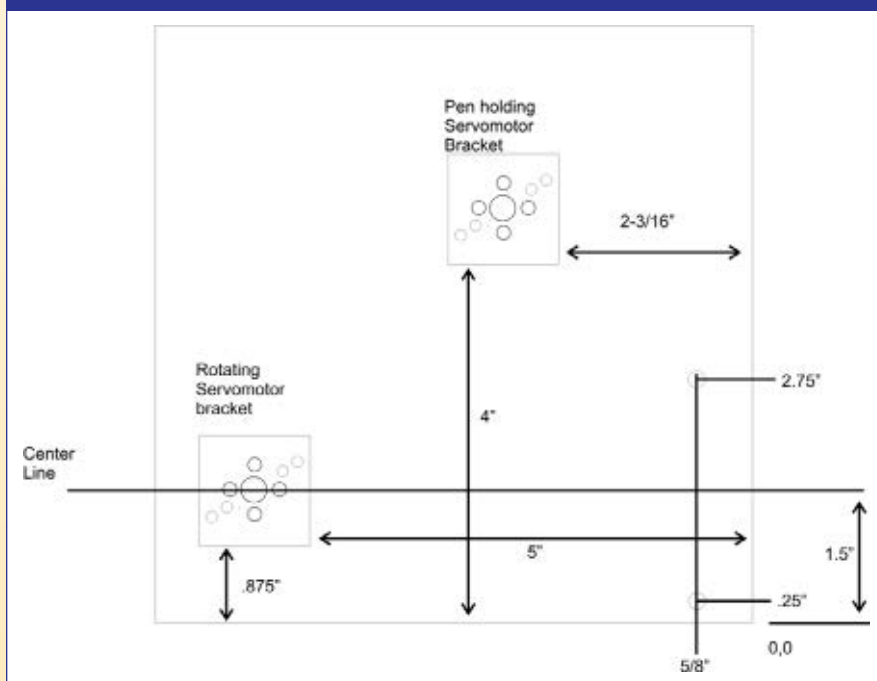




FIGURE 2. Servo bracket.

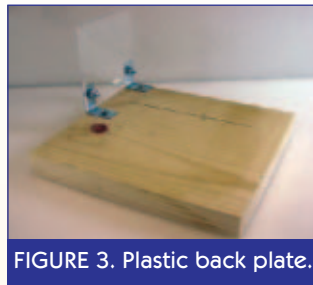


FIGURE 3. Plastic back plate.

and #6 split lock washers. Position the bracketed servo in front of the plastic back plate as shown in Figure 4. Mark the center of the servo shaft on the plastic back plate, as in Figure 5. Remove the back

plate from the brackets and drill a 1/4" hole at the servo shaft position you just marked. Remount the plastic back plate to the brackets. Mount 1/4-20 x 3/4" length machine screws and hex nuts to the 1/4" hole in the back plate and check the alignment with the servo (see Figure 6). This alignment will help keep the egg rotating parallel. Once this is checked, remove the servo from the bracket.

Position each bracket at the marked locations on your platform. Attach the servo bracket in position with two #4 x 3/4" long wood screws. Secure the two servos into their respective brackets. The pen holder servo is inverted so that the shaft is closer to the base platform as shown in Figure 7.

The pen holder is made from two pieces of plastic, measuring 2 1/8" long x 1" wide and 3/8" thick. It is identified as parts A and B. The dimensional drawings for the parts are shown in Figure 8. A standard Hitec servo horn is modified by having one extension clipped off as shown in Figure 9. This modified horn attaches to part B using two 0-80 machine screws and nuts, as shown in Figure 10. A 3/4" long 6-32 machine screw is screwed into the 6-32 insert on Part A of the pen holder. The two halves

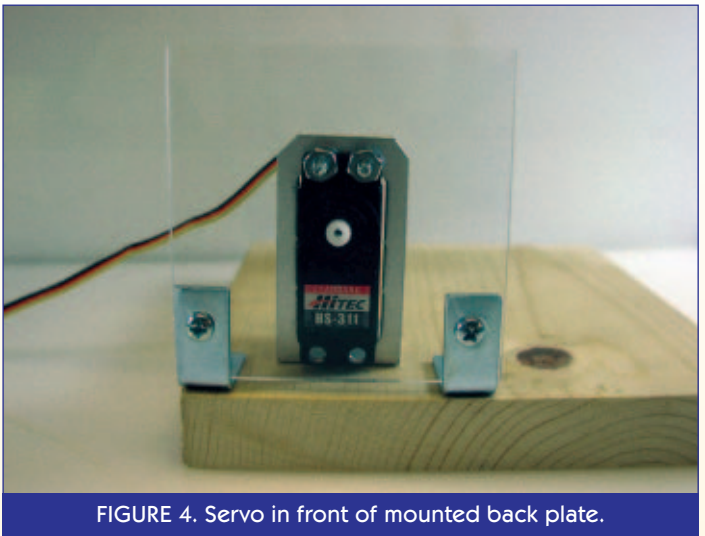


FIGURE 4. Servo in front of mounted back plate.

of the pen holder are attached to one another using a male-to-male 6-32 hinged standoff. The standoff threads are coated with a permanent thread lock compound and screwed into the appropriate ends on parts A and B (Figure 11). The parts are screwed together so they are lying in the same plane. Leave the parts in this position and allow the thread lock to dry. When it has, the pen holder can be attached to its servo as shown in Figure 12.

Egg Holder

The egg holder is constructed from a number of small components: two shaped urethane pads, one round Hitec servo horn, one wood disk, and spring (small and large) plastic black cups (see Figure 13).

The urethane pads are NOT identical. One pad is shaped for the larger end of the egg and the other for the smaller end. You can dry-fit an egg into each pad to check which end the pad is for. As you construct your Egg-Bot, you need to know which end is which.

The wood disk is attached to the round servo horn

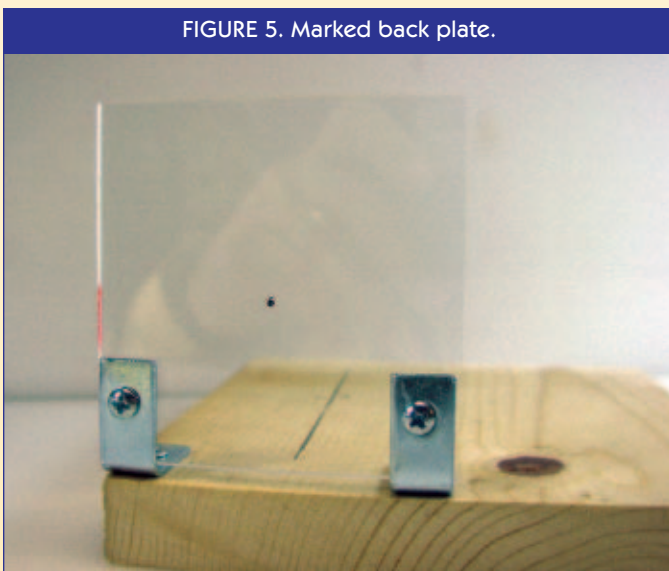


FIGURE 5. Marked back plate.

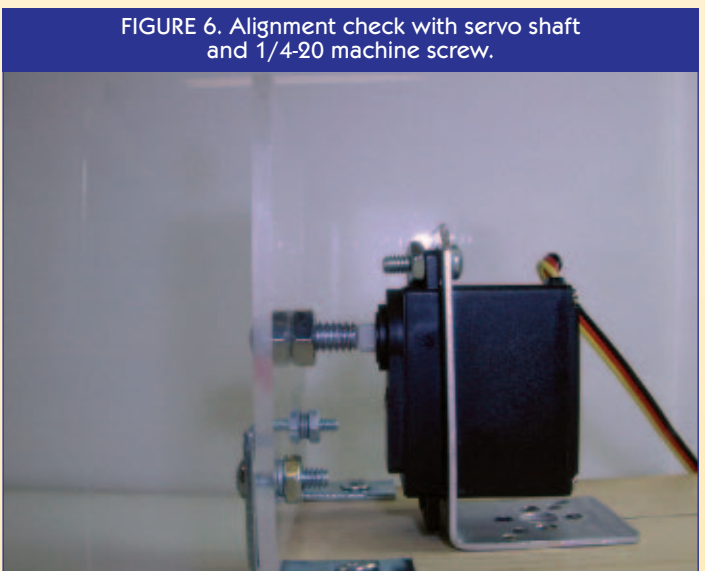


FIGURE 6. Alignment check with servo shaft and 1/4-20 machine screw.

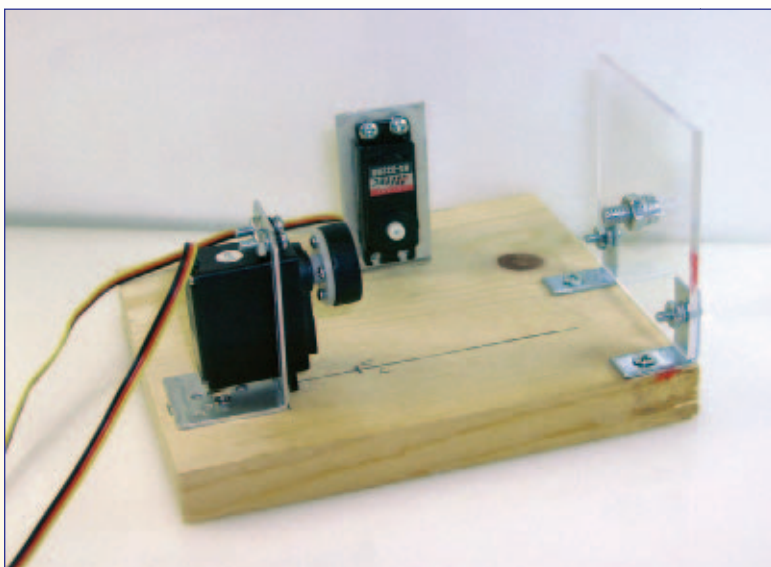


FIGURE 7. Servos in brackets on platform showing the inverted servo that holds the pen.



FIGURE 9. Modified Hitec servo horn.

using two small #2 sheet metal screws. The urethane cup holder is glued to the wooden disk using epoxy (see Figure 14). This is attached to the egg rotating servo horn.

Figure 15 shows the components for the spring backed egg holder. A small amount of epoxy is mixed and placed inside the black plastic cups. The smaller plastic cup is

embedded into the epoxy of the larger cup and the spring is embedded into the epoxy of the smaller cup. Keep everything centered until the epoxy hardens. Glue the second urethane egg pad to the base of the larger plastic cup. When loading Egg-Bot with an egg, the spring on this end is placed over the 1/4-20 bolt on the back plate. The tension from the egg will keep the holder assembly snug against the bolt (see Figure 16).

Once the egg is loaded into Egg-Bot, it should be rotated and adjusted so that it rotates parallel to the base. The egg will probably never be completely parallel (*depends on the chicken -Ed.*) and that's fine. There will always be some variance, considering that one end of the egg holder is a flexible spring that may move up or down by varying degrees. So, basically just try to get the egg to rotate mostly parallel.

Pen Holder

Before attempting to program Egg-Bot, you need to find the range of movement for your pen holder. To do so, you need to use your servo controller. For the example in this article, we will use the SMC-04 USB controller from Servotronics.

The SMC-04 controller can control four servos (see Figure 17). They are attached to three pin headers on the board labeled P1 to P4. Attach the pen holding the servo to P1 and the rotating egg servo to P2. On the SMC-04 board, set the Manual/PC switch to PC interface. Attach the USB cable to the board and start the SMC-04 program (available at www.servomagazine.com or www.imagesco.com/servo/smc04.html). The program starts by rotating the servos to their center position.

Place an egg in Egg-Bot. Attach a fine point marker inside the pen holder hole and then tighten up the 6-32 screw against the pen to lock it snugly in place. I usually set the marker height so that at the center egg position, the pen holder plastic is tilted upwards. This insures writing on the end as the pen holder is moved to the lower portions of the egg.

FIGURE 8. Dimensional drawing of the pen holder plastics.

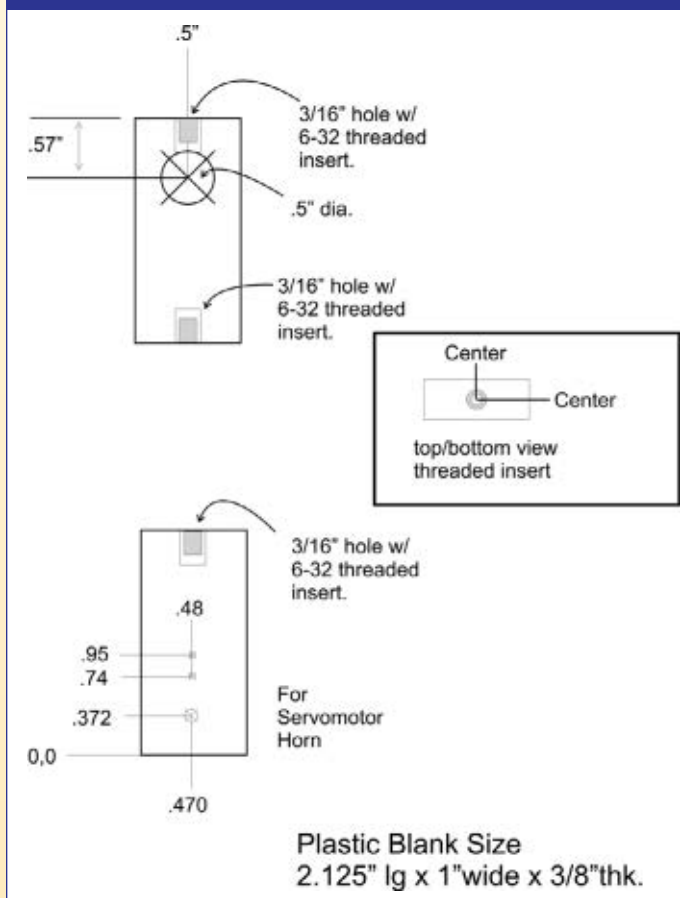


FIGURE 10. Modified horn attached to the bottom plastic.



FIGURE 11. Attaching both plastic pen holder plastics using a 6-32 hinge.

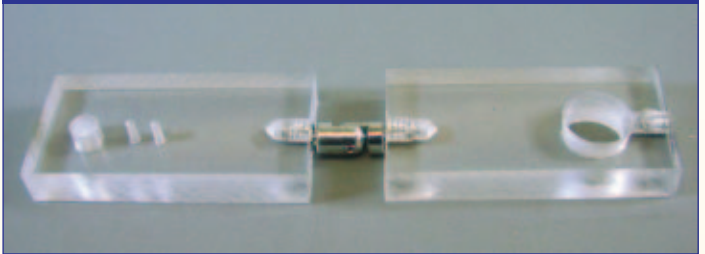
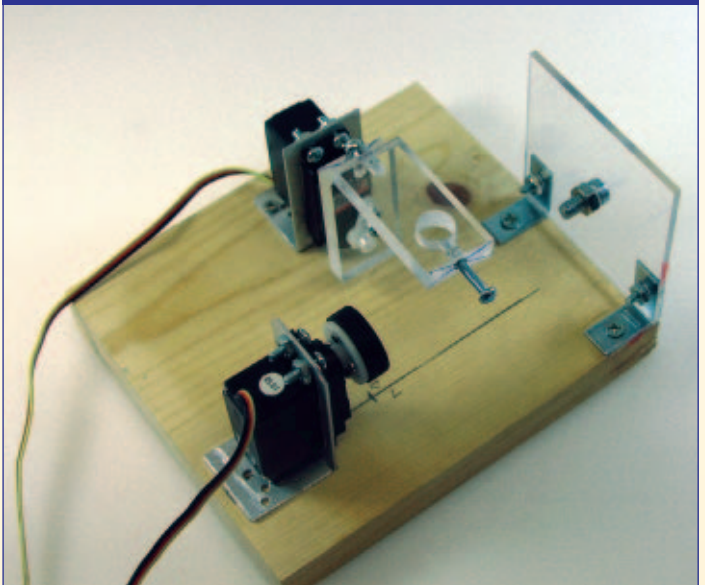


FIGURE 12. Pen holder attached to the servo.



Set the speed control on the program's interface to 10. This will slow the servo's speed forcing the pen to move slowly. Then using the slider controls or text box by the slider controls, slowly increment the numbers until you have written a line close to the end of the egg. When entering numbers in the text box, you must hit the tab key after the number is entered to move the servo to that position.

As you enter numbers or move the slider, the servo will move the pen holder writing a vertical line on the egg. Next, decrement the numbers until you reach the other end of the egg. Keep at least a 1/4" border on top and bottom of the egg. The two numbers you have for the top and bottom will be used to write programs.

In my test, the range of numbers I obtained was 175 to 115. Your mileage will vary depending upon the size and shape of your egg, diameter of pen, height of pen when placed in the pen holder, and, of course, the tolerances in the components.

Rotational Range

The numerical range for egg rotation is 75 to 255, corresponding to a change of about 145 degrees. This taken with the height of the egg defines the area or canvas upon which you can write and/or draw. After the egg is decorated, you can manually rotate the egg 145-180 degrees and write on the opposite side.

Programming Egg-Bot

The first thing to consider is that we do not have a "pen up" option to pick up the pen between letters or graphic designs to stop writing — the pen is always down, always writing.

As a test, I manually wrote incremental movements into Egg-Bot onto a plastic egg to illustrate the differences in distance in the horizontal and vertical movement. In Figure 18, each step represents a incremental movement of 10 in both the horizontal and vertical directions. We can see that the vertical movement of 10 units is approximate 2-3 times greater than the horizontal movement with the same increment.

Keeping this distortion in mind and to simplify my letter writing and graphic designs, I make up a simple template, like the one in Figure 19. The template represents the drawing area on the egg. If we want to print a message on

FIGURE 13. Egg holder components.



FIGURE 14. Finished egg holders.



FIGURE 15. Spring mounted egg holder.



FIGURE 16. Egg mounted in egg holder.

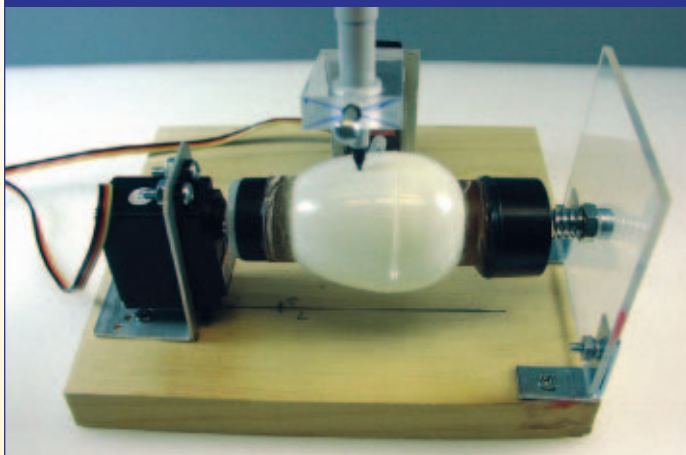


FIGURE 17. The SMC-04 board.

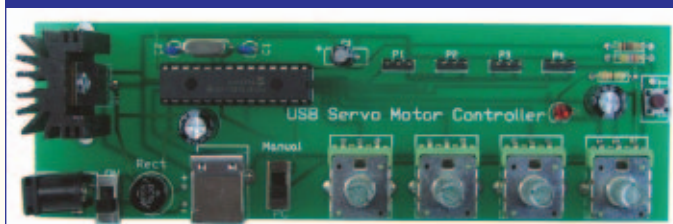


FIGURE 18. Number 10 increments in both the vertical and horizontal plane.

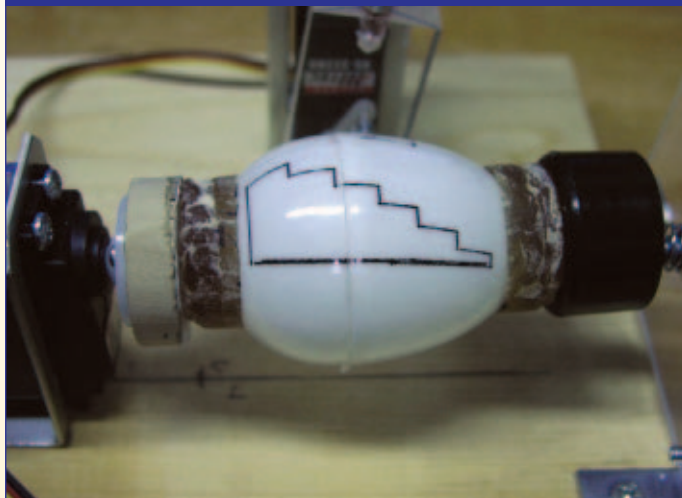


FIGURE 19. Basic graphic template.

175

115

75

255

the egg — like "HAPPY EASTER" — the lettering on the template will look like Figure 20. To see the programming required, let's look in detail at how to program the first two letters in the message. You would then use the same procedure to finish the remaining letters.

The first and second letters are shown in Figure 21. These are close-ups of the letters H and A. My Egg-Bot is setup so that the servo that holds the pen is is connected to P1 (servo 1) and the rotating servo is connected to P2 (servo 2)

We set the starting position for the two servos at 145 and 75, respectively. Set the Speed selector to 10. Set the command mode for Scripting and Step mode. This selection allows you to have two servos move simultaneously. Enter number 145 in the servo 1 text box and hit the Tab key. This will move the pen holder to position 145. Enter number 75 in the servo 2 text box and hit the Tab key. This will rotate the egg to position 75.

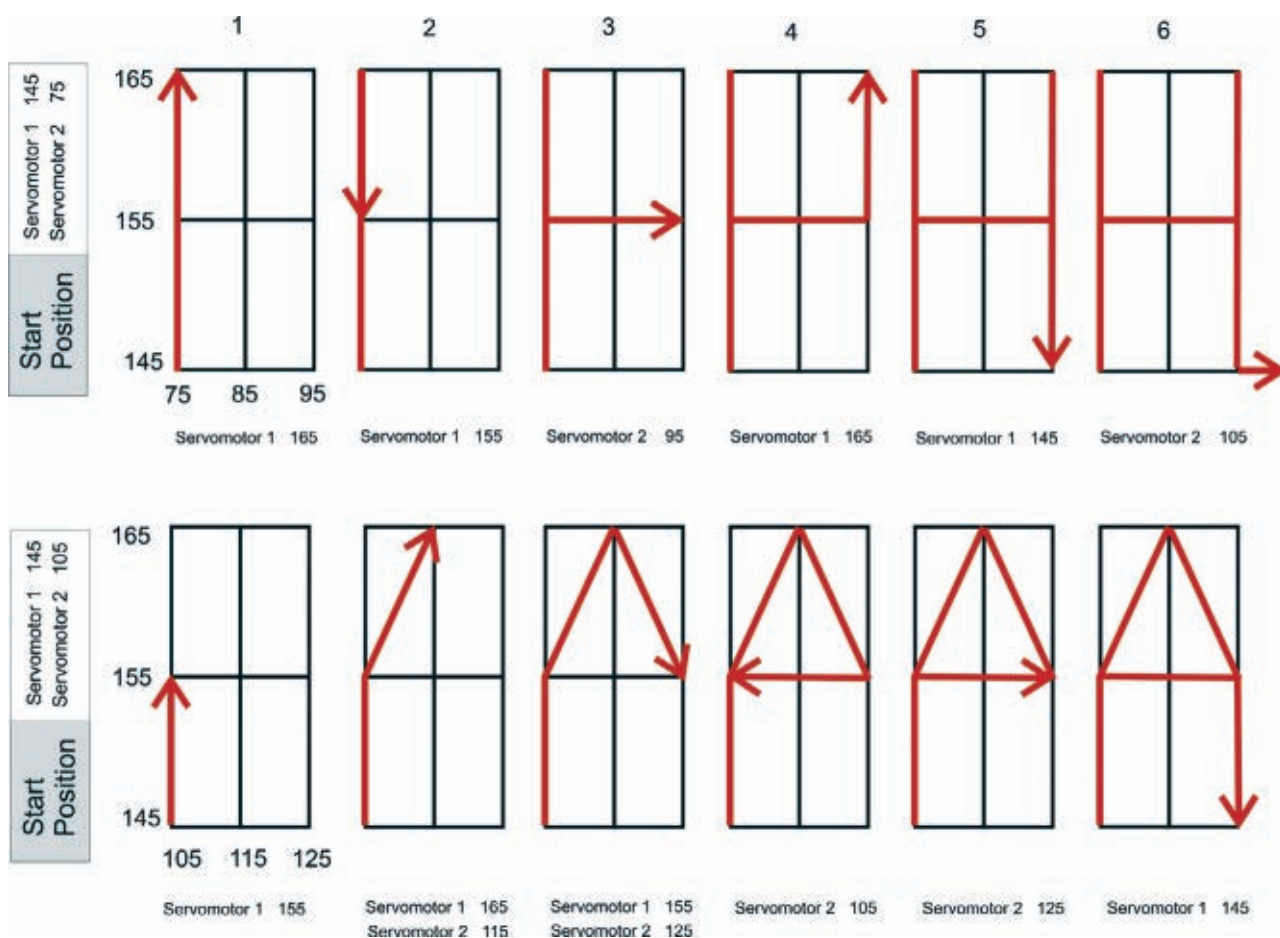
- STEP 1: Enter number 165 in servo 1's text box; hit the Tab key, then click the "add step" button.



FIGURE 20. Writing "Happy Easter" in large bold lettering on the template.

- STEP 2: Enter number 155 in servo 1's text box; hit the Tab key, then click the "add step" button.
- STEP 3: Enter number 95 in servo 2's text box; hit the Tab key, then click the "add step" button.
- STEP 4: Enter number 165 in servo 1's text box; hit the

FIGURE 21. Close-up of letters H and A on the template.



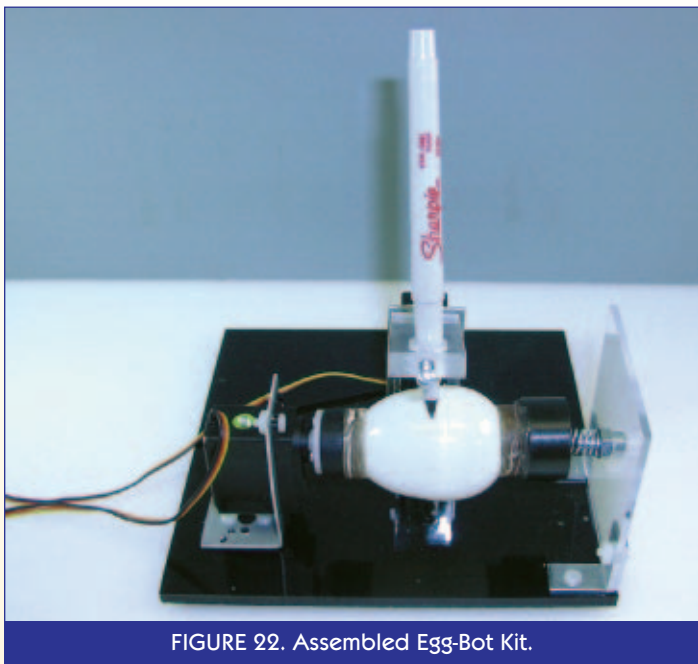


FIGURE 22. Assembled Egg-Bot Kit.

- Tab key, then click the "add step" button.
- STEP 5: Enter number 145 in servo 1's text box; hit the Tab key, then click the "add step" button.
- STEP 6: Enter number 105 in servo 2's text box; hit the Tab key, then click the "add step" button.

Each time you hit the Tab key, the servo responds to the entered command. Each time you click on the "add step" button, the command is written into the script.

We've finished writing the H character and are now in the starting position to write the A character.

- STEP 1: Enter number 155 in servo 1's text box; hit the Tab key, then click the "add step" button.
- STEP 2: Enter number 165 in servo 1's text box; hit the Tab key. Enter number 115 in servo 2's text box; hit the Tab key, then click the "add step" button.

We need a little explanation for this last step. We moved two servo positions before clicking on the "add step" button. What this did is to allow those two servos to move simultaneously during Playback, to write a diagonal line. However, when entering the step, each servo moves independently when the Tab key is entered. So, diagonal lines will only be drawn properly on playback — not during recording. Let's continue.

- STEP 3: Enter number 155 in servo 1's text box; hit the Tab key. Enter number 125 in servo 2's text box; hit the Tab key, then click the "add step" button.
- STEP 4: Enter number 105 in servo 2's text box; hit the Tab key, then click the "add step" button.
- STEP 5: Enter number 125 in servo 2's text box; hit the Tab key, then click the "add step" button.
- STEP 6: Enter number 145 in servo 1's text box; hit the Tab key, then click the "add step" button.

Once you have finished entering your text or graphic program, you can load an egg into Egg-Bot and let it run.

The lettering shown in the templates are large. You can reduce the size of the text 50% and still easily read it (see Figure 22).

Tips

Other than text, graphic designs like boxes, diamonds, lines, and rectangles are also possible. When you have an interesting design, you can have Egg-Bot draw it and then when it's finished, rotate the egg in its holder and redraw the same design again — perhaps changing marker colors between runs.

In most of my tests, I used a fine point marker. You can also vary the width of the marker. The height you set the marker in the pen holder also impacts the drawing of the design. So, it's another variable you can work with.

Going Further with Improvements

When I build something, I usually see ways to make improvements. This project is no exception. One could extend the back plate another inch or two away from the rotating servo. This will allow larger objects to be inserted. The rotating servo could be changed to a larger HS-785HB that would allow full rotations.

Script Files for Egg-Bot

I have made a number of SMC-04 USB script files that draw text and graphics onto eggs using Egg-Bot that are available at www.imagesco.com/eggbot.html. I welcome everyone to submit their own script files and designs to share with fellow Egg-Bot users. **SV**

PARTS LIST

Individual parts

	Price
• (2) Servo holder	\$9.95 each
• (1) Part A plastic pen holder	\$10.95
• (1) Part B plastic pen holder	\$10.95
• Urethane egg holders (pr)	\$8.95
• Wood disc	\$1.00
• Spring large and small plastic cup	\$4.95
• Round servo horn	\$0.50
• Wing servo horn	\$0.70

Egg-Bot Kit I

Includes all the parts above plus base, plastic back plate, servos, back plate brackets, screws, and nuts. **\$89.95**

Egg-Bot Kit II with USB SMC-04

Includes all parts as in Egg-Bot I plus the SMC-04 USB kit*. **\$149.95**

* SMC-04 USB is also a kit that requires soldering and assembly and is available from www.imagesco.com

BUILD THE ULTIMATE ROBOT

Part 6: An Arm for Megabot

by Michael Simpson

Last month, we used Wi-Fi to control Megabot. One of the problems I had was maneuvering the robot in close quarters. The laptop's built-in webcam worked but it just did not give a good view of what was going on around the robot. This month, we will add an external webcam to a fully articulated robot arm. We then will add buttons on our desktop controller program to allow us to control this arm.

STEP 1 - ADD A SECOND DECK

There is just enough room to mount an arm in front of the laptop, but it's a real tight fit. I had always planned on adding a second deck to the base, so now it's time. It will give us plenty of room to add more components and manipulators later.

I wanted to elevate the platform about 1-1/2" above the cradle. This will allow me easier access to the laptop and give me room to mount things on the bottom of the upper platform. To do this, I simply cut two more pieces of wood at 14" x 3" and attach them to the two sides of the cradle as shown in Figure 2. You only need a single wood screw on each end to hold it in place. Just make sure it sticks up 1-1/2" on both ends.

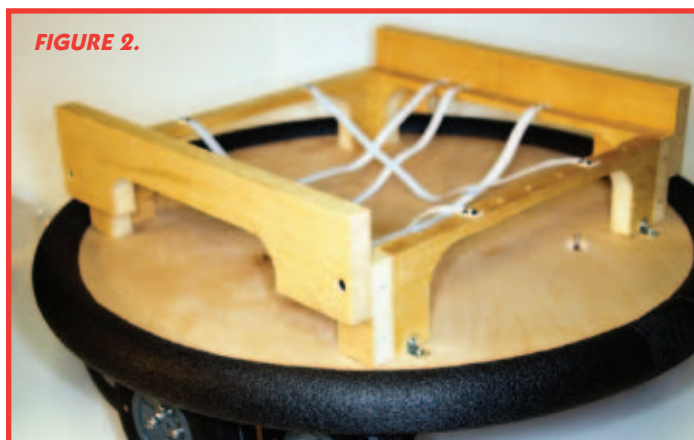


FIGURE 2.

The upper base can be made from any material. You can use 1/4" or 1/8" stock cut to the same diameter as the lower platform. In my case, I used some 1/4" plywood I picked up from my local home center. You don't have to purchase these in full sheets. The one I purchased came in a 2' x 4' section and only cost me about \$6.

To cut out the upper base, I removed the cradle and foam from the edge and just traced it. You can also use the procedure I covered in Part 3. I used a handheld jig saw to cut out the platform.

To assemble the base, I first placed the upper platform



FIGURE 1.
Megabot with
his arm.



FIGURE 4.

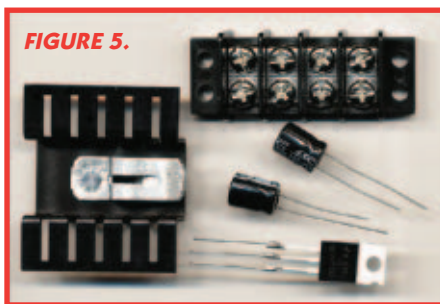


FIGURE 5.

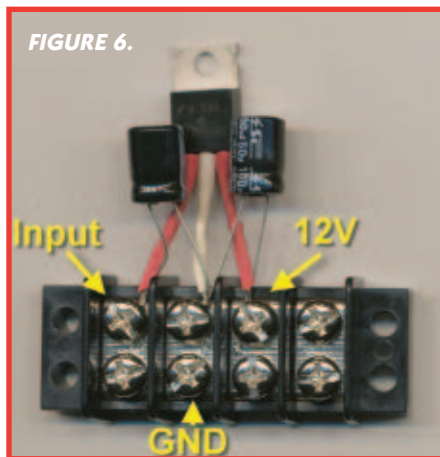


FIGURE 6.



FIGURE 7.

on a flat surface. The surface that will be the top should be face down. I then placed the lower platform — cradle and all — upside down on the upper platform. I used a wood square flat against the table surface and against both the upper and lower platform edges. Just butt the square against the two bases while it's flat on the table surface; first on one side, then the other. Once the two bases are lined up, trace the two 14" x 3" supports where they come in contact with the upper platform. You need to do this for two reasons. First, so that you can place the upper platform back on the supports once you flip the whole thing over. The second reason is so you can drill three pilot holes.

Remove the lower platform from the upper platform. You should now see two thin rectangles where the supports will be. You may not see a complete rectangle, but you should have an idea where the support will be placed. Drill three pilot holes in each rectangle: one on each end about an inch from the edge and one in the middle.

Now with Megabot upright, place the upper platform

on the wooden supports, lining them up with the tracings. You are

trying to get the upper platform lined up as it was when you did the tracing. At this point, you can carefully drill a pilot hole on one of the corners through the hole you drilled. Use a #6 washer and flathead screw, and attach the upper base to the platform as shown in Figure 3. Move to the opposite corner and make sure the markings are lined up, and repeat the process. You can now drill the pilot holes for the remaining four holes and attach with washers and screws. Add a foam bumper to the upper platform as outlined back in Part 3.

STEP 2 - ADD A ROBOT ARM

I am going to use a CrustCrawler AX-12 Smart Arm (shown in Figure 4) for our manipulator. This arm uses the AX-12 Dynamixel actuator. The AX-12 uses the same command structure as the RX-64. Unfortunately, it has a different hardware interface; the AX-12 uses an asynchronous TTL level. The USB2Dynamixel controller has a small switch that allows you to connect it to the AX-12.

For the interface, we will use the USB2Dynamixel but we still have one more obstacle, The AX-12 has a much lower operating voltage than the RX-64s we are using. To solve this, I built a 3A, 12V regulator circuit. You will need the following components:

- Two 100 μ F radial electrolytic capacitors



FIGURE 3.

- TO-220 Heatsink
- Four position barrier strip
- 78T12CT voltage regulator

The first thing you need to do is to extend some leads on the regulator chip. I did this by soldering some 18 gauge wire to each lead as shown in Figure 5.

The easiest way to assemble the regulator is to take the two capacitors and twist them on to the leads of the regulator chip. Start by twisting the positive side of each capacitor on to the outside leads of the 78T12CT chip. Then, twist the negative leads on the capacitors with the center lead on the 78T12CT chip. I also added some heat shrink to help keep the regulator chip's leads from shorting out against the leads on the capacitors.

Once the leads are all twisted together, all you need to do is loosen the first three terminals and insert the leads and tighten them as shown in Figure 6.

Now, take one of the connectors included with an AX-12 (you should have one left after assembling the Smart Arm) and trim 1/2" of insulation about 2" from one end of the connector. Notice how I have marked the rounded end. This end of the connector will be plugged into the USB2Dynamixel.

Wrap the GND lead on the connector to the terminal marked GND as shown in Figure 7. Wrap the 12V lead on the connector to the terminal marked 12V as shown in Figure 8. Tighten the terminals and insert the marked end into the USB2Dynamixel three-pin connector.

Insert the other end of the connector through one of the holes on the left side of the base on the Smart Arm. Then, attach it to the unused connector on the AX-12 inside the Smart Arm base.

Next, attach the Smart Arm to the front of the upper platform as shown in Figure 9. I used some 3/8" #4 machine screws to attach mine. Lay out the terminal strip and USB2Dynamixel as shown in Figures 8 and 9. I attached the controller with double-sided tape and #4 machine screws on the terminal strip.

STEP 3 - FINAL HOOKUP

To power the regulator, you need to run two wires from the switched side of the power source as shown in Figure 10, and connect them to the regulator Input and GND leads (shown in Figure 8).

In order to have a single USB interface to the laptop, I added the hub, as shown in Figure 11. A USB cable is also run from the RX-64 Dynamixel controller. The last attachment is a small webcam to the top of the arm as shown in Figure 12. Plug the webcam USB cable into an open port on the hub (shown in Figure 11).

PROGRAMMING THE BRAIN

Step 1

Let's start with a simple program that will let you

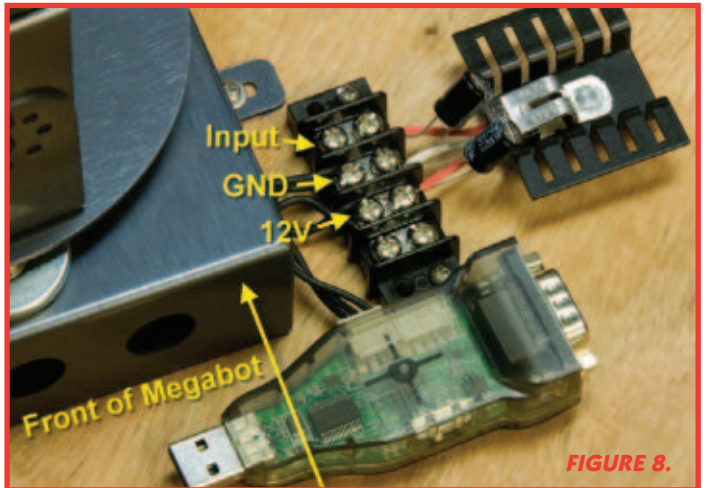


FIGURE 8.

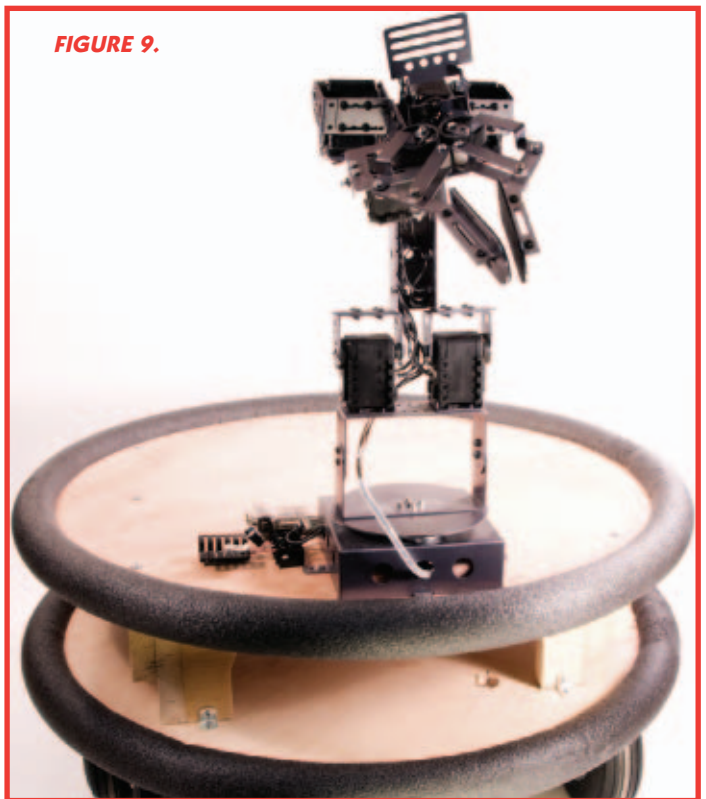


FIGURE 9.

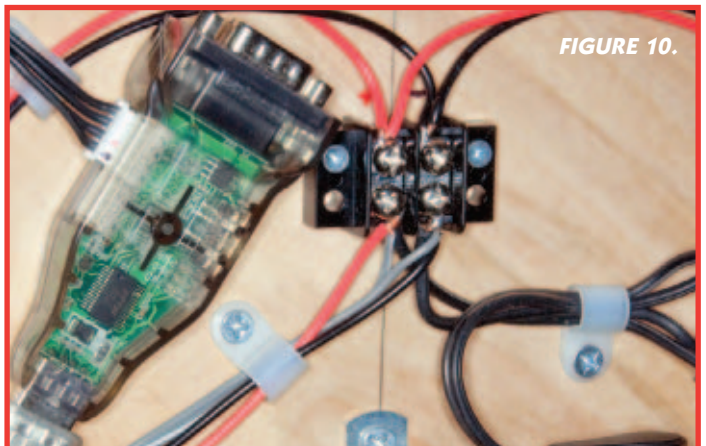


FIGURE 10.



FIGURE 11.

exercise both the RX-64 and AX-12 actuators. The program is called Megabot_Actuator1b_DT.exe (available at www.kronosrobotics.com).

When you run this program on the PC that is connected to the USB2Dynamixel, it will display the output screen shown in Figure 13.

If the program is properly connected to the USB2Dynamixel, the status should return 1 and 1. If you don't get this result, you need to create a file named Port and place the number for both com ports connected to the USB2Dyanmixel controllers in this file. The first line in the file should be the com port connected to the RX-64. The

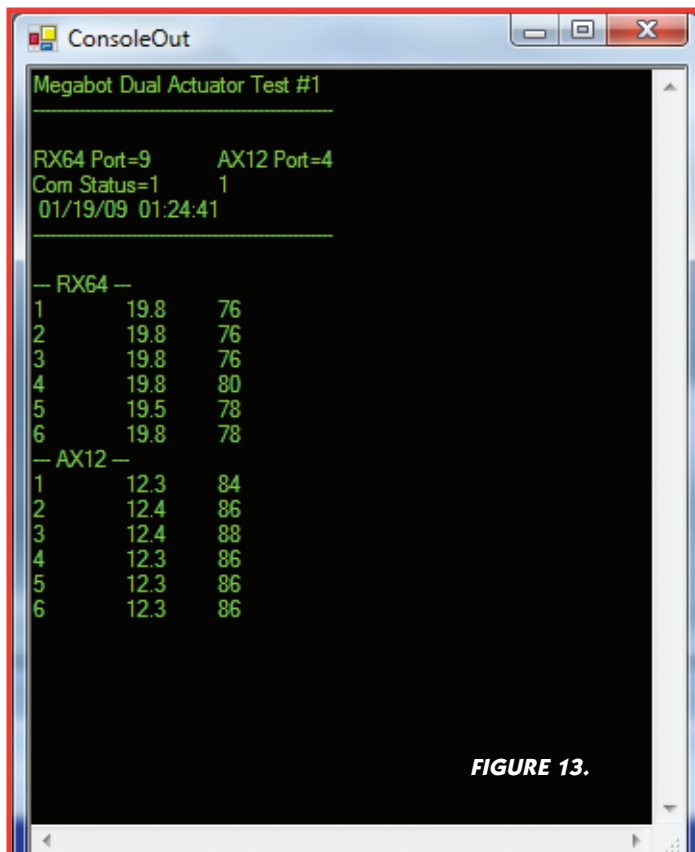


FIGURE 13.

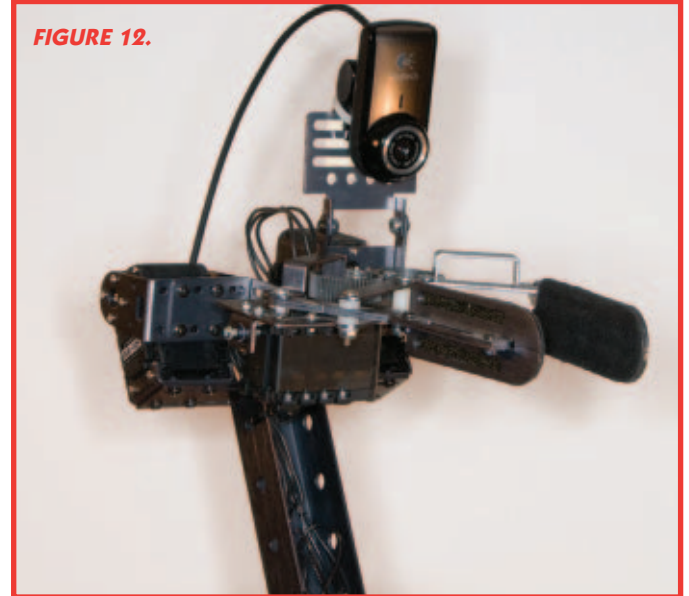


FIGURE 12.

second line in the file should be the com port connected to the AX-12. Once you make the changes, restart the program. If all the actuators are connected properly, you should get a voltage and temperature for each one. If you don't, go back and connect each actuator individually to the PC and program the IDs, as outlined in Part 5.

Before moving on, let me say a few things about the USB2Dynamixel interface using Zeus. To open a communication channel, you use the USB2AXinit command. You supply a channel number so that Zeus can keep track of the com settings you supply. The problem is the USB2Dynamixel library was designed so that only one controller is connected to the PC at a time.

You can initialize multiple USB2Dynamixel controllers by issuing separate USB2AXinit commands. Just supply them with different channel numbers (First Parameter). When you issue one or more commands — like the USB2AXwriteword — you need to set up the correct channel. This is done by setting the AXchan variable with the correct channel. To help clarify this, take a look at the source code for the various code examples.

Step 2

Now that we have verified all the actuators are connected properly, it's time to make them move. Load up the program called Megbot_Actuator2b_DT.exe. This program will cause the wheels to move in one direction, then reverse. At the same time, the arm will articulate as well. If you don't see the wheels and arm moving, go back and check your power connections.

There are a couple functions in the source code that I have provided as shortcuts in case you want to dive in and start programming the Megabot. Figure 14 shows the actual commands and what they control on the arm.

The commands are as follows:

- armgrip

- armwrist
- armarm
- armknee
- armspin

In addition, the armopen and armclose functions will open and close the gripper without having to supply the gripper position.

Step 3

Now it's time for some fun stuff. Load the program called Megabot_Wifibotb_DT.exe on the laptop used in the Megabot. This is the server program shown in Figure 15. Its job is to listen on the network and wait for commands from the client program. When a command is received, it takes an action. Notice how the server program displays the IP address at the top of the form. In this case, it's 192.168.1.201. Keep this in mind, as we will need it later.

In Part 5, we used the built-in webcam on the Aspire One laptop. In this article, we are going to use a small Logitech webcam attached to the robot arm. Load up the program called Megabot_Remote1b_DT.exe. This program is the client and it will connect to the server. It has been modified to include the ability to control the arm as shown in Figure 16.

Before doing anything fancy, I really recommend that you test everything out using a robot stand. First, take the IP address shown on the top of the server (bot) form and place it in the Remote IP: field on the client (remote1) form.

Next, click the Start button at the bottom of the form. This will connect the client to the server. Once connected, both forms will enter state 4 — Receive State Pending. In this state, you can issue commands. Finally, you can now hit the buttons to issue the various commands. Try each to make sure they are all working. Once you are satisfied, you can place the Megabot on the floor without the stand. Give yourself plenty of room and start slow.

Once you know everything is working, it's time to set Megabot free. Remove it from the stand and go for it. I have also added a button labeled "Center." This button will center the arm with the camera facing forward. I recommend you use this position when moving Megabot or you could get disoriented. Notice in Figure 16 how the grip is slightly visible. This will allow you to see items as you pick them up.

GOING FURTHER

I didn't get a chance to add any sensors to the project, but it would be a simple matter to add a microcontroller like the Dios Pro. The Dios Pro has a slave library that will allow you to attach it directly to the AX-12 bus as a device. Once done, you can add almost any kind of sensor to help you automate some of Megabot's actions. A sonar range sensor would be really cool and could transmit telemetry back to your Wi-Fi remote program. If you added a sonar sensor to the rear of the robot, you could tell when an animal or person is sneaking up on your bot.

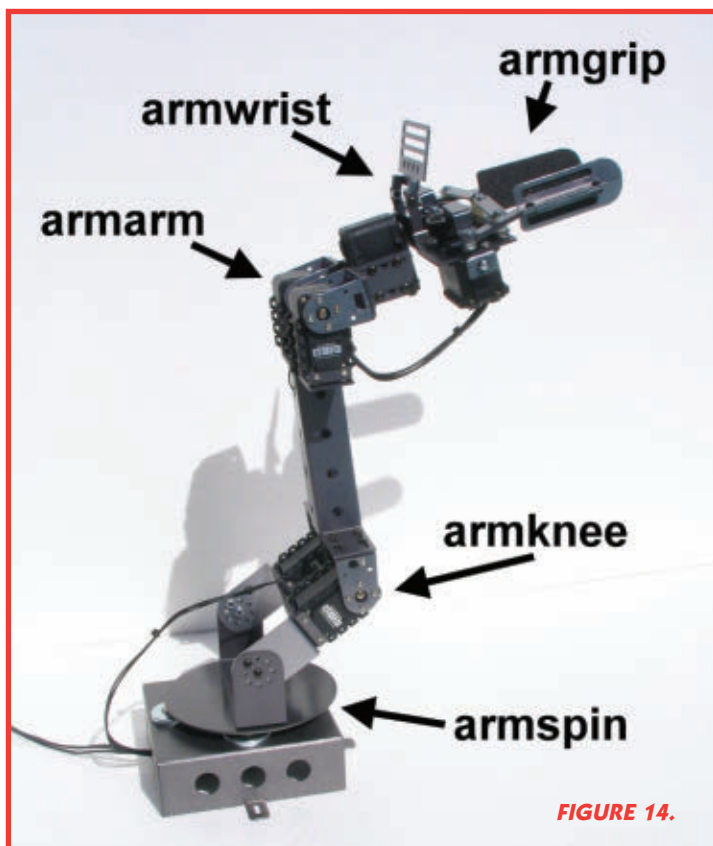


FIGURE 14.

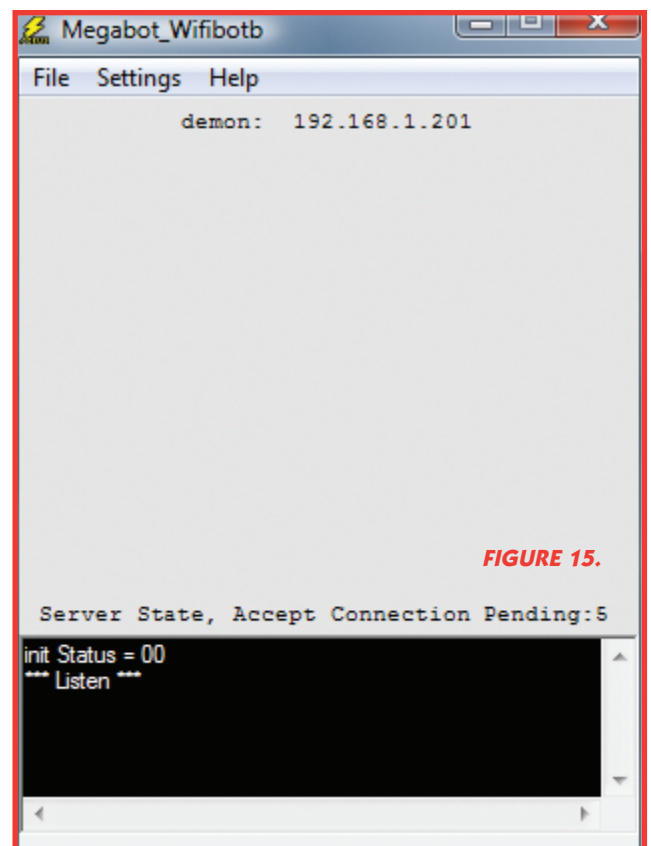


FIGURE 15.

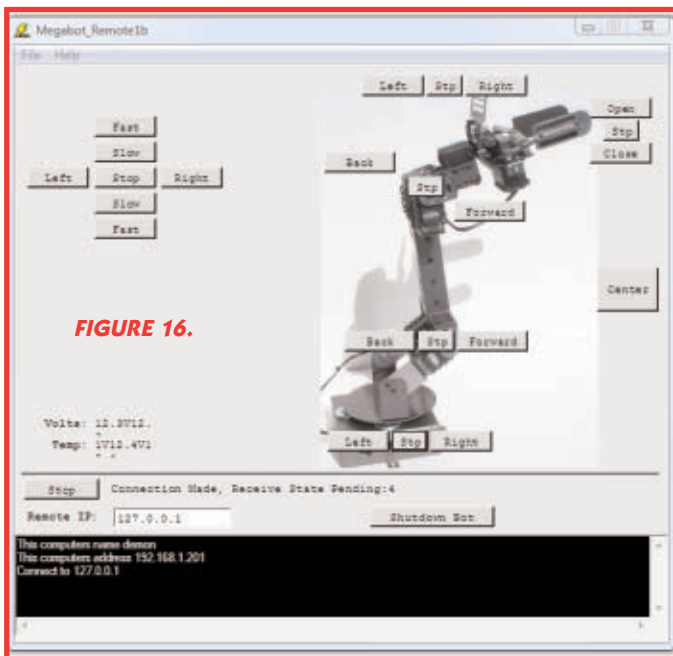
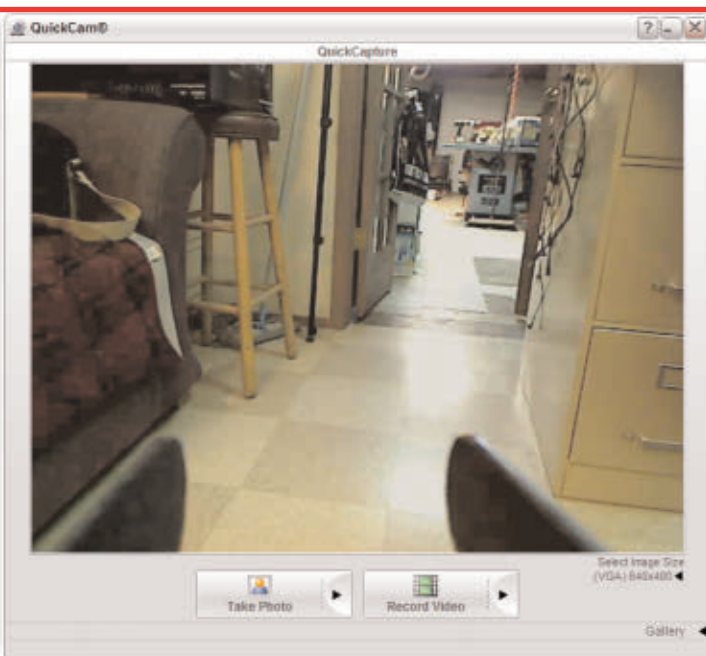


FIGURE 16.

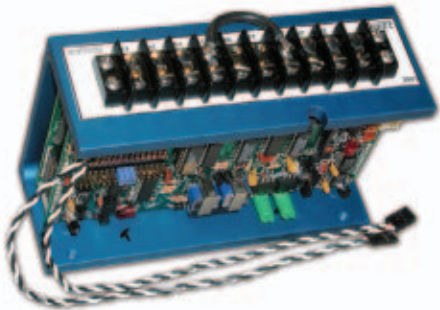


If you added a GPS logger to the robot, you could use the recorded information to help automate navigation. You could place this navigational aid on the laptop, remote, or even both.

I know I had great aspirations when I first started this project. The problem is I just ran out of space and time. That said, enough information has been provided in this series to help you with your own robot projects.

Be sure to check out the Kronos Robotics website for updates, source code, and executables at: www.kronosrobotics.com/Projects/megabot.shtml **SV**

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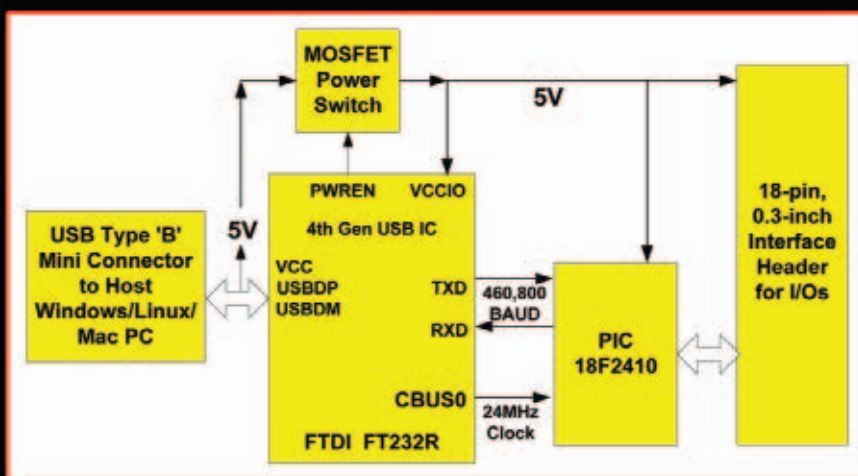
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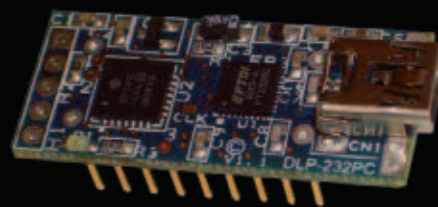
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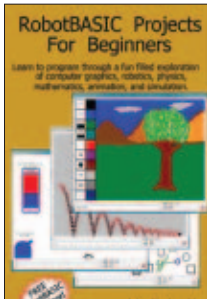
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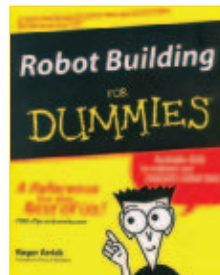


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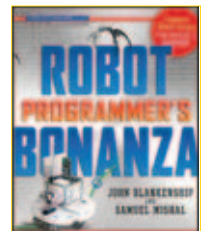
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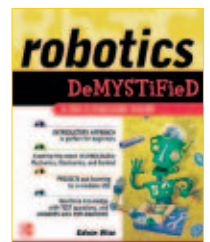
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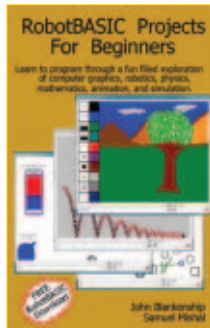
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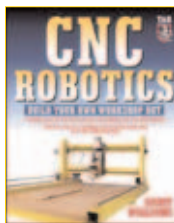
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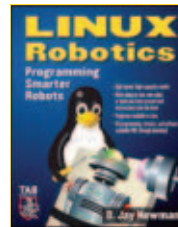
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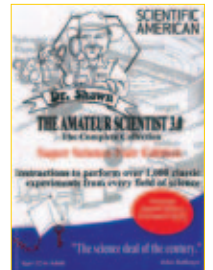
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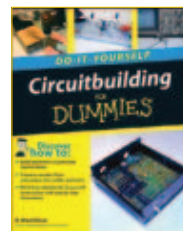


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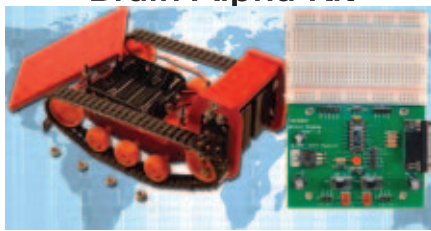
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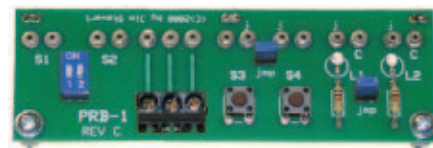
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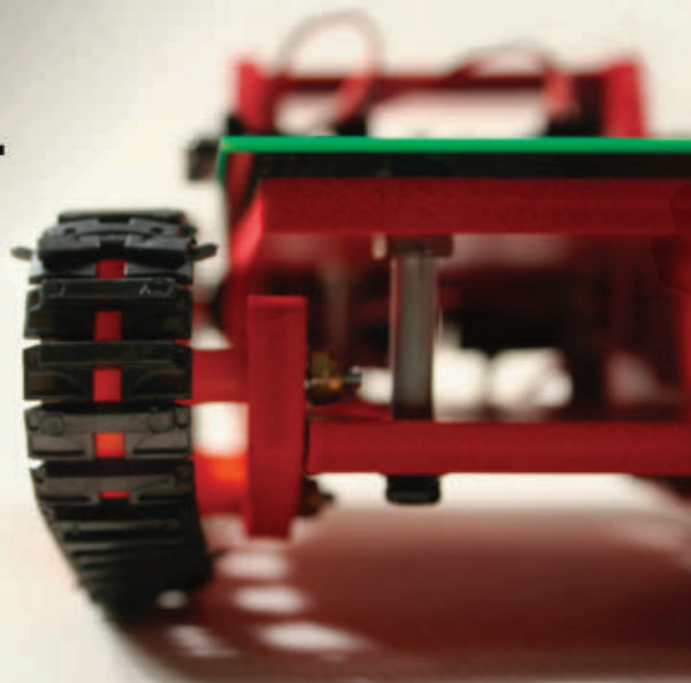
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Emancipating Your SERVO TankBot

by Ron Hackett



If you read the previous TankBot article, you know that the Panasonic PNA4602M IR receiver that we used operates at a frequency of 38 kHz. The SIRC protocol is modulated by the 38 kHz carrier wave, the 4602 demodulates the data, and the PICAXE "infrain2" command returns the appropriate value for any given key-press. As we saw last time, this can be a very simple and powerful combination. This month, we are again going to use the PNA4602M IR receiver, but in a very different manner. Essentially, we're going to generate a 38 kHz IR waveform (with no modulated data) for a brief period of time, turn it off, and immediately "listen" for any echo that might be reflected by a near-by obstacle.

Except for one problem that we'll get to momentarily, the generation of a 38 kHz waveform is a simple task for a PICAXE processor. The "pwmout" command is capable of generating a rectangular waveform over a range of frequencies and duty-cycles. (As usual, refer to the documentation in Part II of the PICAXE manual for the details; we're only going to cover what we need for our purpose here).

In last December's issue of *SERVO Magazine*, we implemented a simple project that enabled us to control the TankBot's movements with any universal TV remote control capable of transmitting the Sony Infrared Control (SIRC) protocol. This month, we're going to take the opposite tack and enable the TankBot to explore the environment on his own, without any intervention on our part. In order for the TankBot to avoid ramming into any obstacles, we're going to implement a simple object-detection system which looks for the echoes of IR waves much the same way that a sonar system detects the sounds that are echoed back to it from nearby objects.

The real power of the pwmout command is that once it is executed, the resulting waveform is continuously generated in the background, freeing up your program to attend to other tasks without the hassle of having to repetitively produce each individual pulse. This is a tremendous code simplification to say the least, and it's the primary reason that the pwmout command is ideally suited for DC motor control.

A PICAXE processor and an H-bridge chip (e.g., the L293D) are

essentially all you need for full PWM control of one DC motor. If you wanted to control two DC motors (and what robot builder wouldn't?), you would need at least a 28X1 processor which has two independent PWM outputs and more than enough computing power to also be your bot's CPU. However, we don't have to worry about that because the TankBot is driven by servomotors rather than DC motors, and the "servo" command is capable of controlling multiple servomotors from one processor.

The problem is that the servo command and the pwmout command depend on the same internal timer in the PICAXE; therefore, these

two commands cannot be used at the same time. The TankBot would have to stop every time it wanted to look for IR echoes, which is an obviously unacceptable situation!

Another possibility would be to use an external circuit to generate the 38 kHz waveform, possibly one based on a "555" timer chip. However, I opted for a third approach that uses a PICAXE-08M as a slave processor for several reasons: The 08M is the same size as the 555 timer and the actual

circuit involved is simpler; the 08M's timing is much more stable than that of the 555 (i.e., it won't drift off frequency over time); and you will find many other uses for a slave processor as you continue to experiment with the TankBot and/or develop your own robot designs in the future.

Programming the PICAXE-08M

Before we get into the details of our IR obstacle-detection circuit, we need to address the mechanics of

programming an 08M on the TankBot's breadboard. Figure 1 shows the schematic of a standard PICAXE programming circuit. The BrainAlpha PC board includes essentially the same circuit with one extra part (a BAT85 diode) that isn't necessary for our current project. Of course, the Brain Alpha's programming circuit is directly connected to the 14M, so we will need another circuit on the breadboard in order to program the 08M. Figure 1 also includes an LED and current-limiting resistor connected to output 1 (pin 6) that we will use shortly to make sure

our circuit is set up correctly.

You will also need a way of connecting the circuit to the PC for programming — see the detailed presentation of the options on my website (www.jrhackett.net/prog cable.htm). Figure 2 is a photo of the 08M and its programming circuit installed on the TankBot's breadboard. (The placement of the parts in the photo may seem a little odd, but if you follow the same layout you won't have to move anything when we construct the IR object-detection circuit a little later.) In the lower right-hand corner, you can see the five-pin connector that I use for all my programming adapter connections. If you prefer, you can simply substitute a female DB-9 connector with three wires soldered to it as described on my site and connected as shown in Figure 1.

If you have a second serial port available on your PC, it's a real convenience to be able to connect two cables at once (one for each processor) and use the "#com1" and "#com2" directives (one for the 14M setup and the other for the 08M setup). With this arrangement, the Programming Editor software automatically downloads the correct program to each processor without the need for switching the cable each time.

Figure 3 presents the "Hello08M" program we will use to test our circuit. As you can see, it's very similar to the one we originally used to test the BrainAlpha board. All it does is blink the LED on output 1, which tells us that the circuit is functioning properly. When you have assembled the circuit, simply type the program into the Programming Editor and download it to the 08M; the blinking LED will tell you when you're ready to move on to the actual project.

An IR Obstacle-Detection Circuit

Figure 4 shows the schematic of our IR obstacle-detection circuit and Figure 5 is the Parts List. All the parts

Figure 1. PICAXE-08M Programming Circuit.

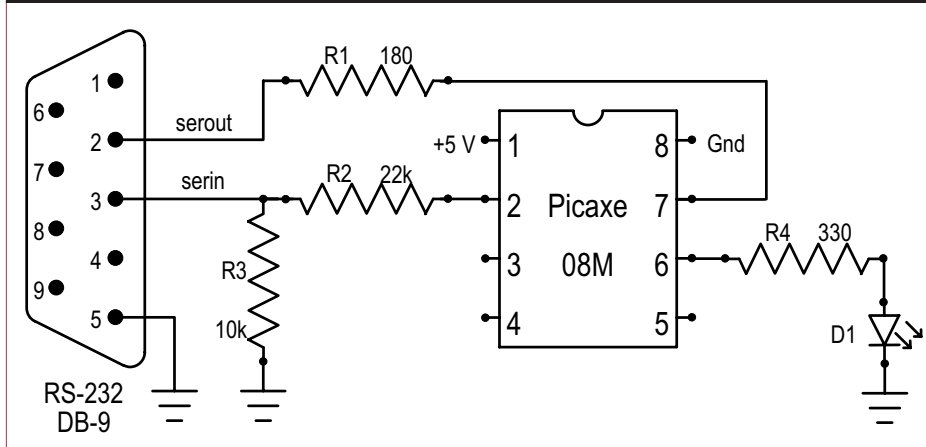
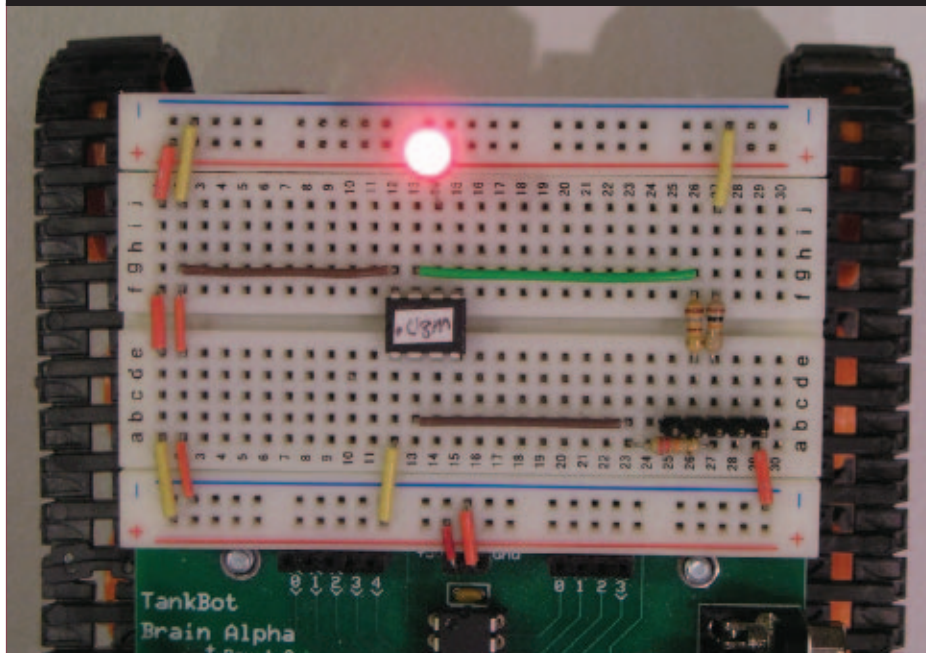


Figure 2. Programming Circuit on Breadboard.



are available on my website. The first thing you may notice in Figure 4 is that there are two IR LEDs connected in series on output 2 (pin 5); while this may seem a little unusual, it's not a problem as long as the current-limiting resistor is sized appropriately. The IR LEDs I use each drop about 1.2 volts, which leaves 2.6 volts across the 330 Ω resistor. Therefore, the current draw is approximately 8 mA. Since an 08M I/O pin is capable of sourcing 25 mA, this may seem a little low, but as we will soon see, the major problem I had as I was developing this circuit was that the PNA4602M IR receivers were too sensitive. I was initially plagued by "false positives" (the indication of an obstacle when nothing was in the TankBot's path), so the reduced power output to the IR LEDs is actually a plus in this case.

The IR LEDs are on output 2 because this is the only 08M I/O pin capable of generating a PWM waveform. The visible-light LEDs on outputs 1 and 0 (pins 6 and 7) are included primarily for debugging purposes — they each light when an obstacle is detected on the respective side. (Whenever you download a program to the 08M, the LED on output 0 will

flicker because output 0 is also the serout line, which the 08M uses to communicate with the PC during a download.) When your system is functioning properly, the LEDs can be removed to conserve power but it's reassuring to have some visual feedback about what the TankBot is "seeing."

Figure 6 is a photo of the complete 08M circuit installed on the TankBot's breadboard. As you can see in the photograph, I have placed heat-shrink tubing around each of the IR LEDs in order to shield the IR receivers from direct IR light. That way, the only IR signals they can receive will come from the echoes that bounce off near-by obstacles.

The software for the 08m (IRecho5_08M.bas) is too long to include here. It's in a zip file (Tank200903.zip) available at the *SERVO Magazine* website (www.servomagazine.com). The same zip file also contains the TankBot's PICAXE14M program that we will discuss shortly. Download the file, unzip it, and print out a copy of each program for reference throughout the following discussion.

Essentially, the IRecho5_08M.bas program consists of an infinite "do...loop" that repetitively outputs a burst of PWM pulses, turns off the pulses, checks for left and right echoes, communicates the results to input4 and input3 on the BrainAlpha PC board, and also displays the results on two LEDs for debugging purposes.

Of course, IR radiation travels at the same speed as visible light and, as

Figure 3. "Hello 08M" Program Listing.

```
' === Hello08M.bas ===

' This program runs on a PICAXE-08M.
' It blinks an LED on output1 (pin 6)
' to show circuit is alive and well.

' === Constants ===

symbol LED = 1 ' LED on pin 6

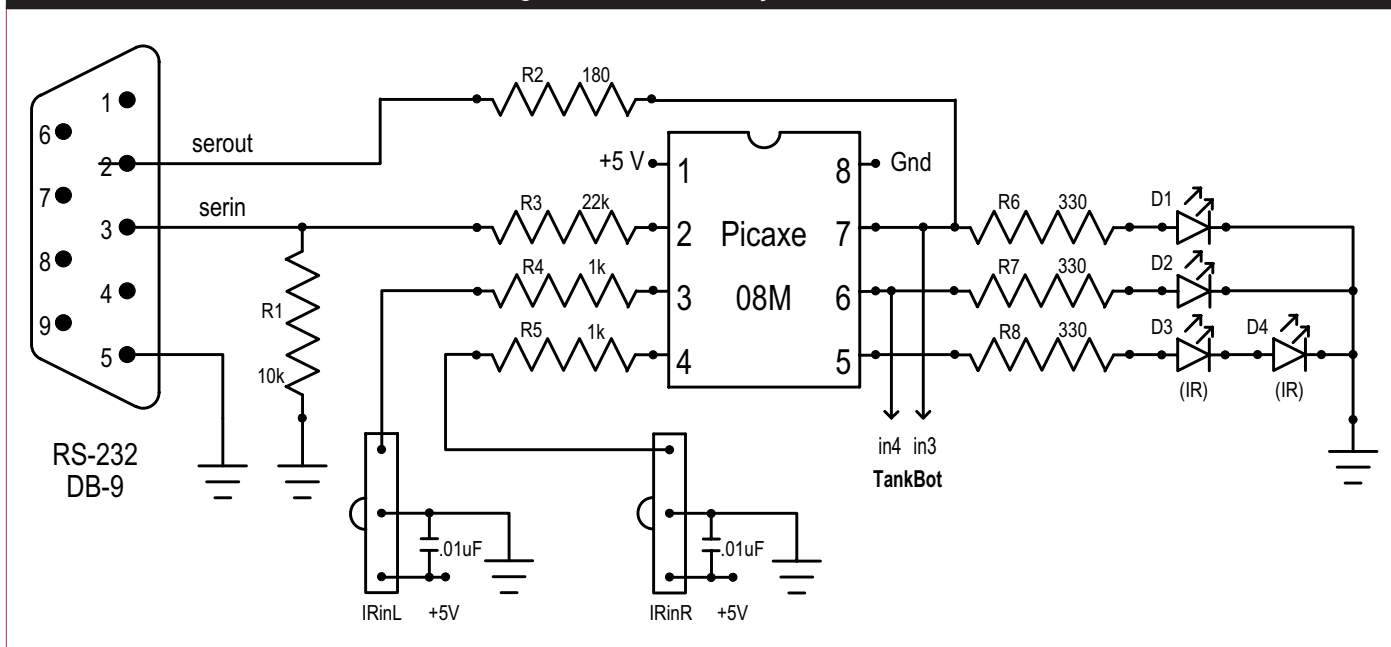
' === Directives ===

#com2          ' specify serial port
#picaxe 08M    ' specify processor

' === Begin Main Program ===

do
  toggle LED    ' change state of LED
  pause 500     ' for 500ms
loop            ' loop forever
```

Figure 4. IR Echo Project Circuit.



you may know, interpreted Basic is a relatively slow language. Consequently, the program includes four different techniques in order to make it fast enough to detect the IR echoes. First, the "setfreq m8" command instructs the 08M to run at 8 MHz rather than its default speed of 4 MHz. This is necessary because 4 MHz is too slow for the program to "see" the IR echoes.

Second, the use of the built-in special function variables "dirs" and "pins" requires some explanation. "Pins" refers to the 08M's inputs, which are automatically separated into

individual bit variables. Only valid input pins are implemented; on the 08M I/O pin 0 is "output" only, so just pins 1 through 4 are implemented (i.e., pins = -, -, -, pin4, pin3, pin2, pin1, -). Similarly, "dirs" refers to the "data direction register," which specifies whether an I/O pin is an input or output. Again, only valid bi-directional pins are implemented. On the 08M I/O pin 3 is "input" only and pin 0 is "output" only, so just pins 4, 2, and 1 are implemented (i.e., dirs = -, -, -, dir4, -, dir2, dir1, -). In a "dirs" command, a 1 indicates that the corresponding pin is an output and a 0 indicates that it is an input.

So, the "dirs %00000111" command specifies that pin 0, pin 1 and pin 2 are outputs and pin 3 and pin 4 are inputs (remember "%" indicates a binary number). Actually, the digit in the pin 0 and the pin 3 position are ignored (because their directions can't be changed) and pins 5-7 are also ignored (because they don't even exist on the 08M!), but you need a digit in each of the eight positions so the simplest approach is to use the one that tells us humans

which direction is specified for the pin.

Further down in the program, when the "echoes = pins" command is executed the current value of the two inputs (pin 4 and pin 3) from the IR receivers is copied into the corresponding bit positions of the "echoes" variables. Since echoes was assigned to variable b0 at the beginning of the program and b0 is also automatically segmented into bit variables "bit7" through "bit0," the bit variables in which we are interested (bit 4 and bit 3) are also automatically available to us. That's why (in the program's variable declarations) we assigned echo_L to bit 4 and echo_R to bit 3.

You may be wondering why all this is necessary, since there are certainly other ways of defining and accessing the variables we need. The reason is that the "echoes = pins" statement executes faster than any other way of doing it and speed is of the essence because we are dealing with IR radiation. Every other approach I tried was not fast enough to see the fleeting echoes.

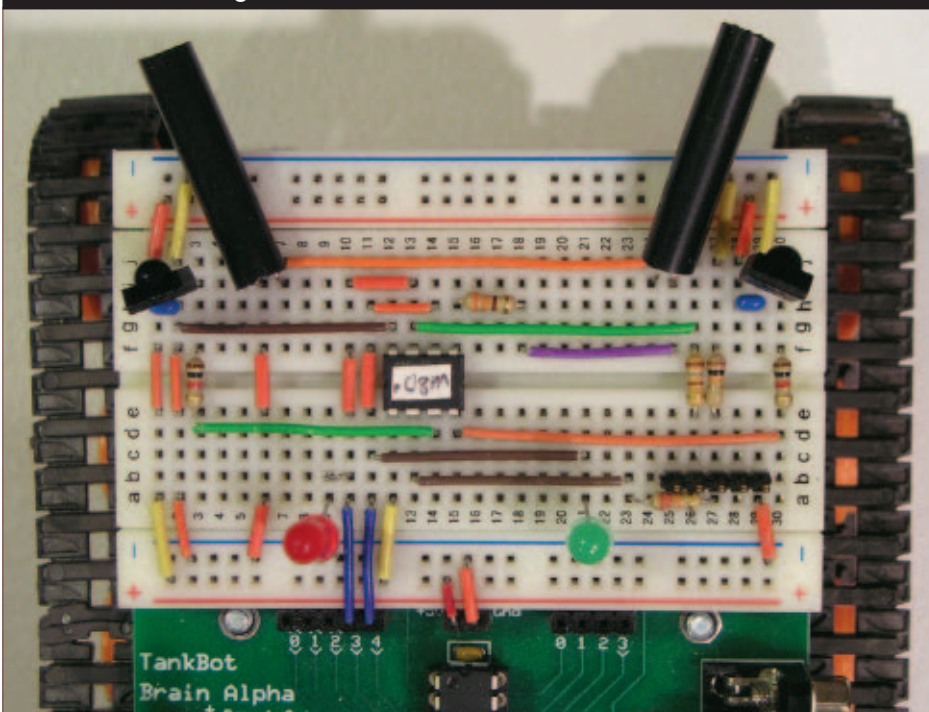
Third, the pwmout command requires some explanation. As the associated comment explains, I used a 42 kHz PWM burst to drive the IR LEDs. As you know, the PNA4602M IR receivers operate at a frequency of 38 kHz so my choice of 42 kHz may seem a little strange at first. When I read the datasheet for the 4602 (available at www.jrhackett.net/IRparts.htm), I discovered that 38 kHz is just the frequency at which the 4602's sensitivity is the greatest. The sensitivity actually varies in a standard "bell-curve" fashion, which means that it decreases as the frequency varies either up or down from the central frequency of 38 kHz. "Detuning" the IR emissions to 42 kHz did help reduce the false-positive echoes, but it still didn't eliminate them entirely.

Before I explain my final solution to the problem (actually more of a "work-around" than a solution), there is one more aspect of the pwmout command to discuss — how to determine the value of the command's two

Figure 5. IR Echo Project Parts List.

Qty.	Part
2	Capacitor, Ceramic, 0.01 μ F
2	IR Detector (PNA4602M)
2	IR LED
1	LED, Green
1	LED, Red
1	PICAXE-08M
1	Programming Setup (See Text)
1	Resistor, 1/4W, 180 Ω
3	Resistor, 1/4W, 330 Ω
2	Resistor, 1/4W, 1K
1	Resistor, 1/4W, 10K
1	Resistor, 1/4W, 22K

Figure 6. IR Echo Circuit on Breadboard.



parameters. In Part II of the PICAXE manual, the documentation for the pwmout command includes formulae for their computation, but there's also an easier way to do it — just use the Programming Editor's "PWM Wizard," located under "PICAXE > Wizard > pwmout" in the menu structure. Be sure to select the 8 MHz option or your parameters will be entirely off base.

As I already mentioned, the above techniques significantly reduced (but did not entirely eliminate) the false-positive echoes. I finally settled on a software approach to coping with the problem, which is implemented in the "for...next" loop in the program. Essentially, the two variables "eLeft" and "eRight" are echo counters. After they have both been initialized to zero, the loop executes five times. In each iteration of the loop, if an echo is detected on the left and/or right side then the appropriate variable(s) is (are)

incremented. When the loop finishes executing, an echo is reported only if the relevant variable equals five. In other words, an echo must be seen five times in a row before it is considered valid. This approach finally eliminated the problem of false-positive echoes.

Of course, your specific setup may have slightly different sensitivity characteristics, so it's a good idea to download the IRecho5_08M.bas program to your TankBot's breadboard circuit and thoroughly test it before allowing your TankBot to roam around freely on his own. If you find that your system reports false-positive echoes, experiment with adjusting the PWM frequency and/or increasing the number of echo reports required to define a valid echo.

Free at Last!

When you are satisfied with the

functionality of your 08M echo-detection system, we're ready to give your TankBot his freedom! The program that will accomplish this goal (TankBotIR.bas) is very straightforward; the included comments clearly explain what it does. Essentially, it moves the TankBot in a forward direction until an obstacle is detected. If the obstacle is on the left, the TankBot should spin clockwise until the coast is clear and then resume moving forward; if the obstacle is on the right, the TankBot should spin counter-clockwise until the coast is clear and then resume moving forward. So, use the Programming Editor to download the program to the 14M on the BrainAlpha printed circuit board (make sure that output 0 of the 08M is connected to input 3 of the 14M and output 1 of the 08M is connected to input 4 of the 14M) and set your TankBot free. **SV**

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The AX2860 is built into a compact 9.0"L x 5.5"W x 1.6"H (228 mm x 140 mm x 40 mm), robust extruded aluminum case, which also serves as a heatsink for its output power stage. The large fin area ensures sufficient heat dissipation for operation without a fan in most applications.

The AX2860 is available now to customers worldwide at \$720 in single quantities, complete with cable and PC-based configuration software. Product information, application examples, and software can be downloaded from the company's website.

For further information, please contact:

Roboteq, Inc.

Website: www.roboteq.com

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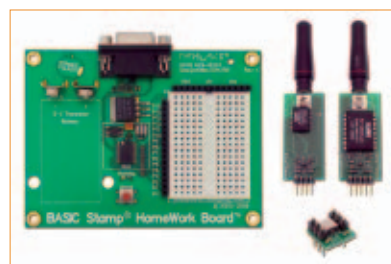
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ROBOT PLATFORMS

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Robotics Squared has a newly released line of animatronic robot eye platforms. The platforms are designed to provide a simple, expandable, low-cost method to give your latest creation that cool factor. Platforms are machined from aluminum and set with realistic plastic eyes. The finished unit has somewhat of a cyborg look to it. Eyelids can be attached to the platform and operate with a scissor-like motion.



Eyes move up/down, left/right simultaneously using two micro servos. A third micro servo operates both eyelids. Eyes range in many colors, from blue, light blue, violet, and hazel brown. Simple design and machining allow users to add servos quickly. Hook the platform to a servo control board and it's ready to animate.

All platforms come fully assembled and ready for servos. The animatronic robot eye platform is available on eBay starting at \$39.99. You can also see it in action at www.youtube.com/watch?v=W2rcVs3luEU.

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ROBOTICS RESOURCES

Tune in each month for a heads-up on where to get all of your "robotics resources" for the best prices!



Setting Up Your Own Robotics Workbench

With the right tools, you can make just about anything. That certainly goes for the fine art of robot building. With proper tools, your robots are more dependable and accurate, and they'll probably look better, too.

In past installments of Robotics Resources, we've looked at tools used to construct robot bodies (see April and November 2008 issues); in this column, we'll take a look at tools used to build the electronic sub-systems of your robot: soldering irons, testing meters, logic probes, and oscilloscopes are among the contenders. As usual, we provide a list of online sources that you can use to further your research.

Soldering Tools and Supplies

If you're doing any kind of custom electronics or wiring on your robot, one of the first tools you'll need is a soldering iron. For the most flexibility, invest in a modular soldering pencil — the kind that lets you change the heating element. For routine electronic work, you should get a 25 to 30 watt heating element. Anything higher may damage electronic components. A 40 or 50 watt element can be used for wiring switches, relays, and power transistors. Stay away from "instant on" soldering irons as they put out too much heat.

Supplement your soldering iron with a soldering stand (for keeping

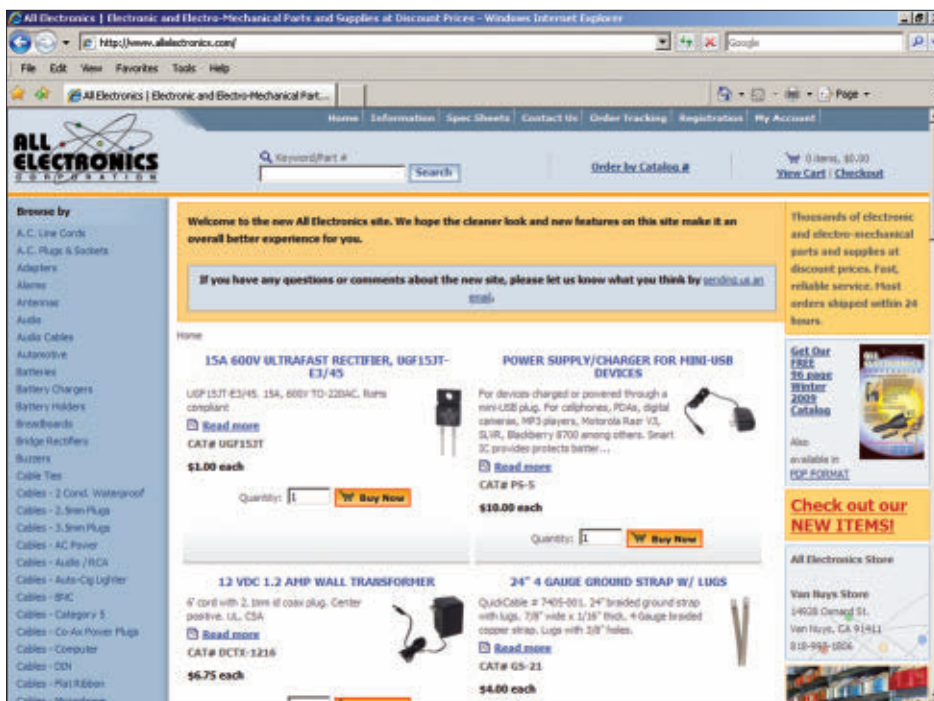
the soldering pencil in a safe and upright position), an assortment of soldering tips of various sizes for small and medium gauge wire, a spool of solder, and a sponge (for keeping the soldering tip clean while you solder).

Additional supplies for a well-rounded soldering kit are: a clip-on heat sink for drawing away the excess heat from sensitive components; a desoldering vacuum tool to soak up molten solder; dental picks for scraping, cutting, forming, and gouging into the work; and a vise or "third hand" to hold parts while you solder.

Volt-Ohm Meters

A volt-ohm meter — also called a multimeter — is used to test voltage levels and the resistance of circuits. This is a moderately priced tool and is the basic requirement for working with electronic circuits of any kind. There are many volt-ohm meters on the market today. For work on robotics, you don't want a cheap model and you don't need an expensive one. A meter of intermediate quality is sufficient and does the job admirably. The price for such a meter is between \$20 and \$50. Shop around

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and compare features and prices.

There are two general types of meters available today: digital and analog. The difference is not that one meter is used on digital circuits and the other on analog circuits. Rather, digital meters employ a numeric display whereas analog meters use the old-fashioned mechanical movement with a needle that points to a set of graduated scales.

Many meters require you to select the range before it can make an accurate measurement. For example, if you are measuring the voltage of a nine volt transistor battery, you set the range to the setting closest to — but above — nine volts (with most meters it is the 20 or 50 volt range). Auto ranging meters don't require you to do this, so they are inherently easier to use. When you want to measure voltage, you set the meter to volts (either AC or DC) and take the measurement. The meter displays the results in the readout panel.

Digital meters vary greatly in the number and type of functions they provide. At the very least, all standard meters let you measure AC volts, DC volts, milliamps, and ohms. Some also

test capacitance and opens or shorts in components like diodes and transistors. These additional functions are not absolutely necessary for building general-purpose robot circuits, but they are handy to have when troubleshooting a circuit that refuses to work.

Meters come with a pair of test leads — one black and one red — each equipped with a needle-like metal probe. The quality of the test leads is usually minimal, so you may want to purchase a better set. The coiled kind are handy. They stretch out to several feet yet recoil to a manageable length when not in use.

Standard leads are fine for most routine testing, but some measurements may require the use of a clip lead. These attach to the end of the regular test leads and have a spring loaded clip on the end. You can clip the lead in place so your hands are free to do other things. The clips are insulated to prevent short circuits.

Logic Probes

Meters are typically used for measuring analog signals. Logic

probes test for the presence or absence of low voltage DC signals that represent digital data. The 0s and 1s are usually electrically defined as zero and five volts, respectively, with most integrated circuits. In practice, the actual voltages of the 0 and 1 bits depends entirely on the circuit. You can use a meter to test a logic circuit, but the results aren't always predictable. Further, many logic circuits change states (pulse) quickly and meters cannot track the voltage switches fast enough.

Logic probes give a visual and (usually) aural signal of the logic state of a particular circuit line. One LED on the probe lights up if the logic is 0 (or LOW); another LED lights up if the logic is 1 (or HIGH). Most probes have a built-in buzzer that has a different tone for the two logic levels. That way, you don't need to keep glancing at the probe to see the logic level.

A third LED or tone may indicate a pulsing signal. A good logic probe can detect that a circuit line is pulsing at speeds of up to 10 MHz, which is more than fast enough for robotic applications — even when using computer control. The minimum detectable pulse width (the time the pulse remains at one level) is 50 nanoseconds (50 billionths of a second); again, more than sufficient.

Although logic probes may sound complex, they are really simple devices and their cost reflects this. You can buy a reasonably good logic probe for under \$20. Most probes are not battery operated; rather, they obtain operating voltage from the circuit under test. You can also make a logic probe, if you wish. A number of project books provide plans.

Successful use of a logic probe really requires you to have a circuit schematic to refer to. Keep it handy when troubleshooting your projects. It's nearly impossible to blindly use the logic probe on a circuit without knowing what you are testing. Since the probe receives its power from the circuit under test, you need to know where to pick off suitable power. To use the probe, connect the probe's

Electronix Express carries all manner of parts and supplies, plus test gear like oscilloscopes.

The screenshot shows the Electronix Express website, a division of RSR Electronics, Inc. The page features a navigation bar with links like HOME, Request Quote, Custom Kits, Hints & Tips, Useful Links, and Ideas Wanted. Below the navigation bar is a search bar and a section titled 'Our Indexes' with links to Product Index, Category Index, Manufacturer's Index, Part Number Index, Web Only Products, and New Products Index. The main content area is divided into 'Popular Categories' (Anti-Static Products, Batteries, Buffers, and Flashlights, Buzzers, Microphones, Speakers, Cable and Wire, Calculators, Capacitors, Cleaners and Degreasers, Clips, Plugs, Jacks, Computers, Connectors, Readers and Cables) and 'FEATURED PRODUCTS'. The featured products include Rigel Oscilloscope Digital Storage - 50MHz, 100MHz (Models DS1052 / DS1102), 01DS1052E 80MHz Scope \$595.00, 01DS1102E 100MHz Scope \$795.00, 01DS1052LCD 80MHz Scope w/Logic An. \$1,195.00, 01DS1102LCD 100MHz Scope w/Logic An. \$1,595.00, and Test Lead Sets (05ALS3 Banana to Alligator \$2.50 \$2.25, 05ALS5 Banana to IC Hook \$3.50 \$2.95). There is also a 'CATALOG 47 Is Now Ready!' section and a 'Web Special' section.



power leads to a voltage source on the board, clip the black ground wire to circuit ground, and touch the tip of the probe against a pin of an integrated circuit or the lead of some other component. For more information on using your probe, consult the manufacturer's instruction sheet.

Oscilloscopes

An oscilloscope is a pricey tool – good ones start at about \$350. For really serious work, however, an oscilloscope is an invaluable tool, one that will save you hours of time and frustration. Things you can do with a scope include some of the things you can do with other test equipment, but oscilloscopes do it all in one box and generally with greater precision. Among the many applications of an oscilloscope, you can:

- Test logic levels
- Check DC or AC voltage levels
- Analyze the waveforms of digital and analog circuits
- Determine the operating frequency of circuit
- Visually check the timing of a circuit to see if things are happening in the correct order and at the prescribed time intervals

A basic, no nonsense model is enough, but don't settle for the cheaper single trace units. A dual trace (two channel) scope with a 20 to 25 MHz maximum input frequency should do the job nicely. The two channels let you monitor two lines at once so you can easily compare the input signal and output signal at the same time. You do not need a scope with storage or delayed sweep, although if your model has these features, you're sure to find a use for them sooner or later.

Scopes are not particularly easy to use; they have lots of dials and controls that set operation. Thoroughly familiarize yourself with

the operation of your oscilloscope before using it for any construction project or for troubleshooting. Knowing how to set the time per division knob is as important as knowing how to turn the scope on.

One of the most important specifications of an oscilloscope is its bandwidth. If 20 MHz is too low for your application, you should invest in a more expensive oscilloscope with a bandwidth of 35, 60, or even 100 MHz. Price goes up considerably as the bandwidth is increased.

The resolution of the scope reveals its sensitivity and accuracy. On an oscilloscope, the X (horizontal) axis displays time and the Y (vertical) axis displays voltage. The sweep time indicates the X axis resolution, generally 0.5 microseconds (millionths of a second) or faster. The sweep time is adjustable so you can test signal events that occur over a longer time period, usually as long as half a second to one second.

The sensitivity indicates the Y axis resolution. The sensitivity of most average priced scopes is about five millivolts (mV) to five volts. You turn a dial to set the sensitivity you want.

When you set the dial to 5 mV, each tick mark on the face of the scope tube represents a difference of 5 mV. Voltage levels lower than 5 mV may appear, but they cannot be accurately measured. Most scopes will show very low level voltages (microvolt range) as a slight ripple.

Over the years, oscilloscopes have improved dramatically, with many added features and capabilities, with these being the most useful features:

- *Delayed sweep* is helpful when analyzing a small portion of a long, complex signal.
- *Digital storage* records signals in computerized memory for later recall. Once in the memory, you can expand the signal and analyze specific portions.
- *Selectable triggering* lets you choose how the scope will trigger on the input signal. When checking DC signals, no triggering is necessary, but AC and digital signals require that you select a specific part of the signal so that the scope can properly display the waveform.

Frys.com is a superstore for all things electronic. Local stores in California, Texas, Arizona, Georgia, Illinois, Indiana, Nevada, Oregon, and Washington.

The screenshot shows the Frys.com website interface. At the top, there's a navigation bar with links like 'PC Computers', 'Apple Computers', 'Notebooks/Tablets', etc. Below this is a search bar and a 'Frys Top Picks' section. The main content area features several promotional banners and product listings. One prominent banner is for 'DESKTOP PC FEATURING' with a price of \$299.99. Another banner mentions 'FREE SHIPPING PROMOTION'. There are also sections for 'Frys Top Picks' and 'Advised Specials'. The bottom of the page shows more product listings with prices and 'Add to Cart' buttons.



ROBOTICS RESOURCES

The probes used with oscilloscopes are not just wires with clips on the end of them. To be effective, the better scope probes use low capacitance/low resistance shielded wire and a capacitively compensated tip. These ensure better accuracy.

Most scope probes are passive, meaning that they employ a simple circuit of capacitors and resistors to compensate for the effects of capacitive and resistive loading. Many passive probes are switchable between 1X and 10X. At the 1X setting, the probe passes the signal without attenuation (weakening). At the 10X setting, the probe reduces the signal strength by 10 times. This allows you to test a signal that might otherwise overload the scope's circuits.

As an alternative to a stand-alone oscilloscope, you may wish to consider a PC-based oscilloscope solution which not only costs less but may provide more features such as long-term data storage. A PC-based oscilloscope uses your computer and software running on it as the active testing component.

Most PC-based oscilloscopes are comprised of an interface card or adapter; the card/adapter connects to your PC via an expansion board or a

serial, parallel, or USB port (different models connect to the PC in different ways). A test probe then connects to the interface. Software running on your PC interprets the data coming through the interface and displays the results on the monitor.

Breadboards

You'll probably want to test any circuit you design or build before you commit to soldering or wire-wrapping (discussed next). This may be done on a solderless breadboard. Solderless breadboards consist of a series of holes with internal contacts spaced one tenth of an inch apart — just the right spacing for ICs. You plug in ICs, resistors, capacitors, transistors, and 20 or 22 gauge wire in the proper contact holes to create your circuit.

Solderless breadboards come in many sizes. For the most flexibility, get a double width board — one that can accommodate at least 10 ICs. Smaller boards can be used for simple projects; circuits with a high number of components require bigger boards. You can choose to buy a breadboard with an integrated power supply or one without. On

most models, the power supply provides one or two amps of regulated five volts; some also deliver 12 volts or can be adjusted to any voltage in between.

When you're buying a breadboard, also purchase a set of pre-stripped wires. The wires come in a variety of length, and are already stripped and bent for use in breadboards. The set costs \$5 to \$7, but you can bet the price is well worth it.

Wire Wrapping Tools

Making a printed circuit board for a one shot application is time consuming, though it can be done with the proper kits and supplies. Conventional point-to-point solder wiring is not an acceptable approach when constructing digital circuits, which represent the lion's share of electronics you'll be building for your robots.

The preferred construction method is to use wire wrapping. Wire wrapping is a point-to-point wiring system that uses a special tool and extra fine 28 or 30 gauge wrapping wire. When done properly, wire wrapped circuits are as sturdy as soldered circuits, and you have the added benefit of being able to go back and make modifications and corrections without the hassle of desoldering and resoldering.

To use a wire wrapping tool, you insert one end of the stripped wire into a slot in the tool and place the tool over a square shaped wrapping post. Give the tool five to 10 twirls and the connection is complete. The edges of the post keep the wire anchored in place. To remove the wire, you use the other end of the tool and undo the wrapping.

A number of different wire wrapping tools are available. Some are motorized and some automatically strip the wire for you, freeing you of this task and of purchasing the more expensive pre-stripped wire. The basic manual tool is recommended for initial use. You can graduate to other tools as you become proficient. Wrapping wire comes in many forms, lengths, and colors, and you need to

Mouser Electronics stocks thousands of parts, tools, and testing equipment.





use special wire wrapping sockets and posts.

Sources

Here are some online retailers that supply electronic testing and construction tools.

All Electronics **www.allelectronics.com**

All Electronics (local stores in Los Angeles; catalog mail order elsewhere) offers soldering tools, meters, and other testing supplies.

B.G. Micro **www.bgmicro.com**

A haven for the electronics tinkerer and robotics enthusiast. Much of the stock is surplus.

Online sales through web catalog; printed catalog available.

Circuit Specialists **www.circuitspecialists.com**

Good selection of solder tools and supplies, prototyping boards.

DesignNotes.com **www.designnotes.com**

Test — including a two-channel PC-based oscilloscope — as well as prototyping tools and supplies.

Digi-Key **www.digikey.com**

Digi-Key is one of the largest mail order retailers/distributors of electronic components in North America. They offer a very fast and efficient online ordering system, complete with links to datasheets (when available).

Electronic Goldmine **www.goldmine-elec.com**

Test equipment, soldering tools, and new and used electronic components (LEDs, potentiometers, resistors, heatsinks, transistors, etc.).

Electronix Express **www.elexp.com**

Test equipment and soldering tools, plus new and surplus electronics, including passive components, motors, relays, and more.

Elenco Electronics **www.elenco.com**

As noted on their website, Elenco is a major supplier of electronic test equipment and educational material to many of the nation's schools and hobbyists (sold through distributors).

Fair Radio Sales **www.fairradio.com**

Though specializing in surplus for ham radio, Fair Radio also offers plenty of general electronics and test equipment. Online sales with web catalog; a printed catalog is available.

Fry's Electronics **www.frys.com**

Fry's is an electronics superstore chain operating primarily on the west coast. They offer a subset of products via their web page, including electronics parts and kits.

Jameco **www.jameco.com**

Large mail order company specializing in electronic parts and supplies for both hobbyists and pros.

Jaycar Electronics **www.jaycarelectronics.com**

Wide selection of components, test gear, prototyping, soldering equipment, and more.

HSC Electronic Supply **www.halted.com**

HSC sells new and surplus electronics, and they have a large assortment of new soldering tools and supplies, as well as other testing gear.

HobbyLab **www.hobbylab.us**

USB-based oscilloscopes, including the low-cost DiSCO and associated PC software.

Marlin P. Jones & Associates, Inc. **www.mpja.com**

MPJA sells both new and surplus electronic and mechanical products, including soldering stations and test tools.

MECI — Mendelson's Liquidation Outlet **www.meci.com**

Surplus electronics, motors, and even a special section for combat robot parts — large motors, batteries, that sort of thing. Online sales with web catalog.

Mouser Electronics **www.mouser.com**

Very large "stocking distributor" of all things electronic. Good source for higher-end soldering stations and tools.

Parts Express **www.partsexpress.com**

Parts Express is an all-around electronics retailer, selling everything from sound systems to test equipment, to stage lighting and electronic components.

RadioShack **www.radioshack.com**

These days, the Shack has fewer components and other electronic items for sale at each store, but they do carry the basics — common value resistors, capacitors, switches, solder, electronics construction tools, that sort of thing. Additional items can be ordered through the RadioShack online store.

Ramsey Electronics **www.ramseyelectronics.com**

Though mostly known for their kits, Ramsey also provides all the soldering tools and supplies you need to build them — and most anything else.

Saelig **www.saelig.com**

Product line includes stand-alone and PC-based oscilloscopes and other test equipment, and electronics construction tools. **SV**

CONTACT THE AUTHOR

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Then and NOW

LARGE ROBOTS

b y T o m C a r r o l l

Last month, we discussed little robots — from nanobots that can't be seen with our naked eyes to the smaller types that many experimenters and hobbyists build. Fortunately for most experimenters, smaller is usually cheaper, but not always. Tiny machines are certainly a lot harder to work on, as anybody who's tried to solder (and unsolder) SMT ICs with a hundred-plus pins knows. The mechanical part is even harder as one might need the skills of a watchmaker experience to accomplish many tasks.

Large robots, on the other hand, are usually easier to construct as basic tools can normally be used to cut metal, fasten fasteners, and position

the parts. However, due to the larger size, large machines of any type need large motors and more beefy power supplies. These larger items cost more money and that may limit going larger. Some of my best robots were in the hundred to two hundred pound class, human-sized anthropomorphic machines. I've always enjoyed building large bots that can move about, pick up things with 'hands,' and look a bit humanoid, but the cost of their components is quite a bit higher. It is this higher cost that keeps most experimenters from building human-size or larger robots.

Honda's Asimo Stars in Rose Parade

I'm not usually wide awake on New Year's Day after staying up the night before, but my wife had the 2009 Rose Parade on TV and I was astonished to see a 49 foot Honda Asimo as the lead-off float. Honda has never been known to build anything second rate and this giant robot replica was no exception. Rising from a 16 foot prone position, the robot stood upright in just two minutes. Honda was celebrating 50 years of doing business in the US and this float was titled "Hats Off in Celebration," in keeping with the parade's theme.

Asimo removed his hat as a gesture to the crowd and a pyrotechnics display emanated from his hat. The whole 35,000 pound float was driven from the rear and visually controlled by a spotter in the front. A 454 cid V-8 engine drove the float and the hydraulic systems, with separate generators to supply electrical power (see Figure 1). Honda has always been innovative with their entries in the Rose Parade. Last year's entry was a floral Honda Ridgeline truck with Asimo at the wheel.

NASA JPL/Caltech Robot in 2005 Parade

Honda was not the only Rose Parade float to feature a giant robot. NASA JPL and Caltech in Pasadena, CA, featured a giant 50 foot tall robot in the 2005 event. "A Family of Explorers" represented a compilation of nine exploration spacecraft built by JPL to explore the Earth's environs and deep space.

Figure 2 shows the robot standing on two of the most famous of these spacecraft — Spirit and Opportunity. On the robot's left leg is the spacecraft Jason-1 (designed to study our oceans) and on the other leg is the Genesis Spacecraft on a mission to explore the Sun. The right arm is GALEX — the Galaxy Evolution Explorer telescope — and the left arm is the Mars Global Surveyor, studying Mars from orbit. On the belly of the robot is Stardust (that grabbed comet samples and sent them back to Earth)



FIGURE 1. Honda float featuring Asimo.

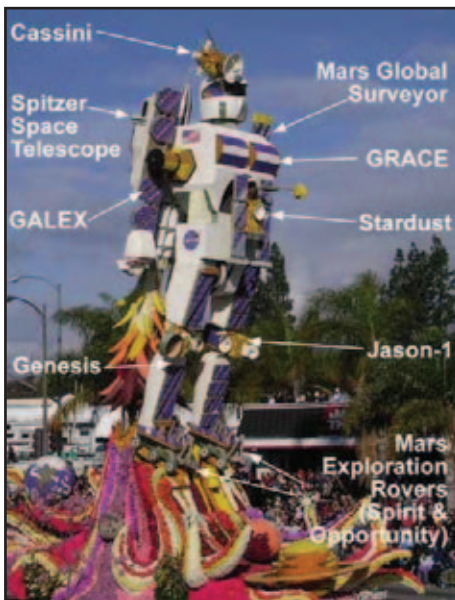


FIGURE 2. NASA JPL float.

and the chest is GRACE (the Gravity Recovery and Climate Experiment that used a new approach to study the oceans' role in climate). On the back of the right arm is the Spitzer Space Telescope, designed to study the universe in infrared light.

The Cassini spacecraft with the Huygens Probe — built by the European Space Agency to study Saturn and its moon, Titan — tops off the robot's head like a hat. The robot's head moved side to side and could lean forward 15 degrees. To go under a freeway overpass, it had to bend down to only 17-1/2 feet and used a 5,000 pound counter-weight to keep it from tipping over in the process.

Dextre Completes the Space Station's Mobile Servicing System

Turning from fantasy to reality, there are some quite large robots designed to operate in space. Another NASA robot that is definitely large and expensive is Dextre, part of the even larger Mobile Servicing System for the Space Station. Figure 3 shows the robot mounted to the Space Station and Figure 4 is a photo of Dextre mounted on the end of Canadarm2. In March of last year, NASA delivered



FIGURE 3. Dextre on the Space Station.



FIGURE 4. Dextre on Canadarm2.

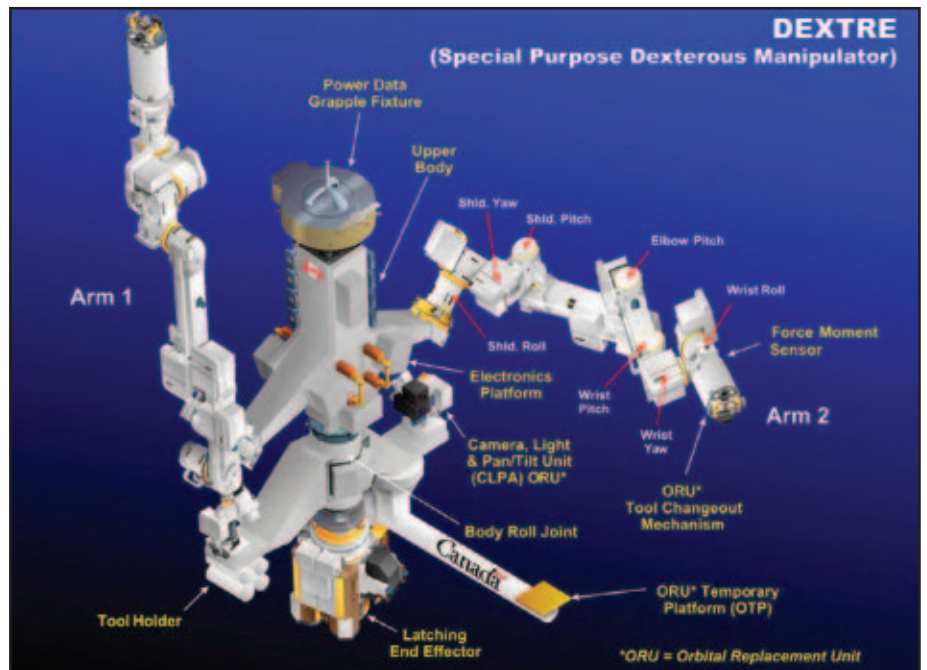


FIGURE 5. Dextre details.

the Canadian Space Agency's Special Purpose Dexterous Manipulator to the Space Station. Dextre cost over \$274 million, weighs 3,664 pounds (1,662 kilograms), and had to have its nine parts assembled on orbit during the Shuttle Endeavour's STS-123 mission.

When latched to a mobile platform attached to the space station, Dextre can perform many of the manipulative tasks that used to require one or more suited astronauts to accomplish. An astronaut inside the shuttle or even earthbound can now do the same tasks without the exposure to the hazards of space. Attached to the new Canadarm2 launched in April 2001, these two unique robotic components and the Mobile Base System offer extreme payload handling capabilities for the

Space Station. The use of robotic techniques allows operation from the safety of a comfortable station and allows astronauts to perform many more manipulative operations in a given time period. The replacement of failed ORUs (orbital replacement unit), assembly of structures, routine maintenance, and manipulation of payloads are just a few of the many tasks Dextre can do. This allows crew members more time to perform science tasks and experiments within the confines of the station. Check out the diagram of Dextre in Figure 5.

Each of Dextre's seven-jointed (axes of freedom of motion) arms can bend in multiple angles to allow gripping and manipulation of required Space Station items. A suite of tools on the robot can be retrieved by one

Technical Detail	Remote Manipulator System	Dexterous Manipulator	Base System
Arm Length	17.6 meters (57.7 feet)	3.5 meters (11.48 feet) linear stroke	5.7 meters x 4.5 meters x 2.9 meters (18.7 feet x 14.76 feet x 9.5 feet)
Mass (approx.)	1,800 kilograms (3,968 pounds)	1,662 kilograms (3,664 pounds)	1,450 kilograms (3,196.7 pounds)
Mass Handling/Transportation Capacity	116,000 kilograms (255,736 pounds)	600 kilograms (1,322.77 pounds)	20,900 kilograms (46,076.61 pounds)
Degrees of Freedom	7	15	Fixed
Peak Power (operational)	2,000 W	2,000 W	825 W
Avg. Power (keep alive)	435 W	600 W	365 W
Applied Tip Load Range	0-1,000 N	0-111 N	N/A
Stopping Distance (under max. load)	0.6 meters (1.96 feet)	0.15 meters (5.9 inches)	N/A

Chart 1. Mobile Servicing System Capabilities.

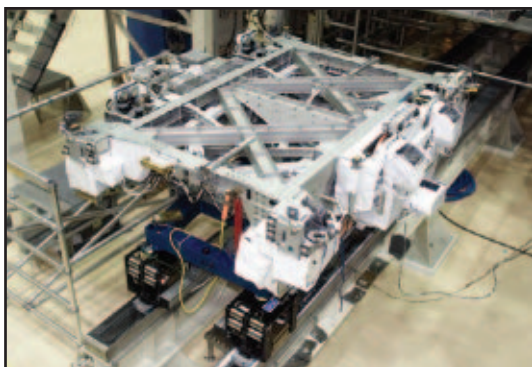


FIGURE 6. Mobile transporter.

or both of the arms, and with the aid of four on-board TV cameras and lights, precise repair and assembly tasks can be performed. On-board force-feedback sensors allow the robot and the controlling astronaut to actually 'feel' assembly and repair tasks. Dextre can also be attached to the Remote Manipulator System robot arm to be directed into tight spaces as necessary for specific tasks. It can also ride the Mobile Base System rails to a position on the Space Station's structure and use the RMS arm to hand it payloads, tools, or act as a separate light, camera, or even a spare hand.

The robot resembles a human somewhat in that it has two arms mounted to a trunk that swivels on a base. These 3.5 meter long arms can delicately handle 1,327 pound payloads. The 'hands' or end-effectors have parallel jaw grippers that can hold, rotate, and manipulate objects. Each arm also has a retractable motorized socket wrench. Software allows only one arm to move at a

time thus preventing collisions between the arms. Chart 1 describes the different parts of the new Mobile Servicing System's capabilities which include the Canadarm2 (installed in 2001), Dextre, and the mobile base.

Figure 6 shows the mobile transporter built by Northrop-Grumman that was delivered to the Space Station back in April 2002. The new Canadian Remote Manipulator Systems

have been so successful on the shuttles and Space Station that NASA is planning on using a variation on the new Orion spacecraft that is slated to replace the Space Shuttle in 2010.

Larger NASA Rovers

When we think of space robotics, many of us tend to think of the lunar and Mars rovers that have quite a bit of autonomy within their mobile bodies. The early Sojourner rover on Mars was a petite 24 pounds; probably close to the size that many of us have constructed for experimentation or even as a combat robot. Sojourner was the key component to NASA's 1997 Pathfinder mission that was so successful, but the need for greater range and sensor/experiment carrying capacity called for a larger rover.

The very successful Spirit and Opportunity rovers weigh in at 384 pounds apiece and managed to travel many miles on the surface of Mars —

even with a jammed wheel that Spirit had to drag behind itself for many months.

Typical of many NASA JPL designed spacecraft, these two rovers long outlasted their life expectancy by many times over, but did suffer lack of usefulness in Mars' winter due to the cold and lack of sun for their photovoltaic cells. Figure 7 shows a comparison of the two different rovers. NASA's Mars Science Laboratory (shown in an artist's conception in Figure 8) is more of the size of a small car. Its design overcomes one of the Mars rover's greatest challenges — the ability to operate in cold environments with no direct sunlight on its surface. Mars has extreme temperature fluctuations from day to night, ranging from a comfortable 86 degrees F in the brightest sun to a -197 degrees at night. You also might note that the rover does not have any solar panels as it is powered by an RTG (radioisotope thermoelectric generator).

An RTG uses the decay of plutonium 238 as a heat source for the rover and also to generate electricity by thermopiles. This technology is not new on Mars spacecraft as it was used on the early Viking landers and as heaters for Spirit and Opportunity. An extensive system of tubing, a radiator, and a pump circulates heated fluid to needed components or excess heat to the radiator on the back of the rover.

Robosaurus

People have been wowed since 1990 by Robosaurus: the 62,000 pound, car-eating 'robot' that was sold in early 2008 for \$575,000 (see Figure 9). Robosaurus is able to reach down and pick up a hapless car in its hydraulically-powered jaws and crush it with 20,000 pounds of force before tossing it aside and setting it on fire. As sort of a celebration afterwards, it lets out a roar and a blast of flames from its nostrils that the crowd can feel in the stands.

Figure 10 shows the very complex cockpit instrument layout. A complete set of show equipment was



FIGURE 7. Spacecraft size comparison.

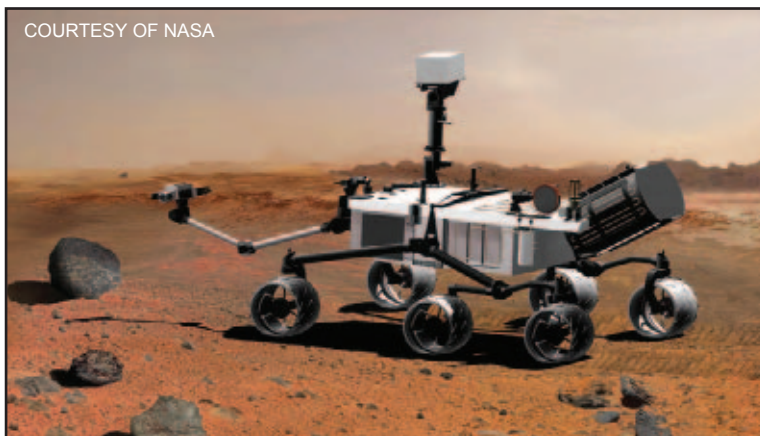


FIGURE 8. Mars science lab rover.



FIGURE 9. Robosaurus.

included that consists of an on-board programming console for the animation features, remote TV cameras, an amateur radio station for broadcasting, special tools, and even a set of operation and maintenance manuals. A separate semi-trailer, spare parts and shop equipment were 'extra.' It would take \$5 million to build another one at today's prices.

What are Large Robots?

Just what exactly is considered a large robot? The term 'large' is just about as ambiguous as the term 'robot.' Add the two together and you have multiple ambiguity. Comparing a parade's float that resembles a real robot to an actual industrial robot such as Kuka's Titan in Figure 11 is like

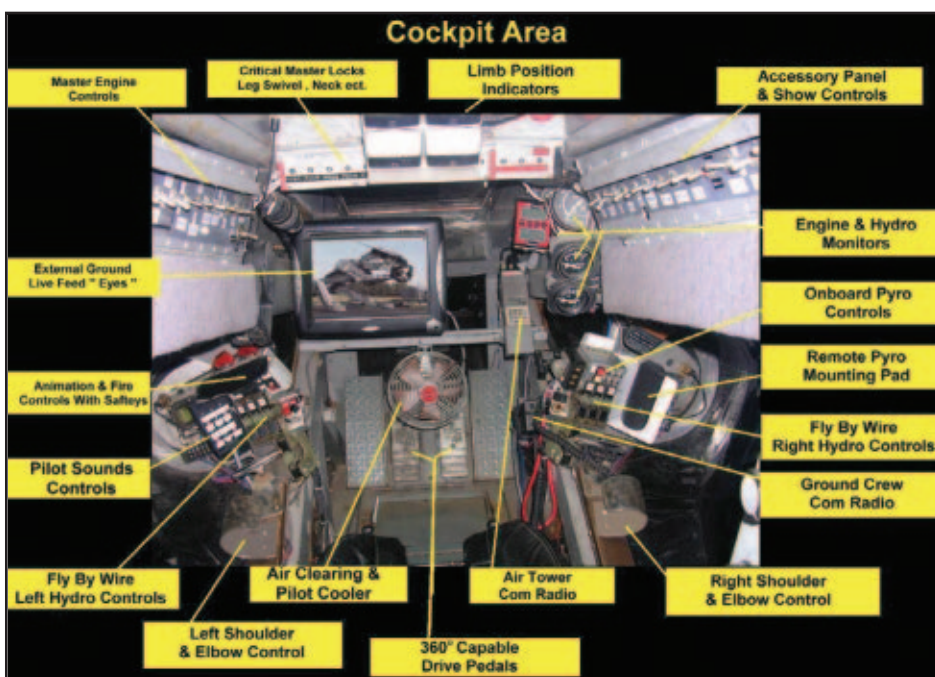


FIGURE 10. Robosaurus cockpit.

comparing apples with aardvarks; they both begin with 'A', but that's where the similarity ends.

Large shipyards in Germany and around the world use very large AGVs (automated guided vehicle) to maneuver one or more fully-loaded containers around a huge yard of thousands of containers without a human in sight. Even huge earthmoving machines in mines are slowly becoming automated. The use of large robotic machines can mean replacing personnel or protecting personnel from dangerous tasks.

The next time you set out to design a new robot, why not scale it



FIGURE 11. Kuka Titan.

up a bit? Your 'large' robot may not fit someone else's idea of large, but you'll open up a new avenue of robotics for yourself. **SV**

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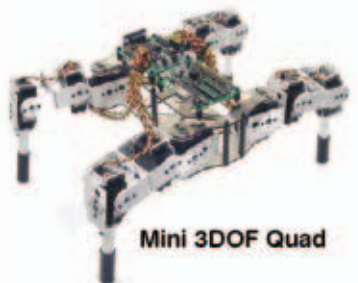
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Mini 2DOF Hex



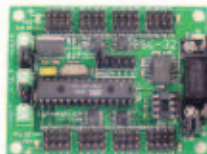
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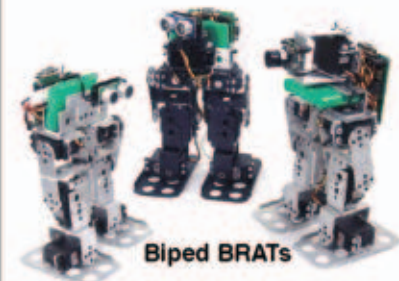
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